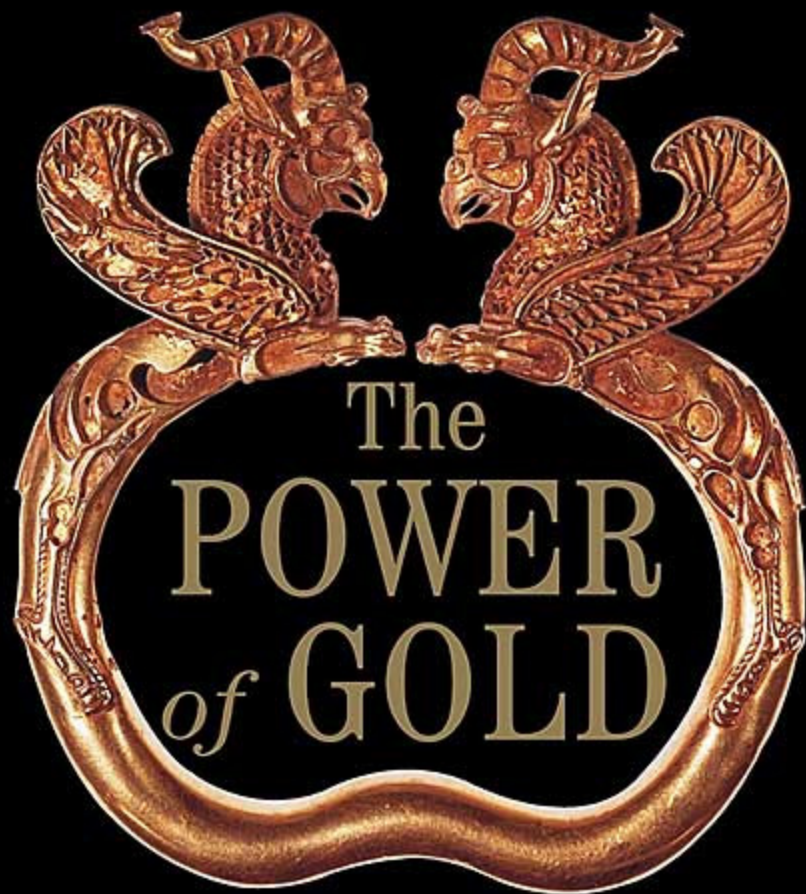


PETER L. BERNSTEIN



The History of an Obsession

ILLUSTRATED EDITION

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The Power of Gold

THE HISTORY OF AN OBSESSION

PETER L BERNSTEIN

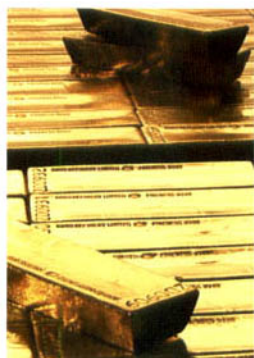
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The Power of Gold

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of Gold

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For Barbara, once again and always.

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Contents



PROLOGUE: The Supreme Possession	I
1. Get Gold at all Hazards	13
2. Midas's Wish and the Creatures of Pure Chance	25
3. Darius's Bathtub and the Cackling of the Geese	45
4. The Symbol and the Faith	59
5. Gold, Salt and the Blessed Town	69
6. The Legacy of Eoba, Babba and Udd	77
7. The Great Chain Reaction	87
8. The Disintegrating Age and the Kings' Ransoms	99
9. The Sacred Thirst	113
10. The Fatal Poison and Private Money	135
11. The Asian Necropolis and Hien Tsung's Inadvertent Innovation	157
12. The Great Recoinage and the Last of the Magicians	173
13. The True Doctrine and the Great Evil	193
14. The New Mistress and the Cursed Discovery	207
15. Strength and Failure	221
16. Breaking the Golden Fetters	235
17. Tearing Loose the Golden Anchor	263
Bibliography	283
Index	295



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PROLOGUE

The Supreme Possession

AT THE END OF THE 19TH CENTURY, JOHN RUSKIN TOLD THE STORY OF A MAN WHO BOARDED A SHIP CARRYING HIS ENTIRE WEALTH IN A LARGE BAG OF GOLD COINS. A TERRIBLE STORM CAME UP A FEW DAYS INTO THE VOYAGE AND THE ALARM WENT OFF TO ABANDON SHIP. STRAPPING THE BAG AROUND HIS WAIST, THE MAN WENT UP ON DECK, JUMPED OVERBOARD, AND PROMPTLY SANK TO THE BOTTOM OF THE SEA. ASKS RUSKIN: 'NOW, AS HE WAS SINKING, HAD HE THE GOLD? OR HAD THE GOLD HIM?'



John Ruskin

This book tells the story of how people have become intoxicated, obsessed, haunted, humbled, and exalted over pieces of metal called gold. Gold has motivated entire societies, torn economies to shreds, determined the fate of kings and emperors, inspired the most beautiful works of art, provoked horrible acts by one people against another, and driven men to endure intense hardship in the hope of finding instant wealth and annihilating uncertainty.

'Oh, most excellent gold!' observed Columbus while on his first voyage to America. 'Who has gold has a treasure [that] even helps souls to paradise.' As gold's unquenchable beauty shines like the sun, people have turned to it to protect themselves against the darkness ahead. Yet we shall see at every point that Ruskin's paradox arises and challenges us anew. Whether it is Jason in search of the Golden Fleece, the Jews dancing around the golden calf, Croesus fingering his golden coins, Crassus murdered by molten gold poured down his throat, Basil Bulgartoctonus with over two hundred thousand pounds of gold, Pizarro surrounded by gold when slain by his henchmen, Sutter whose millstream launched the California gold rush, or modern leaders such as Charles de Gaulle who deluded themselves with a vision of an economy made stable, sure, and superior by the ownership of gold – they all had gold, but the gold had them all.

When Pindar in the 5th century BC described gold as 'a child of Zeus, neither moth or rust devoureth it, but the mind of man is devoured by this supreme possession,' he set forth the whole story in one sentence. John Stuart Mill nicely paraphrased this view in 1848, when he wrote 'Gold thou mayst safely touch; but if it stick/Unto thy hands, it woundeth to the quick.' Indeed, gold is a mass of contradictions. People believe that gold is a refuge until it is taken seriously; then it becomes a curse.



Underside of the Eiffel Tower: If all the gold that has ever been mined in the world were put into a single pile, it would fit in the space beneath the Eiffel Tower.

Nations have scoured the earth for gold in order to control others only to find that gold has controlled their own fate. The gold at the end of the rainbow is ultimate happiness, but the gold at the bottom of the mine emerges from hell. Gold has inspired some of humanity's greatest achievements and provoked some of its worst crimes. When we use gold to symbolize eternity, it elevates people to greater dignity – royalty, religion, formality; when gold is regarded as life everlasting, it drives people to death.

Gold's most mysterious incongruity is within the metal itself. It is so malleable that you can shape it in any way you wish; even the most primitive of people were able to create beautiful objects out of gold. Moreover, gold is imperishable. You can do anything you want with it and to it, but you cannot make it disappear. Iron ore, cow's milk, sand, and even computer blips are all convertible into something so different from their original state as to be unrecognizable. This is not the case with gold. Every piece of gold reflects the same qualities. The gold in the earring, the gold applied to the halo in a fresco and the gold bars hidden away in the United States' official cookie jar at Fort Knox are all made of the same stuff.

Despite the complex obsessions it has created, gold is wonderfully simple in its essence. Its chemical symbol AU derives from aurora, which means 'shining dawn,' but despite the glamorous suggestion of AU, gold is chemically inert. That explains why its radiance is forever. In Cairo, you will find a tooth bridge made of gold for an Egyptian 4500 years ago, its condition good enough to go into your mouth today. Gold is extraordinarily dense; a cubic foot of it weighs half a ton. In 1875, the English economist Stanley Jevons observed that the £20 million in transactions that cleared the London Bankers' Clearing House each day would weigh about 157 tons if paid in gold coin 'and would require eighty horses for conveyance.' The density of gold means that even very small amounts can function as money of large denominations.

Gold is almost as soft as putty. The gold on Venetian glasses was hammered down to as little as five-millionths of an inch – a process known as gilding. In an unusually creative use of gilding, King Ptolemy II of Egypt (285–246 BC) had a polar bear from his zoo lead festive processions in which the bear was preceded by a group of men carrying a gilded phallus 180 feet tall. You could draw an ounce of gold into a wire fifty miles in length, or, if you prefer, you could beat that ounce into a sheet that would cover one hundred square feet.

Unlike any other element on earth, almost all the gold ever mined is still around, much of it now in museums bedecking statues of the ancient gods and their furniture or in numismatic displays, some on the pages of illustrated manuscripts, some in gleaming bars buried in the dark cellars of central banks, a lot of it on fingers, ears, and teeth. There is a residue that rests quietly in shipwrecks at the bottom of the seas. If you piled all this gold in one solid cube, you could fit it aboard any of today's great oil tankers; its total weight would amount to approximately 125,000 tons, an insignificant volume that the

US steel industry turns out in just a few hours; the industry has the capacity to turn out 120 million tons a year. The ton of steel commands \$550 – 2¢ an ounce – but the 125,000 tons or so of gold would be worth a trillion dollars at today's prices.

Is that not strange? Out of steel, we can build office towers, ships, automobiles, containers, and machinery of all types; out of gold, we can build nothing. And yet it is gold that we call the precious metal. We yearn for gold and yawn at steel. When all the steel has rusted and rotted, and for ever after that, your great cube of gold will still look like new. That is the kind of longevity we all dream of.

Stubborn resistance to oxidation, unusual density, and ready malleability – these simple natural attributes explain all there is to the romance of gold (even the word gold is nothing fancy: it derives from the Old English *gelo*, the word for 'yellow'). This uncomplicated chemistry reveals that gold is so beautiful it was Jehovah's first choice for the decoration of his tabernacle.

Inca Gold



RIGHT: *Statue of the
Buddha in Bangkok,
Thailand*

God issued those orders many thousands of years ago. What is the place of gold in the modern world of abstract art, designer jeans, complex insurance strategies, computerized money, and the labyrinths of the Internet? Does gold carry any significance in an era where traditions and formality are constantly crumbling beyond recognition? In a global economy managed increasingly by central bankers and international institutions, does gold matter at all?

Only time can tell whether gold as a store of monetary value is truly dead and buried, but one thing is certain: the motivations of greed and fear, as well as the longings for power and for beauty, that drive the stories that follow are alive and well at this very moment. Consequently, the story of gold is as much the story of our own time as it is a tale out of the past. From poor King Midas who was overwhelmed by it to the Aly Khan who gave away his weight in gold every year, from the dank mines of South Africa to the antiseptic cellars at Fort Knox, from the gorgeous artworks of the Scythians to the Corichancha of the Incas, from the street markets of Bengal to the financial markets in the City of London, gold reflects the universal quest for eternal life — the ultimate certainty and escape from risk.

The key to the whole tale is the irony that even gold cannot fulfill that quest. Like Ruskin's traveler jumping off the boat, people take the symbolism of gold too seriously. Blinded by its light, they cashier themselves for an illusion.

Gold foil







Get Gold

IF GOLD WERE MORE PLENTIFUL ON EARTH—SAY, AS ABUNDANT AS SALT—IT WOULD BE FAR LESS VALUABLE AND INTERESTING, DESPITE ITS UNIQUE PHYSICAL ATTRIBUTES AND BEAUTY. YET GOLD HAS BEEN DISCOVERED ON EVERY CONTINENT ON EARTH.

ONE

at all Hazards



That sounds like a contradiction, but it is not. Although gold deposits are widespread, in one form or another, no one area has yielded its gold easily. Finding and producing gold demands immense effort relative to the amount of glittering yellow metal that makes its appearance at the end of the process.

For example, in order to extract South Africa's annual output of around 500 tons of gold, some 70 million tons of earth must be raised and milled – an amount greater than all the material in the pyramid of Cheops. The South African mines are the worst, but we are all familiar with the tales of the Forty-Niners panning day after day in the waters of California and ending up with nothing but a few dribblets of gold. As Will Rogers put it after returning from a visit to the Klondike, 'There is a big difference between prospecting for gold and prospecting for spinach.'

This radically distorted ratio of effort to output appears to have done little to discourage people from pursuing the worldwide search for gold – perhaps the most telling evidence of how highly prized, vital, essential, and irresistible gold has been from the earliest of times. Even in myths, the quest for gold was gluttonous.

Although gold does not mix with other metals, thin veins of it are scattered throughout the mountains where granite and quartz have filled in cracks in the earth's crust and have been pressed together by fierce heat over millions of years. The elements have washed, blown, and scattered these deposits over the years, but gold has retained most of its purity even as it has suffered the ravages of nature's dynamics. Much of this gold has flowed downward in mountain streams. Gold's high density and weight tend to separate it from the other material in the waters, where it drifts to the bottom as nuggets or flows along as fine as dust.

Relative to the needs for it, gold does appear to have been more plentiful in ancient times, especially in Egypt and the Near East, than it has been since the Roman era. A little bit of gold goes a very long way when it is used only for adornment and decoration and not for coinage or hoarding: mining by the Egyptians produced only about one ton

ABOVE
Red hot ingots of gold

LEFT
St Sophia, Turkey

annually. Until the development of coinage, which put gold into the hands of the masses and greatly expanded the need for it, most of the available gold was owned by monarchs and priests. Its use was ceremonial in large part, a medium for advertising power, wealth, eminence, and proximity to the gods. Whatever was left over was used for jewelry and other forms of personal adornment.

When Moses came down from Mount Sinai to deliver the Ten Commandments to his people, he found the Jews in a delirium worshipping a golden calf. He was so enraged to see them bowing to an icon like those worshipped by the hated Egyptians that he smashed the tablets inscribed with the Word of God – the Ten Commandments – which he had just brought down from Mount Sinai. The story reveals that the Jews, even as slaves, had ample amounts of gold on their persons. It never occurred to them to use their gold to bribe themselves out of captivity in Egypt; as gold was not yet perceived as money, they would have found few takers. Until they melted their gold into the golden calf, they adorned their ears, arms, and necks with it.

The more than 400 additional references to gold in the Bible confirm how plentiful gold was at that time. Poor Job declaims, 'If I have made gold my hope, or have said to the fine gold, 'Thou art my confidence'; If I rejoiced because . . . my hand had gotten much. . . . This also were an iniquity to be punished by the judge; for I should have denied the God that is above.' Abraham, the founder of the Jewish nation, is described in Genesis 13 as 'rich in cattle, and in silver, and in gold.' He furnished the servant who went to fetch Rebecca with vessels of gold, including a nose ring.

When Moses climbed Mount Sinai to receive the Word from God, God gave him a lot more to do than just transmit the Ten Commandments and many associated rules and obligations. God also issued precise directions for the construction of a sanctuary where the Jews were to worship Him, together with a tabernacle to go inside the sanctuary. God began right off by specifying that 'thou shalt overlay it with pure gold, within and without shalt thou overlay it, and shalt make upon it a crown of gold round about.' That is just the beginning: God even ordered that the furniture, fixtures, and all the decorative items such as cherubs were to be covered in pure gold.

Once settled in the Promised Land, the Jews must have accumulated masses of gold, primarily from plundering the tribes they had defeated in battle. Moses and his troops took over 300 pounds of gold from the Midianites, 'jewels of gold, ankle-chains and bracelets, signet-rings, earrings, and armlets.' Gold gleamed from the walls of the interior of Solomon's great temple (located near the Wailing Wall of modern Jerusalem), which was 135 feet long, 35 feet wide, 50 feet high and divided into three chambers. Solomon enjoyed lavishing gold on his personal possessions as well: his shields were made of gold, his ivory throne was overlaid with gold, and he sipped his wine from golden vessels. When the queen of Sheba came to visit Solomon, she brought him an amount of gold (coals to Newcastle?) that has been estimated at as much as three tons – worth over \$20 million at today's prices.

RIGHT
*Worshipping at the
Golden Calf*





The Wailing Wall, Israel

The sanctuary and tabernacle that Moses built to God's protracted specifications have disappeared, and Solomon's massive gold-encrusted temple has been defaced. But in AD 532, after 10,000 men working for six years had used more than 12 metric tons of gold in building the church of St Sophia in Constantinople, the Byzantine emperor Justinian – who supervised the entire operation – could exclaim: 'Solomon, I have surpassed thee!' Justinian was well versed in the uses of gold. He inherited 320,000 pounds of gold, used it all up, and then taxed his subjects to pay mercenary armies, to finance public works, and, most of all, to bribe his enemies to refrain from invading his domains. The process of using gold to proclaim the power of the church would be repeated in gleaming golden mosaics and decoration throughout Italy, in Spain, and even on the wildest steppes of Russia.

Neither Solomon nor Jehovah himself were the first to use gold to inspire reverence. The ancient Egyptians probably set the style for later religions, including the Jews, to emulate. The Jews, with one god, had it easy compared with the Egyptians, who had two thousand deities to worry about, many of whom bore some relation to the all-powerful Sun God. You can consume a lot of gold convincing everyone how powerful and all-knowing 2000 deities are. Christians, with only one god to worship but several thousand saints to pray to, have faced similar problems.

The use of gold in Egypt was a royal prerogative, unavailable to anyone but the pharaohs. That constraint facilitated the way that the pharaohs assumed god-like roles and authenticated their heavenly character by adorning themselves with the same substance that embellished their gods. Creating gold jewelry in Egypt was a high art, lavished upon dead monarchs as well as live ones.

An impressive demonstration of the use of gold to project power was carried out by a fascinating pharaoh who happened to be a woman, described by the Egyptologist James

RIGHT

Tutankhamen's mask. An example of Egyptian use of gold in jewellery and ritual objects



LEFT: *Egyptian jewelry*RIGHT: *Inca jewelry*

Henry Breasted as 'the first great lady of the world.' Hatshepsut was the daughter of Thutmose I, who was the first pharaoh to be buried, about 1482 BC, in the Valley of the Tombs of the Kings at Thebes. After Hatshepsut seized power from her nephew-stepson around 1470 BC, she sat on the throne as king until her death about 1458 BC and was known by approximately 80 titles, including Son of the Sun and Golden Horus (the Egyptian god of light). Although she passed up the opportunity to add the traditional royal title of Mighty Bull, she was nevertheless depicted in most contemporary art as a man.

Hatshepsut was an impressive woman by any standard. She managed a major increase in Egyptian trade with Palestine, Syria, and Crete, which had withered during the preceding 150 years when Egypt was occupied by Asian invaders known as the Kyksos. The explorations for gold during her reign were ceaseless, reaching farther and farther south, probably well into Zimbabwe.

Hatshepsut's demand for gold was enormous, because she was a builder on a scale that would put Louis XIV and his Versailles to shame. She was also fond of gilding her face with a mixture of gold and silver dust. When she decided to erect a great monument for Amon Re, the chief god of Thebes, her original design included two gold pillars 100 feet high that would be seen above the walls of the Karnak complex, which covered an area larger than the Vatican. When her chancellor prevailed on her to be a little more economical, she built the pillars of granite and covered only their peaks with gold. But even that required generous amounts. When the job was complete, she declared, 'Their height pierces to heaven. . . . Their rays flood the Two Lands when the sun rises between them. . . . You who after long years shall see these monuments will say, "We do not know how they can have made whole mountains of gold."'

Most of the gold of biblical times and Ancient Egypt – approximately 4000 years before Christ – came from the bleak and forbidding landscape of southern Egypt and Nubia; nub is the Egyptian word for gold. Nubia continued to supply gold to the Western

world well into the 16th century. According to William Jacob, the output of the Nubian mines 'far exceeded the quantity which was drawn from all the mines of the then known world in subsequent ages, down to the discovery of America.'

The Egyptians had developed these mines from shallow ditches, but in time they cut complex underground shafts deep into the hills. The deeper the mines were cut, the greater the human pain that went on inside. The best description we have of the horrors experienced by the workers in these mines has been provided by Diodorus, a Greek who visited Egypt about the time that Caesar ruled Rome. The air in the shafts was fetid, constantly depleted by the tiny candles that barely illuminated the terrible darkness. The heat was intense, the earth frequently gave way, and subterranean water was a constant hazard. The fires used to crack the quartz in the rock released arsenic fumes that caused excruciating deaths among the many who inhaled them. The slaves had to work on their back or side and were literally worked to death if they were not crushed to death by falling rocks before they expired from exhaustion.

The employment of human labor was the standard mining technique right up to the twentieth century, except for a process that the Romans had devised in Spain, whose gold-stuffed hills served as the backbone of the Roman economy. The Romans originally used human labor to dig as deep as 650 feet to extract the ore from the Spanish countryside, but with a new method, called 'hydraulicking', they used powerful jets of water to break up the rock and expose the gold-bearing earth. The water came from great holding tanks situated as much as 400 to 800 feet above the site. The method, though wonderfully efficient and productive, washed away entire mountains, destroyed farmland, and silted many rivers and harbors.

Hydraulicking was used in spotty fashion in other parts of Europe as well, but its most notable reappearance was in California in 1852, at the height of the gold rush. The Roman technique was faithfully reproduced in the Sacramento area, with water under pressure of up to thirty thousand gallons a minute smashing into the rocky hillsides and mountains. The

Las Médulas mountains in north western Spain, where the Romans mined gold extensively



environmental damage was awful. Forests and farmland disappeared in short order, the detritus even pouring into San Francisco Bay and leaving the landscape dotted with piles of rock and barren mountainsides. Nevertheless, hydraulicking was the primary method of gold extraction in California until 1884, when angry citizens finally had it outlawed.

Today, in the great gold mines of South Africa, the shafts reach down as far as 12,000 feet and the temperature reaches 130°F. As Timothy Green describes it, 'To produce one ounce of fine gold requires 38 man-hours, 1400 gallons of water, electricity to run a large house for ten days, 282 to 565 cubic feet of air under straining pressure, and quantities of chemicals including cyanide, acids, lead, borax, and lime.' The labor force employed in the South African mines exceeds 400,000 men, about 90 per cent of whom are black.

King Ferdinand of Spain coined immortal words in 1511 when he declared, 'Get gold, humanely if possible, but at all hazards – get gold.'

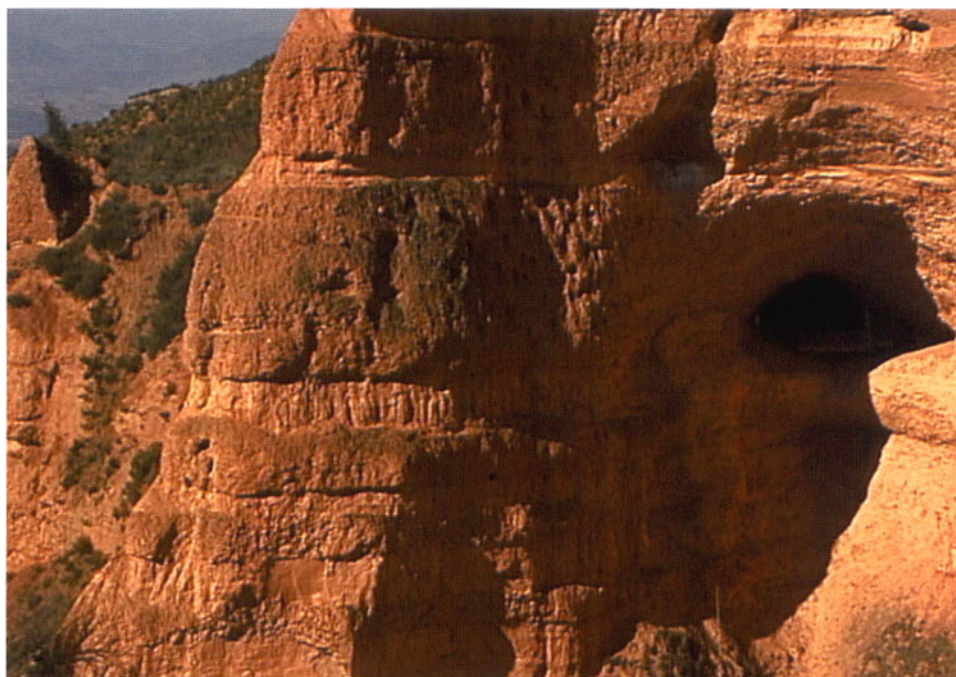
Not all gold has to be mined. When gold is carried down by mountain streams, the prospector can wade in and sieve up the fragments of gold-bearing ore that have broken loose from the mountainside. Gold was collected long ago in this fashion in Asia Minor, where gold coinage first made its formal appearance. Some 3500 years later, the California gold rush of the 19th century began on the banks of the Sacramento River, when the Forty-Niners crowded into the river with their crude equipment to 'pan' the gold out of the rushing waters.

BELOW LEFT

*One of the canals
constructed by the Romans
to bring water from snow
melt to the gold mines to
carry out 'hydraulicking'*

BELOW RIGHT

*One of the mountains
virtually destroyed by
Romans in their search
for gold*





South African miners

They were following a practice that had come down from the ancient Greeks, who used woolly sheepskins for panning gold from the rivers -- the tight curls of the sheep's coat did an excellent job of capturing and holding the fragments of gold as the waters came rushing down the mountainsides. The mention of fleece and gold together immediately evokes Jason and the Golden Fleece, a legend that is worth a brief digression for its moral.

Phryxus, the son of the king of Bocotia, an area in eastern Greece, had been badly treated by his stepmother, so his own mother arranged for him and his sister to escape on the back of a winged ram whose fleece was pure gold, a handsome gift that she had received from Hermes (for services undefined). The trip could hardly have been smooth, because the Golden Fleece must have weighed heavily even on a ram delivered by Hermes. Phryxus's sister, Helle, was apparently susceptible to air sickness, and, lacking the facilities of modern aircraft, became dizzy and fell off the ram into the sea; the point where she landed was named after her as the Hellespont.

Phryxus held on. After a trip of over 1000 miles, he was finally delivered by his ram to Colchis on the far eastern side of the Black Sea. Happy to be safe and alive, he sacrificed the ram to Zeus and presented the fleece to the local king, Aeetes. Aeetes was delighted, as he had been told by an oracle that his life depended upon his possession of this fleece. Consequently, he nailed the Golden Fleece to a tree in a sacred grove and hired a huge, bloodthirsty dragon to guard it.

Meanwhile, back in northern Greece a king named Pelias decided he had better get rid of his handsome and popular nephew Jason, who was trying to assert his family's claim to the throne. Pelias told Jason that he could have the throne if he would first perform a deed 'which well becomes your youth and which I am too old to accomplish... Fetch back the fleece of the golden ram... When you return with your magnificent prize, you shall have the kingdom and the sceptre.' Pelias never dreamed that Jason would succeed and return one day with his magnificent prize; on the contrary, he fully expected Jason to perish along the way or at least in the jaws of the guardian dragon.

Jason did take the Golden Fleece, with the help of his Argonauts, but only after an extensive and prolonged series of hair-raising adventures. Even then, he would have failed had it not been for the assistance he received from Aeetes's daughter Medea, who possessed magic powers. Medea had been hit with a dart thrown by Eros and had fallen madly in love with Jason, so she used all her wiles to catch his fancy. Jason was sufficiently tempted by her to offer to take her back to Greece with him, but on the condition that she support his efforts to take the Golden Fleece. Much as she loved him, Medea was unwilling to yield to what might well have been a seductive ruse. 'O stranger,' she cried, 'swear by your gods and in the presence of your friends, that you will not disgrace me when I am alone, an alien in your land.' Jason swore to make her his 'rightful wife' as soon as they returned to Greece. As such oaths were guarantees as reliable as written contracts in our time, Medea delivered the goods by singing the dragon into drowsiness while Jason seized the Golden Fleece from the tree.

The story does not have a happy ending, because Jason was a compulsive social climber. From the outset, he was determined to become king of his homeland. He risked his own life and those of his friends in search of a sheepskin dusted with gold. He used a king's daughter to bear children and promised to marry her. When he returned to Greece and found that he

A South African mine



could not succeed to the throne, he fled with Medea to Corinth. There he proceeded to woo the daughter of King Creon but he told Medea what he was up to only after Creon had agreed to his betrothal to the princess. When Medea, inconsolable, recalled to him his solemn oath in Colchis, Jason justified himself by saying that their children would be better off because his newly betrothed had better social and political connections in Corinth than Medea did. The only solace he offered her was some gold and a request to friends to provide her with hospitality.

Medea fixed him. With a fine touch appropriate to the occasion, she created a gorgeous gown made of cloth of gold and drenched it in poison. She then presented it as a gift to the bride-to-be. Delighted at the sight of this beautiful garment, the poor young woman wrapped herself in the radiant fabric, twined the golden wreath into her hair, and died a horrible death. Medea then completed her act of revenge by killing her own sons and flying off in a dragon-drawn chariot she had conjured up. Jason threw himself on his sword and died on the threshold of his home. The gold of Aeeetes's fleece had promised Jason power. That power gained him a princess who promised him a throne. But in the end, it was the gold that snuffed out both his bride and his future.





Nidas' Wish and

ALTHOUGH THE CROWNS OF GOLD THAT MONARCHS WEAR ON STATE OCCASIONS MUST WEIGH HEAVY ON THEIR HEADS, NO MONARCH HAS EVER CHOSEN ZINC OR PLASTIC AS AN ALTERNATIVE. RULERS FOR CENTURIES HAVE ALSO BEEN FOND OF STAMPING THEIR LIKENESS ON GOLD COINS, TO CIRCULATE THROUGHOUT THEIR KINGDOM AND BEYOND.

TWO

The Creatures of Pure Chance



The tension between gold as adornment and gold as money developed early in history and has continued up to the present time. The everlasting radiance of gold, together with its scarcity, suggested such exceptional value that its route from the golden calf, the gilded phallus, and the Golden Fleece to its use as money was probably inevitable. The process works both ways: gold's massive purchasing power adds to the lustre we see when we look at gold jewelry or a gilded dome.

Value alone is insufficient for a substance to qualify as money. Lots of things have value that do not serve as money. In fact, the most effective forms of money have developed from objects that were otherwise quite useless, such as paper and computer blips.

In early Britain, cattle and slaves served as money. Their value was set by law – although the church, eager to discourage slavery, refused to accept slaves in payment of penances. Pepper was popular in medieval times. In some areas, hoarding cattle to serve as wealth instead of as a food supply is a practice that has continued into modern times; this practice has led to serious ecological degradation in parts of Africa, where the sheep and goat population shrank by over 66 million from 1955 to 1976.

That is a rare case. In modern times, nothing useful has ever functioned as money for very long. For example, the cigarettes accepted as currency in Germany in the early days after World War II ultimately went up in smoke. Gold, in contrast, has always been useless for most practical purposes that call for metal, because it is so soft. With only 125,000 tons of it in existence, gold is also too scarce to have many uses.

But gold has clear advantages as money compared to other kinds of useless substances that people have used for the purpose. Unlike cowrie shells, which were the main form of money for centuries in parts of Asia, gold is remarkably durable and does not easily

Book illustration of the legendary King Midas, after turning his daughter into a gold statue

fragment. Every single piece of gold, no matter how small or how large, is instantly recognizable everywhere as a receptacle of high value. Furthermore, every piece of gold is valued only by its weight and purity, attributes that are inconveniently applied to cattle.

Seen from the perspective of uselessness, the electronic blips on computer screens that comprise most of the money in the modern world are the best form of money – we have no other use for them, they are readily recognizable as money, they weigh a lot less than gold or even paper, they are easily transferable, they can be broken down into any amount we choose from a penny to trillions of dollars and even beyond, they are as durable as we wish them to be, and they have a kind of magic that commands our respect.

Yet gold endures as a standard of value. From the Golden Rule to Olympic gold, it has commanded far more respect than any other substance in history.

But we should hesitate before admiring the sophistication of our contemporary currency while snickering about currencies in societies supposedly more primitive than ours. Consider the monetary system on the small island of Yap in the Caroline Islands, as charmingly described by an American anthropologist named William Henry Furness III, who spent several months on Yap in 1903.

Furness points out, 'In a land where food and drink and ready-made clothes grow on trees and may be had for the gathering, it is not easy to see how a man can run very deeply in debt for his living expenses.' Nevertheless, people like some tangible representation of the labor they have expended that can be accumulated as wealth.

The medium of exchange, or, more properly, the store of value on Yap at that time was called *fei*. *Fei* consisted of thick stone wheels with diameters ranging from saucer-size pieces to 12-foot millstones. The stones from which these *fei* had been fashioned came from limestone quarries found on the island of Babelthuap, one of the Pelao Islands about 400 miles away, and brought to Yap long ago, piece by piece, in canoes and on rafts by some venturesome natives described by Furness 'as persuasive as . . . the most glib book-agent.'

The smaller and more portable *fei* served as a medium of exchange and were handed around in payment for fish or pigs. The larger *fei*, however, received different treatment. The natives punched holes in the center of these *fei* to facilitate moving them about, but most of these big stones weighed so much that they remained permanently in one spot. On the rare occasions when a major transaction took place, the process went through with a simple acknowledgment of change of ownership while the 'coin' continued to sit undisturbed wherever it happened to be.

In fact, the wealthiest family in the community owned an enormous *fei* that no one could see or had ever seen. According to this family, their *fei* lay on the bottom of the sea. Many generations past, while an ancestor was towing it on a raft attached to his canoe, a terrible storm came up. Unlike the protagonist of Ruskin's story, this man had decided that life came first and money second: he cut the raft adrift and watched the huge stone

sink below the waves. But he survived to tell the tale and to describe to everyone the extraordinary size and quality of the stone he had lost. Nobody had ever doubted the veracity of his testimony. As Furness described it, 'The purchasing power of that stone remains, therefore, as valid as if it were leaning visibly against the side of the owner's house.'

Furness goes on to tell what happened when the German government bought Yap from Spain in 1898 and wanted to transform the island's rocky coral paths into proper roads for modern transport. The natives had no interest in spending their time doing that kind of work, despite repeated commands from the Germans to get busy. The Germans finally decided to levy a fine that would be lifted only when the task was completed. A German official went through the island, marking the most valuable *fei* with a black cross that confirmed the government's claim to that stone. According to Furness, 'This instantly worked like a charm: the people, thus dolefully impoverished, turned to and repaired the highways . . . that they are now like park drives.' Then the government erased the crosses, and 'Presto! the fine was paid, the happy *failus* resumed possession of their capital stock, and rolled in wealth.' In other times and other places, we call this sequence of events taxation and government spending.

This story reminds me of an experience of my own early in my career in 1940 when I went to work in the research department of the Federal Reserve Bank of New York in the heart of the city's financial district. One day, as a treat, my boss took me down to see the gold stored in the antiseptic vaults of the bank, five stories underground – sunk below bedrock to discourage thieves from tunneling through the outside walls. We entered the area through a ponderous airtight and watertight cylindrical door of stainless steel that unlocked automatically at nine in the morning and locked automatically at five in the afternoon. Just inside was a lunchbox, replenished daily with fresh sandwiches, to provide for any hapless member of the staff who got stuck inside when the automatic locks slammed shut at the end of the day. A little further on, there was a scale for weighing the gold, a scale so sensitive that a pea would send it rocking. With gold, even dust matters.

The gold was stored in oversized closets, about 10 feet wide, 10 feet high, and 18 feet deep. The closets were filled to the ceiling with towering piles of gold bricks, each brick the size of three large candy bars. The bricks weighed about 30 pounds apiece – 400 troy ounces – and were worth \$14,000 in those days, when gold was officially priced at \$35 an ounce. At those prices, \$2 billion was stacked up there, a sum of money that was sufficient to buy four days' worth of the *total* production of goods and services in the United States at that time but was crowded into just one small space five stories below the busy New York City streets. Seeing over 100,000 gold bars, stacked to the ceiling and ablaze under the electric lights, is an unforgettable and chilling sight.

That gold did not belong to the United States. It belonged to France, Britain and Switzerland, and to many other countries as well. Those countries had for a long time

stored their official gold holdings at the New York Federal Reserve for both safekeeping and convenience. Each bar consigned in this manner was impressed with its owner's seal or a similar mark for identification. This process was known as earmarking gold, an expression that may date back to a method of indicating ownership of domesticated beasts. Earmarking enabled each nation to avoid all the care and expense of moving gold cross-country or across the seas when one country had cause to transfer gold to another. For example, if Britain lost gold to France, a guard at the Federal Reserve had merely to bring a dolly to Britain's closet, trundle the gold to the French closet, change the earmark, and note the change on the bookkeeping records.

These movements of just a few feet from one closet to another often reflected a major change in wealth between countries, with broad ramifications on economic well-being. Yet the citizens of each country never saw the gold to which their government held a claim. If the gold had sunk into the Hudson River but the bookkeeping had progressed just the same, the economic and financial consequences to each nation would have been just as far-reaching as when the gold was shifted from one closet to another.

This procedure bears a striking resemblance to what went on at the island of Yap, with its transfers of ownership of assets that never moved and with the agitated economic activity that resulted when the Germans marked a black cross on the *fei*. As we shall see, the resemblance between so-called primitive and so-called modern uses of money did not stop at the shores of Yap and the cellars of the Federal Reserve.

The *fei* of Yap were stores of wealth. Stores of wealth sit. Money moves. It travels from one pocket to another. A store of wealth is mass; money is measurement of wealth.

Gold's durability, density, and glow made it a natural choice as a store of wealth long before people thought about using it as money. Like everything that has served as a store of wealth, gold in ancient times was a passion, a blatant expression of power, a means to provoke envy among enemies or people of lower status, or a vehicle for currying favor — as when the queen of Sheba showered gold on King Solomon.

Gold deployed as money becomes something different. People who go out to spend or lend money have to be cool-headed, calculating, precise, strategic in their vision. Before gold could be used as money instead of as a store of wealth, people had to become sufficiently productive to have something to trade, travel had to become more routine, and measurement had to be defined for the purpose.

In short, money comes into being when people are doing business. Not much business was transacted at Yap, where economic life was communal rather than commercial. We need money when we want to hire someone or because we want to offer the money to someone else in exchange for something we do not own. We use money when we want something today rather than tomorrow. Then we borrow from someone willing to wait until later to spend their money. Money moves from buyer to seller, from



lender to borrower, and from borrower to lender. It seldom sits still very long and someone else is always involved.

Weighing gold

When gold was only a store of wealth, payments from one party to another were infrequent. The process was cumbersome and time-consuming. Like cattle and the stones of Yap, no two gold bars or rings in ancient times were ever precisely the same size and fineness. As a result, every transaction involved testing for purity and putting the gold on a scale to determine its exact weight.

Coins were an ingenious innovation designed to get around the tedious business of weighing and checking purity, but they did not come upon the scene until around 700 BC, a good 2000 years or more after gold was first launched on its monetary career. Although coins enabled people to skip the measurement process and get right down to business, coins could serve this purpose only if they were genuine – they had to be worth precisely what their inscriptions represented them to be worth for this purpose.

Even at the very beginning, therefore, a widely accepted method for gauging the purity of gold and determining its weight was essential before gold could be used as money. Gold has acquired its own measuring system for these purposes, although versions of this system are now used on other precious metals and the most valuable jewels.

We define the purity of a piece of gold in terms of its carats. For example, 24-carat gold is 100 per cent pure. Carat – the word derives from the Greek word *keration*, *qirat* in Arabic, and *carato* in Italian – was originally a measure of weight rather than purity,

however, and for a delightful reason. Carats are the fruit of the leguminous carob tree, every single pod of which weighs one-fifth of a gram.

Today the carat has been replaced by the grain as the conventional unit of weight. Grains of barley or wheat in the middle of the ear have the same remarkable attributes as the carat – a standard weight regardless of the size of the ear. The troy ounce, which comes from the French town of Troyes where the measure was first put into use, weighs 480 grains, and 12 troy ounces equal one pound, which is the same as one 16-ounce pound avoirdupois. Thus, troy ounces are heavier than the ounces we are used to employing. The modern convention is to express the weight of gold in grains, but the price is expressed in troy ounces.

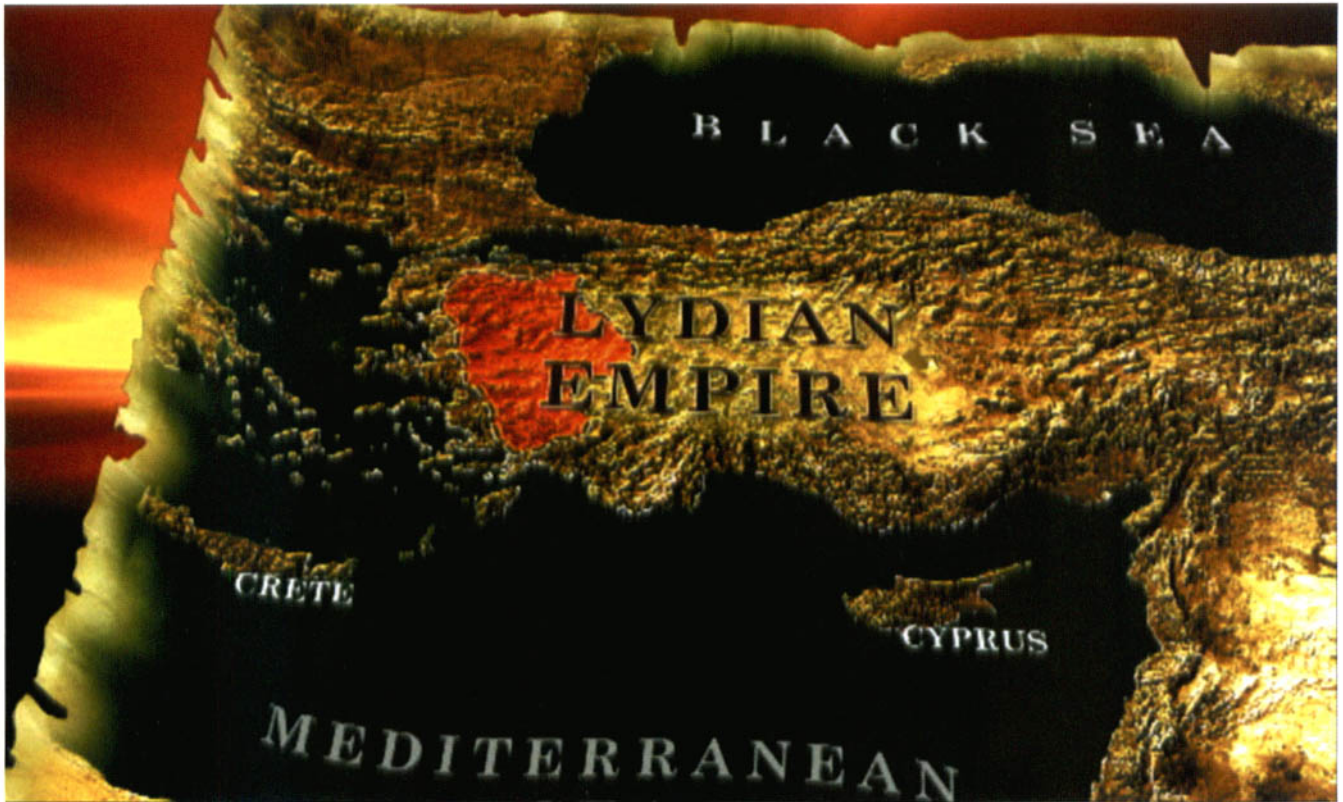
The Egyptians were casting gold bars as money as early as 4000 BC, each bar stamped with the name of the pharaoh Menes. The Egyptians even had a defined ratio between gold and silver. Throughout most of history, silver has been valued at only 5 per cent to 8 per cent of gold's value – ratios of 12 to 20 parts of silver to 1 part of gold – but the Egyptians set silver equal to 10 per cent as much as gold because they had no indigenous silver supply. It is also possible that the arithmetic was easier at that ratio, but we have no evidence of that. In any case, this step was the beginning of a complex, incestuous, and occasionally violent cohabitation of gold with silver in the money stock, a battle that haunts most of the history of gold as money.

The awkward process of weighing gold and checking its purity in every transaction sounds like more of a nuisance than it was in reality. These ancient civilizations bore a greater resemblance to the island of Yap than to an industrialized society like ours. When most property belonged to the monarch, when economic activity was primarily agricultural, and when transportation was so difficult that most communities were self-sufficient, long-range trade and commercial transactions were either rare or of minor importance.

As the need for money grows, it rapidly inspires innovation to make it function more efficiently and conveniently. The Assyrians and Babylonians were more active traders than the Egyptians, and they developed more elaborate and uniform gold bars. They stamped lions on the heavier bars, about 30 pounds in weight, and put ducks on the smaller bars that weighed about half as much. The lions and the ducks were a help in signifying value, but until about 600 BC people still wanted to weigh each piece of gold rather than accept the stamped indications at face value. The Mesopotamian peoples also broke their gold monies into smaller denominations known as talents, minas, and shekels; these denominations soon became common throughout Asia Minor and the Greek cities and settlements throughout the Mediterranean basin. The shekel has survived to this day in Israel.

A shekel from the 1st century BC





The Lydian Empire

The process of weighing the precious metals in each transaction was indeed a nuisance for everyone concerned, but these ancient arrangements had one great advantage that would vanish once coinage arrived upon the scene. When money was just pieces of metal of varying weights, it had no nationality. Even the Egyptian bars traded on the basis of their weights, not because they carried a pharaoh's name. In Genesis, we read that Joseph's brothers sold him to total strangers from another land for thirty shekels of silver, without any concerns about rates of exchange or acceptability of the silver in the foreign country. Thus, while our ancient forebears functioned with the one form of money that was acceptable everywhere, modern experts – having experienced the original sin of national monies – dream of a supranational currency but have no idea how to implement it.

The prosaic sequence of events that led from crude gold bars to a full-fledged system of coinage developed from a romantic and dramatic sequence of events that took place in the eastern part of Asia Minor, now Turkey. This story, which is admittedly part legend, begins in Phrygia, a kingdom whose capital city by that name was located on the banks of a small mountain torrent called the Pactolus. The first king of Phrygia, about 750 BC, was Gordius, a poor man with nothing to his name but a pair of oxen. Gordius was succeeded by his son Midas, thereby initiating a curious tradition for the Phrygian dynasty, who named themselves alternately Gordius and Midas.

This first King Midas was poor like his father, but we are told that he was a good man who wanted to be generous to others despite his poverty. One stranger whom Midas invited into his home turned out to be the foster father of Bacchus. Bacchus was so impressed with Midas's hospitality to his foster father that he granted the king any wish of his own choosing.

That irresistible offer was what got poor Midas into trouble. Midas's wish to have everything he touched turn to gold is usually held up as illustrating the dire consequences that stem from being overly greedy. Money isn't everything, as the saying goes. We should hesitate, however, before assuming that money was an obsession with Midas. If, according to the story, Midas inherited from his father Gordius nothing more than two oxen, he must have been a poor man, especially for a king. If he was a good man, why then should we assume that he was greedy? Perhaps his wish simply reflected a desperate desire to find a shortcut out of his dire poverty, a choice made without regard to the consequences.

Midas discovered his error in short order. When his food turned to gold as he tried to eat it, and even his beloved daughter became a golden statue when he embraced her, Midas begged Bacchus to throw the damned wish into reverse. Bacchus must still have held a high opinion of Midas, for he immediately obliged by instructing Midas to bathe in the Pactolus River. Midas thereby transferred his golden touch to the Pactolus, the legend continues, which is why that river turned out to be such a rich source of gold for the Phrygians and their close neighbors the Lydians. Midas thereby ended up with the best of both possible worlds – the gold in the Pactolus made him rich, but he was once more able to eat and touch his loved ones without turning everything and everybody into solid gold. The actual location of the Pactolus is no longer visible, but geographers believe that it was a stream carrying alluvial gold from the slopes of Mount Tmolus in Anatolia. By the time the Romans took over this area, perhaps half a millennium later, the mountain had been eroded by rushing water and had no more gold to yield.

*Burial mound of the kings of
Phrygia in western Turkey,
perhaps including Midas
himself*



Midas did not live happily ever after. The Cimmerians, a powerful nomadic tribe from southern Russia, invaded Phrygia and overthrew Midas, who committed suicide by taking poison to escape the savage hordes at his gates. Midas was not forgotten, however, for his chariot remained tied to a post by a complex knot in the main temple of Gordium for three hundred years. An oracle predicted that whoever could untie the knot would become king of Asia; this was none other than the Gordian knot that young Alexander of Macedon would cut through with his sword in 334 BC, on his way to conquer the lands all the way from Egypt to India.

Most of the reliable history about this area of Asia Minor, as opposed to blends of fact and fiction, comes down to us from Herodotus, the Greek historian who lived around 500 BC. Herodotus's *Histories* comprised the first extended narrative in prose in Western civilization and set a tough act for later historians to follow. He emerges from these accounts as consistently perceptive, wise, and entertaining, with a sharp eye for gossip as well as the foibles of the characters whose chronicles he chose to record.

Herodotus's history begins in about 700 BC in Lydia, an area to the northwest of Phrygia; Lydia occupied most of the center of Asia Minor from the Aegean Sea inland approximately 200 miles. Sardis, the capital city, had the good fortune of sitting on great supplies of alluvial gold, most of which streamed down from the mountains into the Pactolus River – thanks presumably to Midas. Lydia also mined a metal called electrum, often referred to as 'white gold,' which was about two-thirds gold and one-third silver. The word derives from the ancient Greek word *Hlektwr* (elector), which means 'he who shines' (the Greek word for sun is *Helio*, as in heliotrope) and is the root from which we derive the modern word *electric*. With all that wealth bestowed upon them, the Lydians frequently engaged in wild orgiastic dances in honor of Cybele, the goddess of the mountains and guardian of ores and metals.

According to Herodotus, the kings of Lydia traced their ancestry from Hercules and had ruled for 22 generations, or 550 years, at which time their king was named Candaules. Candaules was madly in love with his beautiful wife. He was also a show-off. One day he hid with his favorite bodyguard Gyges to give Gyges the opportunity of observing the lady undress and display her lovely body. Unknown to the two men, the queen noticed what had happened. She called Gyges to her the next day and told him that either the man who had planned this violation must die or the man who had illegally seen her nakedness must perish. She let Gyges choose between making the event legal by killing his king and then marrying her and leading the kingdom, or being killed immediately by her instead. That choice is what is known today as a no-brainer. And so began the dynasty known by the tongue-twisting title of Mermnadae.

Although the Lydians were outraged at the murder of their king, Gyges persuaded them to wait to hear what the oracle at Delphi had to say on the matter. The oracle

declared in favor of Gyges, perhaps not coincidentally because of the generous gifts of gold and silver that he subsequently lavished upon her, including six golden bowls that weighed about 1800 pounds (and worth over \$6 million at today's prices). Nevertheless, the oracle also predicted that Gyges's dynasty would perish in the fifth generation, when Candaules's descendants would finally claim their revenge on the Mermnadae. Gyges and the Lydians took little notice of her prophecy – at that moment.

The first three descendants of Gyges – Ardys, Sadyattes, and Alyattes – ruled for a total of 118 years, of which 57 were accounted for by Alyattes alone. These three kings of Lydia spent most of their time making war on their southern and western neighbors in an effort to extend their domain to all of western Asia Minor out to the Ionian coast of the Aegean, although Ardys (660–637 BC), like Midas, had his hands full holding off the invading Cimmerians. Unlike most empire-builders through history, however, the Mermnadae refrained for the most part from destroying the homes and holy places of the defeated peoples, who were also left to enjoy loyal autonomy. The Lydian kings simply wanted monetary tribute and assured supplies of food and other materials, reasoning that they would be better off with a peaceful empire than one filled with people eager to take revenge against them.

Croesus, son of Alyattes and the great-great grandson of Gyges, ascended the throne in 568 BC at the age of 35. This Croesus was the man who most people wished they were as rich as, which was a good thing, but he was also the fifth generation of the Mermnadae, which was an unfortunate thing. Regardless of the double-talk usually offered by the

*Ruins of the Lydian
capital, Sardis*



oracle at Delphi, the prediction the oracle gave to Gyges about the fifth generation being the last would turn out to be correct. Nevertheless, during his reign, Croesus completed most of the conquests that his predecessors had begun. He succeeded in occupying nearly all of western Turkey, including Phrygia, and even made an alliance with the Spartans on the Peloponnesus.

Herodotus tells some entertaining stories about Croesus. The most revealing involved a visit by Solon, who had just written a code of laws for the Athenians, who promised to obey them for ten years. Solon took those ten years off to go sightseeing. When he arrived at Sardis, Croesus was impatient to show him the treasury with its immense wealth in gold. Then he turned to Solon, asking whether Solon had so far seen anybody, in all his far-flung travels, whom he considered to be 'more fortunate than all men.' Solon mentioned a great war hero of Athens and a couple of prizewinning athletes and their devoted mother. Dumbfounded, Croesus exclaimed, 'As far as you are concerned, our prosperity amounts to nothing, and you do not even consider us on a par with private citizens!'

Solon agreed. 'When you ask me about human affairs,' he replied, 'you ask someone who knows how jealous and provocative god is. . . . My dear Croesus, humans are the creatures of pure chance.' He admitted that rich men can gratify their desires and have the resources to absorb misfortune, but he then pointed out that the lucky man does not have to concern himself with misfortune: 'He suffers no bodily harm, he doesn't get sick . . . he has good children, and he is handsome.'

Herodotus tells us that the Lydians 'are the first people we know of to mint and use gold coins and silver coins, and they were the first retail tradesmen.' Sardis had a marketplace with a cluster of small shops offering a wide variety of goods ranging from meat and grain to jewelry and musical instruments. Herodotus had a word for this: *kaphloi* (*kapeloi*), which translated literally means 'merchant' or 'seller'; in Greek slang, it means 'man with a big hat' and could be read in more modern terms as 'huckster'. The Lydians were so busy converting almost everything into salable merchandise that, as Herodotus reports, 'Except for prostituting their female children, the Lydians observe the same customs as the Greeks.' As the women accumulated coinage, however, they created their own dowries and as a result had unusual freedom in choosing their husbands.

These Lydian innovations in the development of money and trade were no coincidence. In addition to its location on the banks of the Pactolus, streaming with alluvial gold, the Lydian capital of Sardis sat astride the great east-west highway that linked the Aegean Sea to the Euphrates and more distant Asia, a span of nearly 1700 miles. Trade and commercial activities were a natural development, and they brought with them the need for weights and measures and, most important of all, money in a convenient form for doing business. Money, in turn, created a demand for goldsmiths,



money changers, and ultimately bankers. Sardis grew into a major urban center filled with wealthy families living in the highest luxury.

One ingenious Lydian innovation was the use of a local black stone, similar to jasper, for testing the purity of the lumps of gold received in payment for commercial transactions. This stone came to be known as the touchstone, because goldsmiths rubbed gold objects against it and then compared the mark against a set of 24 needles containing varying proportions of gold and silver, gold and copper, and all three metals. The 24th needle was pure gold, just as 24 carats measure pure gold. All of this contributed to the development of a well-functioning coinage system, but we cannot appreciate what the Lydians accomplished and what Croesus in particular achieved without a brief step backward about 150 years.

At the beginning of the 7th century BC, Lydian money consisted of bean-shaped lumps of electrum, called dumps. These dumps were too heavy for easy exchanges, as they had no uniformity in size or weight and bore no stamp to indicate their value.

Gyges, the first of the Mermnadae, made a revolutionary reform in Lydia when he suppressed private issuance of metallic money (primarily electrum) and established a state monopoly over the issuance of dumps. The official monopoly of the state over the creation of money has persisted throughout history. Article I, Section 8 of the Constitution of the United States, for example, declares that 'The Congress shall have power to coin money, regulate the value thereof, and of foreign coin.' These concepts dominated the control of money supplies – note the US reference to 'coin' – as long as money was hard, but they began to diminish in importance as we progressed to modern times. The development of negotiable credit instruments during the late Middle Ages and the increasing use of commercial bank liabilities as money – the modern checking account – bypassed the state's monopoly over the creation of money and diluted the importance of gold as a means of payment for daily transactions. The role assigned to gold gradually changed into a kind of governor of the monetary system, a backing that was intended to set limits on the issuance of all other forms of money.

When Ardys succeeded Gyges on the throne in 660 BC, he too was interested in creating a more efficient monetary system. He began stamping the electrum ingots with marks to guarantee their weight and value, providing different ingots for different folks: Lydia had one set, the Babylonian towns to the east had a different set, and the Ionian coast towns to the west still another. In time, however, the dumps became more uniform in size, and less than 50 years passed before the lumps and dumps became recognizable coins: round, uniform, and clearly stamped. A lion's head – the logo for the dynasty launched by Gyges – appeared on every one of them. The innovation spread rapidly in a western direction toward Greece, where coinage soon became an integral part of a system of rapidly developing trade all around the Mediterranean basin. If the Lydians were the first people to invent and use coins, the Greeks were the first to make coinage an art form; for the



One of the first coins in the world, made of electrum by the Lydians and now in the British Museum

Greeks, beauty was an aim in designing money as much as it was in virtually everything else they touched.

The story is probably only a rough approximation of what actually happened, for nothing that took place that long ago is ever beyond controversy. Some modern experts had believed that full-fledged Lydian coinage originated before 700 BC, perhaps 50 years earlier than that, even though Herodotus had set 687 BC as his estimate of the date. But in 1951, a group of archeologists working in the great Ionian city of Ephesus came upon a huge hoard of Lydian money buried under the ruins of the temple of Artemis, which had been built about 600 BC. Over 3000 items came into view, including unstamped dumps, stamped dumps, and a mass of coins with the lion's head struck upon them, in addition to a substantial pile of jewelry and statuettes fashioned of gold and silver. Careful examination confirmed that the first true coins dated from around 635 BC, which in turn confirmed that Herodotus was right in the first place and should not have been doubted. This dating would place the beginning of coinage around the end of the reign of Ardys, the son of Gyges, or at the beginning of the reign of his son, Sadyattes.

Croesus played the climactic role in this process. Although we shall see that he turned out to be a disaster as a military strategist, thereby fulfilling the oracle's prophecy about the fifth member of the Mermnadae, he was a master innovator when it came to monetary affairs and in his appreciation of the economic and political power packed into the precious metals. He was not kidding with Solon: he was convinced that money and happiness were inseparable.

Croesus's father Alyattes had been the first of the line to issue gold coins, which developed into a lucrative source of exports for Lydia and paid for much of their imports; the Lydian standard of living thereby enjoyed the advantages of trading something useless for something useful. Recognizing the value of these gold coins to his country's prosperity, Croesus called in all the outstanding electrum coins, melted them down, and minted new coins in the new style of pure silver and gold. In 1964, modern archeologists succeeded in uncovering the fire-resistant pots where Croesus's men extracted the impurities from the gold and silver from electrum by heating the metals with a mixture of lead and salt – a method that has not been found in excavations elsewhere.

The coins of Croesus were stamped on one side with the foreparts of a lion and a bull, the arms of the city of Sardis. The opposite side had oblong and square punch marks, or depressions – the technical numismatic expression is that the coins were 'incused' – to show their value. Most important, Croesus made the denominations and weights of his new coins conform as closely as possible to the weights and denominations of the old currency. The basic denomination, already familiar to everyone in that area of the world, was called the stater, which was subdivided into smaller denominations of thirds, sixths, and twelfths. The coins were minted with great care in order to maintain the uniformity of their size and weight. As a result, they were immediately acceptable throughout his

kingdom. The division of the staters into twelfths carried forward to the development of the troy ounce, composed of 24 carats of pure gold, and reappears again in the British shilling, which – until the relatively recent conversion into the decimal system – consisted of 12 pennies.

In the process of putting his reform into action, Croesus had launched the bimetallic currency system that would prevail in most countries over most of subsequent history. The silver coins were needed to serve as the denominations that were too small for the use of gold; most of the gold was used to finance foreign trade. Like the Egyptians, Croesus set the ratio of gold to silver at 10:1 as a matter of convenience, although he made no legal ruling to that effect. This bimetallic system had its useful features, but, as we have seen, monetary systems based on two metals were seldom stable, because changing supplies of the two metals over time caused their relative values to fluctuate.

Nevertheless, when his reform was complete, Croesus had established the first imperial currency in the history of the world. His beautiful coins of gold and silver were immediately accepted – indeed, demanded – throughout Asia Minor and were circulated in Greece on the western side of the Aegean as well. This universally accepted currency played a critically important role in adding to the entire area's prosperity and economic development: it

Sardis



stimulated trade both within the Lydian Empire and with the nations to the east, west, and south, which in turn encouraged the free interchange of people and ideas. Croesus's accomplishment was equivalent to the establishment of the euro in western Europe in our time. If that revolutionary step of creating a common currency for communities that had always had their own money can succeed, the euro will have achieved precisely what Croesus had consummated: increased trade within Europe and with the rest of the world, with populations more mobile, and enjoying a robust rate of economic growth.

*Delphi, where Croesus plied
the oracle with gifts of gold*

By the time he was done, Croesus had created a great innovation that has reverberated through history up to our own era. It was not just the establishment of a rational, systematic, and widely acceptable form of money, a step that was momentous in its own right. There were many other materials that he could have used as a base for his monetary system – copper, shells, or beads, for example. The focus on gold and silver, however, transformed those metals into the ultimate standards of wealth and money. In time, these attributes would prove to be more valuable than the reverence accorded them as objects of religious worship or as articles of beauty.

Fifteen years into his reign, Croesus began to worry about the growing power of the Persians, whose king Cyrus had already led his troops into the eastern parts of Asia Minor along the shores of the Black Sea. Croesus was well aware of Cyrus's expectation to gain great economic power as well as valuable territory by subduing the Lydians.

Croesus decided that he should take the offensive and cut down the Persian power before it became invincible. Under similar circumstances, most leaders throughout history have sat down with their generals and other advisors and mapped out a strategy to confront the approaching enemy. Not Croesus. Rational and ingenious when it came to



*Ruins of the world's first
gold refinery at Sardis*



money and gold, Croesus worked out his military strategy by consulting oracles; he sent messengers to the oracle in Delphi, to six other Greek oracles, and even to one in Libya. He tested the forecasting accuracy of the oracles by instructing each messenger to count 100 days from their departure, visit the oracle, and ask the oracle what Croesus was doing on that day. He then chose something to do that he was convinced no one would be able to guess: he chopped up a turtle and a lamb and boiled them together in a bronze pot.

When all the messengers had returned with the responses of the various oracles, he was astonished that one of them had actually guessed right: the oracle in Delphi. Croesus had always been partial to that oracle, because it had legitimized the reign of his great-great grandfather Gyges. The Delphic oracle prophesied that Croesus would be eating 'strong-shelled tortoise seething in bronze with the flesh of lambs.'

Croesus lost no time in plying the oracle with gifts, including 117 ingots of pure gold, each weighing 150 pounds, to say nothing of a lion of pure gold that weighed 600 pounds, plus a golden vat of 522 pounds that could hold 5000 gallons for mixing wine and water. He also ordered all Lydians to make a sacrifice for the oracle. The oracle, in return, conducted business with Croesus in thoroughly modern fashion. Croesus received, according to Herodotus, 'rights of first consultation without a fee, front-row seats at Pythian games and festivals, and the right, in perpetuity, for any Lydian who so desired to become a citizen of Delphi.'

The oracle also advised Croesus that if he made war on Cyrus, he would 'destroy a great empire.' Happy and confident, Croesus took off to do battle with the Persians, even though they outnumbered his forces. The first engagement was a fierce one but ended in a standoff. Croesus figured he had better withdraw to Sardis and wait until he could gather his allies before attacking Cyrus a second time. Cyrus, aware of Croesus's intentions, hurried toward Sardis, forcing Croesus to face him on the great plain that lies before the city. When Cyrus saw that Croesus had placed his powerful cavalry in the front ranks, he transferred his own horsemen to the camels usually employed in carrying food and equipment. Horses are afraid of camels and cannot stand the sight or the smell of them. The Lydian cavalry was thrown into confusion by the camel charge and the whole Lydian army had to retreat into the city, where they suffered a siege that lasted 14 days before the Persians finally broke through and claimed victory. The Delphic oracle had got it right again: a mighty empire had been destroyed, but it was the empire of Croesus that fell, not the empire of the Persians.

Cyrus decided to celebrate his victory by burning Croesus at the stake as an offering to the Persian gods. As the flames were mounting, the soldiers heard Croesus call out the word Solon three times. When asked what he meant, Croesus told them that Solon was 'a man I would give a great fortune to see talking with all the tyrants of the earth.' Cyrus was so moved by what Croesus told him about Solon's visit that he ordered the fire put out and Croesus untied. As they sat together as friends, Cyrus pointed out to Croesus that

the crowd they could see in the distance was 'looting [his] city and carrying off [his] wealth.' Croesus still had his wits about him, despite all he had been through. 'It's not my city or my wealth they are looting,' he pointed out. 'None of it is mine anymore. What they are looting and leading away belongs to you.'

With those poignant words, Croesus fades from view.

Without the Midas touch, the Lydians might have been nobodies. As the economists like to say, the gold was probably a necessary even if not a sufficient condition for their ascendancy. The origins of Lydian power and dominance – indeed, their views of their earthly mission – were rooted in the golden fruits of the Pactolus River and the electrum ores that coursed through the mountains around them. Other nations in other times have set out on conquests to become rich; the Lydians achieved their empire in large part because they were rich – as rich as Croesus.

The library at Ephesus



A close-up photograph of a stone relief carving. The carving depicts a reclining figure, likely Darius the Great, in a relaxed pose. The figure's head is turned slightly to the left, and a small, circular feature, possibly a nose or a decorative element, is visible. The stone is light-colored with some darker, possibly carved, areas. The text "Darius's Bath" is overlaid in a white, cursive script across the upper right portion of the image.

Darius's Bath

THE DEFEAT OF THE LYDIANS BY THE PERSIANS GREATLY ACCELERATED THE TRANSFORMATION OF GOLD FROM A MEANS OF ADORNMENT INTO A CENTRAL ROLE AS MONEY. DESPITE THEIR REMARKABLE INNOVATIONS IN COINAGE, THE LYDIANS THOUGHT THAT IT WAS THEIR GOLD, NOT THEIR MONEY, THAT MATTERED.

THREE

and the Cackling of the Geese

When Solon came to visit, Croesus did not say, 'Look how much money I have!' He showed Solon his 'treasure.' Nor did he use his handsome golden stater coins to compensate the oracles whose services he so voraciously consumed. He sent them gifts of magnificent objects crafted from gold. Nevertheless, the services the oracles offered in return, such as rights of first consultation without a fee and front-row seats at the Pythian games, would normally be forthcoming today only in response to a monetary accommodation.

That viewpoint was soon to come about. Croesus's ingeniously designed monetary system would lead to a profound social transformation, although he was totally unaware of the genie that he was letting out of the bottle. The radiant purity, malleability, and density that made gold so appropriate for objects of worship – including its general uselessness for anything other than adornment – were precisely the attributes that made it such an extraordinarily convenient material for coins.

The result was the development of a curious kind of feedback in the role of gold in society. Gold would never have attained its position as supreme monarch of monetary systems without its unique physical attributes, yet the demand for gold became so insatiable over time because it was used as money. The demand for gold to embellish religious objects and our own bodies has a limit somewhere; the demand for gold as money is infinite. This has caused all kinds of havoc throughout history, as men were



*A gold daric (450-330 BC)
showing 'The Great King'
with bow, spear and crown*



provoked to risk the wildest deeds in search of new sources of gold or to plunder the supplies that belonged to others.

In a sense, the transformation of gold into money democratized it. Thanks to coinage, the ownership and use of gold after Lydia was no longer a royal prerogative. It was now literally in the hands of common citizens, even if only the most wealthy, who could touch and feel it, hoard it in their homes, buy things with it, and pay their debts with it – even as they continued to put their gold through their ears and noses and to wrap it around their necks, wrists, and fingers. Before long, they would be paying their taxes with it. The notions of power and wealth thus blended into one.

The Greeks provide a moving example of the merging of gold as adornment and gold as money. In accord with the tradition that gold imparts godliness to inanimate objects, Phidias clothed his great statue of Athena in the Parthenon in a cloak of gold, even though gold was much rarer than silver in Greece. This cloak was, in effect, the official treasure of the city-state of Athens, in full display for all to see, unlike the US gold stock buried in Fort Knox – that is, until the wars with Sparta, when the Athenians had to peel the golden treasure from their goddess and convert it into money to finance their military operations.

Although Greek coinage was primarily silver, the use of Lydian-style gold coins proliferated elsewhere and ultimately provided the model for the coinage system that would function throughout the entire Roman Empire. Long before Rome, however, Cyrus (558–529 BC), the victor over Croesus at Sardis, and his successor, Darius (521–485 BC), had promptly adopted Croesus's international system of coinage and put it into operation throughout the great Persian Empire. Darius also outdid Croesus and his staters: instead of a local logo, Darius stamped his own likeness on his coins and labeled them *darics*. But then a monarch who calls himself the 'King of Kings' would be likely to do something like that.

Darius's money – and his likeness – spread far and wide. His coinage has been found from the Baltic to Africa and throughout central Asia. In addition to using their gold to mint coins that financed trade across such a wide span of lands, the Persian government was also the first in history to collect taxes in coin rather than in kind. The world has never been the same. In fact, the kinds of money that governments have been willing to accept in payment of taxes throughout history have been a primary influence on what forms of money become most acceptable to society at large. We shall see instances in which even debased money with minimal purchasing power has enjoyed some broad acceptability when governments have permitted their citizens to use it to discharge their tax liabilities.

The Persians emulated the Lydian system of coinage with great success, but they also knew how to use gold as a manifestation of power. For example, after Alexander the Great



had finally smashed the Persian Empire by defeating the King of Kings at Issus in 331 BC, he entered Darius's huge field tent and examined the golden chariot, the golden throne, and the golden bathtub, all elaborate objects of great beauty – and these were only Darius's traveling equipment. 'So this is what it meant to be a king,' observed Alexander.

Alexander the Great

Impressed as he may have been by Darius's golden bathtub, Alexander was already fully familiar with the use of gold as money as well as for adornment. Some time after 360 BC, Alexander's father, Philip II of Macedon, had opened up rich sources of gold and silver in Macedonia and Thrace – northward in the Balkans toward modern Bulgaria – and was soon minting both metals into coins in sufficient number to meet all his current needs and to finance his plans for future military conquests.

Establishing substantial monetary wealth was just one of this brilliant man's achievements in the course of transforming the backwater countryside of Macedonia into the greatest power of his age. If Philip were alive today, he would be in constant demand as an expert consultant on both the economics and the politics of developing nations. He had an innate sense of priorities, of how to set his ideas in motion in the most effective and efficient manner, and of how to deploy power to maximum advantage.

When Philip ascended the throne of Macedon in 359 BC at the age of 23, his mountainous kingdom consisted of a small population of mixed tribes, who were poorly fed and occupied their time primarily by making war against one another. By the time Philip was through, he had established Macedonia as a great power and himself as the dominant personality in all of Greece, even though many Athenians considered him a provincial upstart.

He began where he should have begun: with agriculture. Through irrigation, canal construction, land drainage, and flood control, he turned the alluvial plains of his kingdom into a breadbasket. His ample and growing food supply helped him make peace

among his pesky constituency, attracted a growing population of southern Greeks, led to the building of new towns, and provided an expanding pool of manpower for his armies. In addition to feeding the increasing populations, Philip's agricultural reforms significantly enhanced Macedonia's military power by generating feed and pasture for horses and cattle. Like the German panzers, Philip's abundance of horses would give his army fearsome mobility in combat. In addition, the meat from the cattle gave his men strength and endurance over his enemies, a feature that attracted Napoleon's attention many centuries later. His local campaigns around Macedonia also augmented that absolutely essential economic resource: slaves – slaves to work the mines, slaves to work the fields, slaves to keep the whole economy humming.

Philip also made full preparation for his succession, hiring Aristotle from Athens to tutor his son Alexander from the ages of thirteen to sixteen, the equivalent of sending the young teenager to Harvard as an undergraduate and Cambridge or Oxford for graduate work. When Philip was assassinated in 336 BC, after a reign of 23 years, Alexander reminded his people that "My father took you as nomads and paupers, wearing sheepskins, pasturing a few sheep on the mountains...he made you inhabitants of cities and brought good order, law, and customs into your lives."

Philip set the value of his gold coins at ten times the value of his silver coins, a convenient piece of arithmetic that differed from the unwieldy Persian ratio of 131/2:1; his choice also probably reflects the relative increase in gold supplies from newly discovered mines. He decorated his highest-value coins on one side with a chariot to celebrate his victory in the chariot race at the Olympic games of 356 BC, and with a head of Zeus on the other side; many wondered whether that head was Zeus's or really Philip's.

Philip's keen financial instinct led him to produce more coins than he needed for current transactions and army pay. He hoarded the balance as a reserve to finance the wide-ranging military campaigns against the Persians that were on the drawing boards at the time of his assassination.

After Philip was killed and Alexander inherited the throne from his father, he kept all of Philip's Macedonian mints busy pouring out coinage, in addition to the output of mints in Greece, Asia Minor, Syria, Egypt, and Mesopotamia. Supplying gold and silver to feed the mints was no problem. In addition to Philip's hoard and the current output of the Macedonian and Thracian mines, Alexander captured huge amounts of treasure during the course of his victorious campaigns to the east. Alexander followed his father's policy of making gold the prime monetary standard, and, being a man of action and splitter of Gordian knots, he stayed with the 10:1 relationship, adjusting the supplies of each metal from his substantial reserves in order to make the simple ratio function without difficulty throughout his immense empire.

Although Alexander had so much gold available to him that he could afford to be prolific in minting coins, the demand for coins ran high. Alexander wanted to be certain



TOP

One of Alexander's coins, with Hercules on one side. Is this a portrait of the conqueror himself?

BELOW

On another of his coins, Alexander had Nike – representing victory – portrayed on one side, and the warrior goddess, Athena, on the other

that his men would find his coinage acceptable, no matter where they were serving. The troops under his command were primarily mercenaries, and he paid them well to discourage them from looting. He paid many of their civilian debts, gave wedding presents of money to some troops, and distributed separation bonuses when they went home. In his vision of himself as a bearer of Greek civilization rather than as a conqueror, Alexander also brought along with him scientists, engineers, and explorers, and these people had to be paid as well. In addition, he expanded his civilian rule over conquered territories by establishing more than 70 new towns in a great arc from Egypt to India; this, too, created an additional demand for acceptable money. Finally, his vision included prosperous trade among the many areas of his diversified empire, an objective that needed a common currency. Alexander was convinced that trade would promise higher living standards for all.

Alexander and Philip both understood the propaganda and public relations value of gold coinage. Whereas Philip's *philippeioi* had the head of Zeus stamped upon them, Alexander stepped down a notch. He used Hercules, a god ranking below Zeus but symbolizing the greatest physical strength; like Philip's Zeus, the portrait of Hercules stamped on the coins bore a striking resemblance to Alexander himself. Alexander did not change the name of the gold coins established by his father: they remained *philippeioi*. His followers, however, kept Alexander's designs but changed the name of the coins from *philippeioi* to *alexanders*!

Alexander's coinage system prevailed for over 150 years, from India in the east to most of the Greek and Egyptian areas to the west, until the Roman general Quinctius Flaminius defeated Philip V at Cynocophalae in 197 BC and brought closure to the Macedonian hegemony. Flaminius had learned his monetary lessons well: his first step to commemorate his victory was to transform some of Philip's tribute into new gold coins bearing his portrait – the first time a living person had appeared on Roman coins.

The Romans had been using metallic money for a long time. During the 4th century BC, they were in the habit of keeping their treasure in the temple of Jupiter. This choice was made for the purpose of security, but it was also an interesting blend of temporal wealth and heavenly religion. In 390 BC, so the legend runs, the cackling of the geese that lived around the temple alerted the Romans to a surprise attack by the Gauls, who were invading Italy at that point. The Romans were so grateful for this notice of impending danger that they constructed a shrine to their goddess of warning, whose name was Moneta; *moneta* in turn became the derivation of 'money' and of 'mint'.

That was not all in the way of money that we have inherited from the Romans. The Romans also gave us the monetary denomination of pound – *libra* – which is why the pound sterling is identified by the symbol £. In addition, the Latin word *denarius* came to stand for penny and was conventionally abbreviated as d in British usage. Finally, the term

solidus, which meant that a coin was pure gold or silver, was worth one-twentieth of a pound of silver and was equivalent to 12 *denari*. British money was built up from these ratios. A pound was equal to 20 shillings and a shilling was equal to 12 pence – a system that lasted from Norman times until the 1970s, when Britain finally yielded and adopted the decimal denominations long in use everywhere else.

Although the Romans may have started using the word *moneta* in 390 BC, their gold stock at that moment was small. The Elder Pliny mentions an amount in the Roman treasury that contained less than half the gold that Phidias had lavished 50 years earlier on his statue of Athena in the Parthenon and only one-seventh as much as Croesus had delivered to the oracle at Delphi 150 years earlier. Even as the Romans opened up additional gold supplies by extending their territories, and even when their victory over Carthage in the Punic Wars brought them control over the massive reserves of gold in the Spanish mountainsides, they still perceived gold as a reserve but not as something spendable.

The Roman need for gold grew rapidly after about 150 BC as the empire expanded at an accelerated pace, which meant rising military requirements at the same time. Edward Gibbon, in his *The History of the Decline and Fall of the Roman Empire*, reports: 'The golden eagle which glittered in front of the legion, was the object of their fondest devotion; nor was it esteemed less impious, than it was ignominious, to abandon that sacred ensign in the hour of danger.' Later, he tells us, 'The Emperor Domitian [ruled 81 to 96] raised the annual stipend of the legionaries to 12 pieces of gold, which, in his time, was equal to about 10 of our guineas.' Ten guineas in Gibbon's time was the equivalent of about \$53, or roughly \$2500 in 1999 purchasing power – but there was far less to buy in Roman times and a soldier received shelter, food, and medical care in addition to compensation. Hence, \$2500 was a generous annual stipend.

Indeed, the Romans used coinage-money to a far greater extent than any of their predecessors in history. Thousands of soldiers throughout the empire had to be paid, and some Roman generals even minted their own gold coins to distribute to their troops. Furthermore, bread and circuses did not come for free, but promoting domestic tranquility among the Roman *populi* was essential if emperors hoped to remain in power. The doles were distributed in cash on occasion, but even the more frequent payments in kind, the *alimenta*, or bread rations, were largely imported from outside Italy and had to be paid for with coinage.

These recurrent and growing needs for coins were accompanied by an increasing demand for replacement coinage as many coins simply disappeared, some worn beyond usefulness as coins, some in shipwrecks, and some due to plunder by barbarians. A significant amount of gold went to the East in exchange for spices from India and silks that took a circuitous route but that originated in China; once the metal arrived in India,

it stayed there and did not return to the channels of trade. At the same time, the highest-quality ore was being depleted, so the level of mining activity had to expand even more rapidly than the need for metal to mint.

There seemed to be no limits to the demand for gold. After Caesar conquered Gaul, the Romans imported over 100,000 slaves from that territory to work the mines in Italy, to say nothing of the slaves they employed to work away their short lives as miners within Gaul itself. We have already seen how the Romans used slaves to exploit the mineral riches of Spain, at a level of cruelty and disdain for the environment that readily matched the appalling record of the Egyptians.

Wealthy Romans showed off to one another by generously lavishing gold on their bodies, their women, and their homes, but they measured their wealth by their accumulations of gold coins. In the Roman Republic, and the empire that followed, golden money was essential to grease the way to political power. Unlike all the monarchies that had ruled nations since the beginning of time, in Rome it was how much gold you had, rather than who your father was, that defined how much say you had in the affairs of state. How much you had to say, in turn, defined how much bribery and other loot came your way from others in similar pursuit of power and riches.

For example, when Julius Caesar returned from service in Spain as quaestor (provincial official for financial affairs), he had harvested sufficient Spanish gold to buy him attention as a leader, but not enough to take him as far as he had hoped to go. He therefore combined his interests with two other ambitious Roman citizens, one a fabulously rich man named Crassus and the other a military commander named Pompey.

Crassus had begun accumulating his fortune by organizing a fire brigade that put out fires only if paid in advance. In those cases where the owner failed to pay and the building was destroyed by fire, Crassus would buy up the burned-down ruin at a fraction of its worth as a standing structure. He acquired a large number of tenements in this fashion, restored them, and let them out at fancy rents. In addition, Crassus lent money at interest and acquired ownership of silver mines, agricultural estates, and slaves in great numbers. He even educated his slaves to become readers, stewards, and cooks. The huge income that accrued to Crassus from all this wealth enabled him to bribe officials so that he could buy up additional confiscated estates at depressed prices.

Although Pompey ended up with his head cut off, in all likelihood as a result of a contract on him taken out by Caesar, Crassus was destined for an even more horrible end. Crassus was eager to show that he was more than a moneybags and that, like Pompey and Caesar, he could successfully command troops in battle. Accordingly, he provoked a war with the Parthians in Mesopotamia and set off on his campaign with 44,000 troops under his command, foot soldiers for the most part. At the battle of Carrhae in 53 BC, the Parthians attacked the Romans with 10,000 horse archers and a corps of 1000 Arabian camels, making quick work of the job at hand. Crassus attempted to negotiate a surrender,



but the Parthians set upon his troops with such ferocity that less than a quarter managed to escape. For Crassus, the Parthians reserved a special fate that expressed their disdain for the money-mad Roman civilization that he represented. They finished him off by pouring molten gold down his throat.

Up to this point in our story, the supply of gold has been taken for granted or the opening up of new supplies more or less kept pace as the demand for gold expanded. The Jews escaping from slavery, the Egyptians, the Lydians, the Persians, and Philip and Alexander all appear to have had enough gold to do with it whatever struck their fancy, from crafting objects of worship and beautification to coining elegant coins as means of exchange and stores of wealth. From today's perspective, we can see that they were in the happy position of owning an indefatigable printing press bestowed upon them by nature, whose output, because it happened to be shiny, dense, and malleable into beautiful things, was accepted without question everywhere.

Now everything changes. With an empire that reached from the Mediterranean to the Black Sea, and from the border of Scotland to the southernmost areas of Egypt, the Romans found that their supply of gold for coinage constantly fell short of their needs despite mining output of at least five tons a year. Quite aside from governmental expenditures that had to be financed, the emperors spent money on themselves with a degree of abandon that their citizens could only envy. Yet nature sets the ceiling on the supply of gold and silver: you cannot create metal out of nothing. The alchemists in later times were to learn that lesson over and over again.

A society that uses metal for money will always be constrained by the supply of that metal. The random location of mineral deposits makes countries such as Lydia rich as a



matter of good luck and other countries greedy for gold as a consequence of bad luck. History teaches us that natural advantages are not an automatic formula for success, but having a head start endowed by nature never did anybody any harm.

When a nation's supply of metal is insufficient to meet the needs for coinage, and, in many instances, even when coins are no longer the only acceptable form of money and paper substitutes are in use, there are three ways out. One is to live with an insufficient supply of money, so the demand for goods at current prices chronically falls short of the supply of goods offered for sale, and downward pressures on the price level persist over extended periods of time. This painful process has often occurred as a default solution, with dire political and social consequences. The Great Depression of the 1930s is the most vivid but by no means the only example of this policy. A second method of overcoming a shortage of monetary metals is to import gold from other areas, either by plunder or by trade. These solutions have motivated both great adventures and complex economic policies, not always with happy outcomes. The third method is the simplest but is unlikely to be successful over the long run – namely, to use the same amount of metal to produce a greater supply of coins.

That solution is known as 'debasement of the currency', which used to mean literally reducing the metallic base from which coins are minted, or mixing base metal with the precious metal, while leaving the face value unchanged. Over the years, debasement has come to mean any irresponsible, or at least ill-advised, effort to create new money out of nothing – a process at which governments have become increasingly ingenious with the passage of time.

Hadrian's Wall, the northernmost border of the Roman Empire in Britain



*Mosaic portrait of
Constantine the Great*



The monetary innovation of debasement has a long history. For example, Dionysius of Syracuse (405–367 BC) had borrowed heavily from his citizens and was hard put to figure out how he could pay them back. He ordered all coins in the city brought to him, under penalty of death. He restamped the coins so that each one-drachma coin now read two drachmas. After that, paying off his debts was easy. Dionysius's methods were drastic, but the essence of the process – like many things Greek – was classic in its execution.

The Roman emperors learned to make debasement a routine procedure. One might argue that the Romans had no choice, given the dynamics of their society and their empire. Even though they succeeded in developing abundant supplies of gold throughout their empire – and, in fact, expanded their empire in some directions primarily to acquire new sources of gold – their financial requirements and their insatiable demand for adornment in gold grew so rapidly that they simply never had enough gold to satisfy their needs. Most important, they never acquired a sufficient sense of rectitude to choose between government spending and the good things in life.

The Romans set an example for debasement that later rulers throughout history have followed in many different formats, but on a scale that few have matched. The usual method of debasement was to mint coins with an unchanged face value but smaller in size and with a reduced metallic content, thereby stretching the available supply of metal to produce a larger number of coins. Debasement worked best when people were fooled into thinking that nothing untoward had occurred with newly issued coins, but you can't fool all of the people all of the time. Debasement on many occasions did lead people to melt down the old coins and bring the unadorned metal to the mint; they would walk away with more coins of a given face value than those that had been melted down. It was the state that benefited from the increased inflow of precious metals that this process brought about. In view of the primitive nature of tax systems in those days, the debasement process was an important source of governmental revenue.

Nero was the first emperor to take the route to debasement – a development that should come as no surprise. Nevertheless, Nero was a piker at the task, despite his mindless pursuit of luxury. The heavy spending by his successors on personal goodies and the maintenance of the armies and bureaucrats over the wide stretches of the empire strained government finance to the limits. As paper money and bank credit had not yet been invented, debasement was the only available method to create enough purchasing power to satisfy the constantly expanding needs.

By the time that Gallienus became emperor in AD 260, silver coins had 60 per cent less metal than they had when Augustus became emperor. Gallienus then threw discretion to the winds. He ruled for only eight years, but he managed to cut the silver content of the coins down to a mere 4 per cent. The outcome was inevitable: wild price inflation. One expert has estimated that prices increased at the negligible pace of 0.4 per cent a year over the 250 years between the reign of Augustus and when Gallienus was emperor; but during

the 34 years after Gallienus began his machinations with the coinage and Diocletian became emperor, prices rose over 9 per cent a year – which means that, in ad 304, prices were 20 times higher than in AD 260. Roman money was now not just a financial wreck; it was a physical wreck as well. The copper coins had so little metal content and had become so thin and frail that their imprint could be placed on only one side.

Although the small-denomination Roman coins had become essentially worthless, the gold coins had fared better. The Romans did reduce the gold content and size of the coins over time so that more coins could be produced with a given amount of gold, but they resisted the temptation to mix the gold with alloys, the technique that destroyed the ability of Roman copper coins, and even some silver coins, to function as anything more than curiosities.

After Diocletian came to power in AD 284, he spent some 20 years trying to reform the currency and, under a bewildering variety of price and output regulations, to bring inflation under control. Diocletian was also wary of how gold could lead a nation into trouble. According to Gibbon, in AD 296 Diocletian caused a diligent inquiry to be made 'for all the ancient books which treated of the admirable art of making gold and silver [i.e. alchemy], and without pity committed them to the flames; apprehensive, as we are assured, lest the opulence of the Egyptians should inspire [the Romans] with confidence

*Constantine's solid gold
solidus, or bezant – the
'dollar of the Middle Ages'*



to rebel against the empire.' What with everything, Diocletian was exhausted by the burdens of being emperor. In 305, he retired voluntarily and set himself up in a lovely palace on the Dalmatian coast, where he lived more or less happily for the rest of his life.

Diocletian's successor was Constantine, who reigned from 306 to 337 and who immediately set out to improve the acceptability and respectability of the Roman currency by issuing a new gold coin called the gold solidus, which later became known as the bezant. When Constantine issued the first bezants, they weighed 4.55 grams – heavier than any other gold coin in existence -- and were 98 per cent pure. At \$300 per ounce of gold, the bezant in today's money would be equal to \$42.66, but the purchasing power of gold in Constantine's time was much greater than it is today. Clearly, this was a coin with high current value. The bezant continued in production, with unchanging weight and purity, for about 700 years, long after Rome had fallen to the barbarians. The gold bezant thereby deserves a place in the *Guinness Book of Records*, as no other coin in all of history can match its longevity.

The supply of gold for minting was not a problem for Constantine. His conquests eastward brought massive inflows of tribute. Building in part on what he had learned about government fiscal policy from Diocletian, Constantine also levied new taxes payable only in gold or silver and employed the proceeds to feed his mints for transformation into the new coinages.

But the richest source of gold came about as a result of Constantine's conversion to Christianity, which he established as the state religion in 313. Inspired by his vision of a shining cross and the words 'In hoc signo vinces' ('In this sign, I shall conquer'), Constantine proceeded to strip all the pagan temples throughout the empire of the gold and other treasures they had accumulated over hundreds of years. Some of this gold resided almost permanently on top of his head: he wore his bejeweled golden crown at all times.

The possession of gold from this point forward is no longer a matter of right, privilege, or hierarchical position in society. It is earned or it is plundered or discovered anew in the rivers and mountains. Whatever the source, an increase in the stock of gold provokes high excitement, because that gold is an instant path to money – and to power.

A low-angle photograph of several massive ancient Egyptian columns. The columns are made of dark stone and are covered in hieroglyphs. The scene is bathed in a warm, golden light, likely from the setting or rising sun, which creates a dramatic silhouette effect on the left and highlights the textures of the stone and the carvings. The columns rise steeply towards the top of the frame, creating a sense of height and grandeur.

The Symbol

IN AD 200, WHEN THE ROMAN EMPIRE WAS ABOUT THE SAME AGE AS THE UNITED STATES IN AD 2000, THE CAPITAL OF THE EMPIRE WAS LOSING CONTROL OVER ITS OUTER TERRITORIES, BUT ROMAN COINS WERE STILL IN CIRCULATION EVERYWHERE.

FOUR

and the Faith



Like the US dollar, Roman money had lost purchasing power and, in some areas, the deep respect it had once deserved, but it was nevertheless the only money that circulated throughout the empire.

Then, as Rome's European dominions succumbed to the invasions and depredations of the barbarians, Rome's common currency vanished. It did not stop circulating because of any decree or agreement among the new rulers who proliferated within the borders of the old empire. Roman money dropped out of sight because money itself virtually dropped out of sight.

There was little use for money in the terrors and ravages of the early years of the Dark Ages. Trade and travel throughout Europe fell to a trickle. Urban life disintegrated as people huddled as close as possible to food supplies; the barbarians themselves came from rural communities and the way of life in cities and towns was unfamiliar to them. As the magnificent Roman roadways deteriorated into ruts, even bricklaying skills grew scarce. The need for money for soldiers' wages also dried up as government-supported armies were replaced by roving bands of ruffians who lived off the land.

But coins are hard. Unlike dollars on a computer screen, they were not about to disappear into thin air. The Roman coins of gold continued to exist as though nothing had happened, although they seldom circulated as money. So where did they go?

People hoarded the coins and other golden objects against the terrifying insecurity and loneliness of the times. Archeologists have discovered substantial amounts of buried treasure from the Dark Ages throughout Europe, even in far north Scandinavia. Sometimes as crudely as this, sometimes in more elaborate forms, hoarding gold at times of uncertainty and fear would persist throughout most of history. There is little difference in principle between the burying of gold in the backyard during the Dark Ages and the desperate effort to build and preserve gold reserves at the Bank of England in 1930 and 1931.

The events that followed the fall of Rome were so diverse in both place and time that they fail to fit into neat segments. The so-called Dark Ages had many intermittent gleams of light. Furthermore, even though the history of the eastern Roman Empire based in



The Church of St Sophia in Constantinople (now a mosque). Its dome was lined with 12 tons of gold

Constantinople was strikingly different from what happened in Europe, the interaction between East and West persisted to such an extent that an effort to sharply distinguish the two chronicles would be more confusing than illuminating.

The preoccupation with gold was one of the primary forces that tied the two areas together. Gold as both money and adornment played a central role in Byzantium. In Europe, brand-new kingdoms whose leaders had never heard of Croesus or Crassus launched their gold coinage as soon as they could; the jeweler's art flourished as well, even under the most primitive political conditions; and the traditional marriage between golden ornamentation and religion – though bumpy, like many marriages – never quite came to a state of divorce.

The Byzantine emperors ruled from Constantinople for over 1000 years. By and large, they were a decadent, corrupt, conspiratorial, cruel bunch. But while they may have debased their regimes morally and politically, the integrity, purity, fame and acceptability of Constantine's golden bezant was an overriding preoccupation with all of them. The entire history of the Byzantine Empire is marked by its obsessive focus on gold, not only as money but also as an advertisement of unrivaled opulence. Gold served as the key instrument that the emperors used along with cruelty and repression to bind together their sprawling, disparate territorial domains. The golden bezant financed the empire's imports, its armies, and its alliances with other nations.

Justinian's effort to surpass Solomon when building the church of St Sophia out of an inherited hoard of some 300,000 pounds of gold was just one instance of the lavish use of golden displays to shout power. All the labyrinthine palaces that the emperors maintained on the banks of the Bosphorus were overlaid with copious decorations of gold and precious jewels; understatement was not their style. Emperor Theophilus deserves the prize for ostentation for the tree of gold he created to shade the gold throne. The tree and the throne were flanked by golden birds, lions, and griffins; at the arrival of a visitor, the lions would swish their tails and roar as the birds warbled a welcome.

The uses of gold were so varied and so extensive that Constantinople's skilled goldsmiths were in demand throughout Europe, especially in Italy. The goldsmiths were the principal artists of the Dark Ages, during the centuries before painting, sculpture, and architecture became the predominant forms of art. They worked on the mosaics that cover the portals of St Mark's in Venice, on the breathtaking mosaics on the Church of San Vitale in Ravenna, south of Venice, and as far off as Monreale outside of Palermo. When the European goldsmiths saw the beautiful and delicate work of their counterparts in Byzantium, the Byzantine style became the high fashion of the early Middle Ages. In fact, the patron saint of goldsmiths, St Eloi (641–660, also known as St Eligius), was a monk and mintmaster from 7th-century Gaul who learned his skills in Constantinople. His frequent appearance in paintings right through to the 15th century testifies to his importance and prominence. The English goldsmiths had their own patron saint, St Dunstan, a Benedictine monk and skilled artist who was archbishop of Canterbury from 960 to 988. He must have been quite a man: a golden embroidery of 1470 shows him in his workshop tweaking the nose of the devil.

The gold for display was just on the surface, however. Behind the scenes, the emperors amassed huge hoards of gold coins and gold bars. Basil Bulgaroctonus (ruled 976–1025) had over 200,000 pounds of gold, much of it hidden away in subterranean chambers. Around 530, Emperor Anastasius owned a store of some 300,000 pounds. And Empress Theodora, who ruled in the mid-9th century, died with 100,000 pounds in her possession. These were huge sums for their time but make a dramatic contrast with our own era in which gold stocks are measured in thousands of tons rather than in thousands of pounds.

The gold was a security blanket. The Byzantine rulers frequently went to war, when troops had to be paid in gold, but the emperors never reached a point where their enemies ceased to threaten. The empire was constantly menaced by Bulgarians and Germanic hordes to the west and, after the 7th century, to the east and south by militant Muslims attacking every infidel in sight. Since the Byzantines could not fight on all fronts simultaneously, they paid out a never-ending stream of gold tribute to keep their enemies at bay by buying off potential aggressors directly and by bribing European allies to provide protection. Today, we would call it outsourcing security.

That approach was especially important in the case of the Lombards, with whom the Byzantine emperors carried on a long but unstable relationship. The Lombards, originally known as the Langobards (long-beards), came from what is now modern Hungary. During 568–9, with the aid of Saxons and Slavs, they invaded Italy 100,000 strong under the leadership of their king, Alboin – which is why northern Italy is still known as Lombardy.

The imposing Lombard invasion of northern Italy threatened the priceless Byzantine possessions of Venice and Ravenna. Without gold, all would have been lost. First, the emperors bribed some Lombard factions to fight with other Lombard factions. Then they paid out gold to recruit as allies the Franks, another Germanic tribe that had established



ABOVE
Constantinople



itself in Gaul, or modern France. Gold continued to move from Byzantines to Franks, from Byzantines to Lombards, and even back and forth between Lombards and Franks, as each group used the gold to buy off or bribe the others.

Internecine warfare among the Lombards continued at a high level until the middle of the 8th century, at which point they finally got their act together. The monarchy the Lombards established was so firm and threatening that the Byzantines were not alone in deciding the job had to be finished, once and for all – even the pope for once took the side of the Byzantines. In 754 and 756, they joined in supporting a major invasion of Lombardy by the Franks, greased along by a bribe of 50,000 gold solidi. Having beaten back the Lombards, the Franks then proceeded to collect gold tribute from both their allies and their defeated enemies – at one point they were collecting an annual tribute of 12,000 solidi. The accumulation of all that gold in what would one day be France explains why it was the Merovingian monarchs who led the way back to gold coinage in Europe after the fall of Rome. In 773–4, Charlemagne, Merovingian king of the Franks, conquered the Lombard kingdom outright and annexed it to his own empire. In return, the pope crowned him emperor of the Holy Roman Empire in the year 800. For centuries thereafter, the French coveted Lombardy as part of their natural realm, until Francis I reclaimed it for France in 1515. On this occasion, however, Francis was given only a gala meal by the pope – and not long afterward he lost Lombardy again to the current emperor of the Holy Roman Empire.

Where did all this gold come from? The towering position of the bezant in the medieval world did not depend on domestic sources of gold. The gold supplies from the Pactolus River that seemed like a bottomless well to Croesus had long since been exhausted, and no other important sources were known to exist in Byzantine territory. Although some gold came in from beyond the eastern borders from as far as Russia, the richest source of mined gold came from the old Nubian mines in southern Egypt and Sudan.

Unfortunately, Nubia was a source for the Byzantines for only a short time. The Arabian Muslims conquered those territories in the 7th century, formed lasting relationships with the Nubians, and, in one fell swoop, cut off this seemingly bottomless source that Byzantium had enjoyed for so long.

The shock was profound. As one historian, Robert Sabatino Lopez, a distinguished scholar of monetary history, has described it, 'The empire that amazed the world by the profusion of its riches and by the abundance of its gold coinage was constantly threatened with exhaustion of its stock of precious metals.' Now trade, military campaigns, taxation, and raw plunder were the only means of sustaining the golden passions of the emperors that were essential to the illusion on which so much of their power depended. Military victories brought more than booty from defeated enemies (when Nicephorus captured the Bulgarian royal treasury, he planted the imperial seal upon every object). Not unlike modern privatization schemes, confiscated lands could be distributed to officials, soldiers,



ABOVE
Statue in Vienna

LEFT
St Mark's, Venice

and sailors, a move that created new income from taxation. Nevertheless, triumphs on the field of battle were an uncertain and interruptible source. Taxation was more predictable, and the ruthless, cold-blooded, and pitiless efforts of the emperors' agents on this score would be the envy of an IRS agent in our own time. But even taxation has limits.

Consequently, after the loss of Nubia, trade and commerce became the prime means of bringing gold into the coffers of the emperors as well as to the merchants and manufacturers on whom the stream of tax revenue was so heavily dependent. Byzantine trade followed an essentially triangular pattern involving Europe in one direction and the Muslims to the south in the other. Byzantium imported hardly anything from the Europeans but sold them luxury items including the finest of textile products and the arts of the goldsmiths. The silken fabrics woven in Constantinople were as highly prized in Europe as the jewelry and adornments of the goldsmiths and were so much in demand that the Byzantines finally started to grow their own raw silk. These net exports earned sufficient gold to balance Byzantium's chronic import surplus with the Muslim countries. The Muslims also maintained a positive trade balance with Europe, selling luxuries, olive oil, and horses in exchange for timber, iron, and slaves.

Immediately after the loss of Nubia, the emperors established a wide variety of arrangements to facilitate commercial activity with other countries, including a significant improvement in port facilities where officials could protect the merchandise from thieves. The emperors also imposed restraints on trade, such as severe restrictions on the export of gold and essential foodstuffs, and made a sustained effort to limit imports to food and raw materials not produced in the empire. All of this kept customs officials busy inspecting the luggage of travelers, making surprise raids on shops, checking on movements of gold bars and gold bezants, and preventing forged coins from entering into circulation.

Nevertheless, the Byzantine government failed to develop any systematic plan to encourage industry or even to keep book on the balance between imports and exports. Keeping track would have been virtually impossible in any case, without a central bank or other institution through which all the money flow could ultimately pass. Each merchant and trader settled up his own balances with his counterparts in other countries, so the gross flows of coinage in both directions were large but the net inflow or outflow was obscured.

The whole process would have been a failure without the bezant. In both military affairs and in trade and commerce, the foundation of Byzantine wealth and financial clout was this wondrous golden coin. The emperors used every means to pursue the Lydian tradition of using gold coinage not just as money but also as public relations and advertising to emphasize their power and wealth. As a contemporary of Justinian described it, the gold money of the empire 'is accepted everywhere from end to end of the earth. It is admired by all men and in all kingdoms, because no kingdom has a currency that can be compared to it.' Another contemporary proclaimed, 'It is not right for the Persian king or for any other sovereign in the whole barbarian world to imprint his own



*Interior of the church
of St Sophia in
Constantinople*





The iconoclasm of the 8th century AD also affected the designs on coins. Those shown here are quite simple, based on the cross

likeness on a gold stater, and that, too, though he has gold in his own kingdom; for they are unable to tender such a coin to those with whom they transact business.' One hundred years later, Justinian II (685–711) went to war against the Arabs because the Caliph had struck gold coins with his own portrait. Justinian lost, but defeat did not

discourage the Byzantine chronicler, who insisted, 'It is not permissible to impress any other mark on gold coins but that of the Emperor of the Romans.' Furthermore, Justinian won a moral victory in that struggle, for the Muslims later replaced the portraits and religious figures of the bezants with quotations from the Koran, thus distinguishing the Arab dinars for hundreds of years.

The fame of the bezant has not faded with the passage of centuries: as recently as 1951, Robert S Lopez characterized the bezant as 'the dollar of the Middle Ages'. Maybe even better than the dollar: 'The bezant outstripped the dollar in stability and intrinsic value. Indeed, its record has never been equaled or even approached by any other currency.' He goes on to assert that 'The bezant was more than a lump of gold. It was a symbol and a faith, the messenger of the divine emperor to his people and the ambassador of the chosen people to the other nations of the world.'

The early issues of bezant coins portrayed the image of the emperors, occasionally their spouses, and frequently their sons. Then Justinian II decided to proclaim not only his importance but his piety. He took the revolutionary step of exhibiting the haloed bust of Christ on his coinage. For its time, this was too much. Justinian's presumption helped ignite the Age of Iconoclasm, when populace and emperors rose up to eradicate all image-venerating doctrines and practices in the churches. The coinage was incidental. The primary focus of the movement was on the golden-haloed icons that decorated the walls of the churches and that had become objects of a cult-like veneration as channels from which superhuman power descended to humanity. The result for the emperors was far from a total loss: they filled their coffers with the gold torn from the places of religious worship by the iconoclasts. Thus, the gold from the Byzantine icons did not disappear any more than the gold in the Roman coinage had disappeared; it merely showed up in a new location and form.

Iconoclasm became official policy in 730 under the reign of Leo III (717–741), leading to more than 100 years of savage persecution of reluctant clergy, accompanied by military battles with all who dared to harbor the old ideas. Leo, as it happens, had been a favored protégé of Justinian II, the ill-behaved coin-maker. In an echo of the story of Jason and the Golden Fleece, Justinian grew suspicious of Leo and sent him off on a highly risky assignment to the far reaches of the eastern frontier. Leo surprised Justinian by accomplishing his mission and returning to Constantinople in one piece. Two emperors later, in 717, Leo marched on the capital with his troops and seized the throne for himself.

The Age of Iconoclasm finally came to an official end in 843, by order of Empress Theodora. The icons were returned to their previous locations and their veneration was



restored to orthodox dogma. Coinage design rapidly followed suit. Busts of helmeted emperors now appeared holding a cross; occasionally Christ showed up on the opposite side from the emperor's portrait. Emperor John I Tzimiskes went so far as to issue coins showing him being crowned by the Virgin herself, with the hand of God above.

All of this was wonderful in its own way, as long as it lasted, but no hegemony in history has lasted forever. After Constantinople fell to the Crusaders in 1204, the bezant began to lose its purity to debasement and, as a result, its wide acceptability. Fifty years later, the rising Italian trading powers of Florence, Genoa, and Venice began issuing gold coins that gained as much reputation in their own time as the bezant had enjoyed in its heyday. Indeed, by the middle of the 14th century, citizens of Byzantium were paying their taxes in Venetian gold ducats. Constantine XI, the unfortunate man who was reigning when Constantinople finally fell to the Turks in 1453, was the only emperor who appears never to have issued any coins at all. There is a saying that 'The empire disappeared when it had spent its last sou.'

The ruins of the walls of Constantinople. Just before it fell to the invading Turks in 1453, Constantine IX was the only Byzantine emperor not to have issued any coins

Turkish skyline showing St Sophia





Gold, Salt

THE RULERS OF THE ARAB DOMAINS MAY HAVE BEEN DEEPLY RELIGIOUS, BUT THEY STILL TOOK THE BYZANTINE EMPERORS AS THEIR ROLE MODELS WHEN IT CAME TO THE GOOD THINGS IN LIFE. ONE HAS ONLY TO READ A THOUSAND AND ONE NIGHTS TO CAPTURE THE FLAVOR OF THIS SOCIETY.

FIVE

and the Blessed Town

Despite Mohammed's admonition that 'He who drinks from gold or silver drinks the fire of Hell,' the Caliphs had an immense appetite for gold and the romantic and bizarre types of display that gold could provide. At the wedding of the son of Harun al-Rashid, who was the protagonist of the Arabian nights, the groom's father-in-law threw gold balls around for the pleasure and possession of the wedding guests. He bestowed 5000 gold pieces on a poet and paid 400,000 pieces for a robe of honor for a courtier. Golden trees and singing golden birds in the palace at Baghdad were the inspiration for Theophilus's extravagant throne in Constantinople. One king's sister left 2.7 million dinars and 12,000 robes woven in gold thread and jewelry. Cairo in the 11th century had thousands of shops selling gold, jewelry and luxurious textiles.

The Arabs had no difficulty accumulating a massive golden treasure. Their creativity at the task was impressive. They ravaged their defeated enemies, outsmarted their competitors at trade, and opened up a major source of gold that had contributed a mere trickle over the centuries before their efforts came into play.

The piles of gold collected as the prize of warfare were enormous. The booty came from Persia, Syria, Egypt, Palestine, the great westward sweep across North Africa, Spain, and from as far away as Poitiers in France before the Arab armies were finally halted there by Charles Martel in 732. The Arab invaders of Egypt in particular amassed a huge heap of treasure from ravaging the agglomeration of gold that had lain for thousands of years in the tombs of the pharaohs. They also reopened old gold mines in Egypt, Nubia and Ethiopia while they carried out exhaustive searches for new alluvial supplies in the mountain streams of those areas.

The economic consequences of these conquests were profound. It was not just the booty and the reopened mines. The Arabs soon succeeded in eating deeply into the heart of Byzantine economic power by setting themselves up as traders of extraordinary acumen and persistence. In time, they dominated the major commercial contacts that had served the Byzantines so well for so long, throughout all of the Byzantine sphere of



A dinar

influence, even as they built new commercial relationships all along the southern Mediterranean. The Arab ships plied the seas down the east coast of Africa and across the oceans to India and China in search of profit. They even traveled northward, through the river highways of Russia, to the Scandinavian countries, trading merchandise acquired from across the seas for furs, amber, honey and slaves.

Trade requires money. Money conveys power. Gold serves more purposes than conspicuous consumption. Less than 50 years after the death of Mohammed, the Arabs emulated the great rulers of the past with the debut of their own gold coinage – the dinar – issued by the Caliph Abd el-Melik at Damascus. These coins, 97 per cent pure gold and minted in great quantity, gradually displaced the bezant as the major international currency, circulating throughout the Arab domains and everywhere in Christian Europe as well.

The first dinars were imitations of the Byzantine coins, which gave them immediate acceptance: people are always hesitant to accept money that looks funny, regardless of whatever other attributes it may have. As we have seen, however, the portraits and religious figures of the bezants were replaced by quotations from the Koran.

The appetite of the Arabs for gold was so voracious that by the 9th century even the fruits of conquest, even the revitalization of the East African sources, and even the gains from trade were insufficient to meet their needs. There never seemed to be enough gold for the elaborate forms of luxury that the Arabs devised or to maintain the hectic pace of the mints that poured out the dinars in such volume.

Luck was with the Arabs. As a result of their conquest and settlement of the northern coast of Africa, they made contact with a source of gold that had fed the fortunes of Carthage more than 1000 years earlier. The Arabs never actually possessed the West African gold mines, but their genius for trade did the job for them. For several hundred years, they enjoyed a virtual buyers' monopoly over the gold that lay hidden far to the south, below the farthest reaches of the Sahara, in an area of approximately 600 square miles, with its southern border defined by the east–west coastline that stretches from the Ivory Coast to Nigeria. This area has also been known as the Gold Coast, although the wealth it gained later from exporting slaves may have exceeded the gold that thousands of camels had hauled so faithfully over the vast Sahara for so many years.

Although the Romans and the Byzantines had both held sway on the Mediterranean coast of Africa at one time or another, their primary purpose in occupying the area was military. They hugged the coast and its seaports, ignoring the riches that lay to the south across great unmapped wastes of pure desert. The Arabs, however, meant business when they occupied North Africa. They set up trading posts such as Tunis on the sea; they also opened up centers such as Fez and Marrakesh at significant distances inland. Ultimately, their traders would appear in the heart of the Sahara itself.

Sijilmesa, where the road to Morocco crossed the main north–south route to the gold country, was the place where all the caravans met. It was described by one Arab trader as 'the gateway to the Sahara. . . . One of the greatest cities of North Africa and the most famous of the whole universe . . . whither traders take goods of no value and return with their camels laden with coarse gold.' The city grew rich simply by taxing the huge volume of traffic that passed across its borders. Deeper into the interior were towns with exotic names, for example, Taghaza, Taodeni, and Gadames, and, most famous, the major commercial center of Timbuktu. Timbuktu was located well over 1000 miles south of the Strait of Gibraltar, on the banks of the Niger River, which, together with the Senegal to the east, enclosed much of the gold mining area.

The great abundance of West African gold had been known to people around the Mediterranean for centuries. Around 500 BC, Herodotus himself provided a lively description of the territory, which subsequent visitors over the years never failed to confirm. Reaching the gold supplies across the great unmarked and arid wastes of the desert involved a risky, complicated, and lengthy voyage, in which navigation by the stars was every bit as essential as for ships at sea. E W Bovill, the most authoritative contemporary historian of the Sahara, has observed that 'Outside the polar regions there are few parts of the world less encouraging to human occupation.' Nevertheless, Herodotus provides sufficient information to tell us that active communication between the coasts and the interior Sahara existed even in his day. Ancient rock drawings reveal that the bullock was the primary means of transportation.

*Camel train in the Sahara,
carrying gold to trade for
salt*



The camel first appeared in the Sahara some time around AD 100, perhaps introduced by the Roman legions involved in military campaigns that demanded speed. The camels probably came from Egypt, where the Persians had brought them about 500 years earlier. This remarkable innovation in the art of transportation – equivalent in some ways to the introduction of the automobile or even the airplane in modern times – greatly shortened the time spent moving between watering points, thereby permitting a much wider range of travel. Bullocks can come close to matching camels in their ability to do without water – a maximum of about ten days – but most camels can carry two to three times as much load as a bullock. Furthermore, the best camels can cover twice as much mileage in a day as the typical bullock, which is no minor consideration when the time needed to reach the next watering hole is the difference between life and death. The innovation of the camel was remarkable in another sense. The introduction of the camel, according to one authority, 'contradicts the most basic Western metaphor of technological progress: here

Camel rider in the Sahara



the wheel – present in North Africa and the Sahara from Phoenician times – had to be “disinvented” to make possible the linking of the Sudan and the Mediterranean.’

The impact of the camel on the potential volume of trade was revolutionary. As Bovill describes this development:

[The introduction of the camel] marked the dawn of a new era for the northern half of the continent. . . . The camel gave man freedom of movement he had never known before and brought within his reach the remotest pastures. The caravan routes lost half their terrors and new roads were opened for the flow of trade and culture.

The landscape that led to the gold fields was not the only feature of the region that people from Europe and the Near East would find strange. Herodotus cites the Carthaginians as the source for the following story. The Carthaginians described to him a place on the west coast where they would neatly arrange the merchandise they wanted to trade, return to their ships, and ‘raise a great smoke’. At that point, the natives would come down to the shore carrying gold, would leave as much gold as they believed the Carthaginian merchandise was worth, and then would withdraw from the scene. The Carthaginians, in turn, would come ashore and look the situation over. If they were satisfied, they would take the gold and sail away; if not, they would return to their ships and wait patiently. The process would continue until both sides were satisfied – but they would never see each other face-to-face or exchange a word. This ‘dumb barter’ was characteristic of how business was transacted throughout much of the gold-bearing areas. It continues to exist in some parts of Africa to this day.

We can only speculate on why dumb barter as a method of doing business should have persisted for so long. Perhaps the natives insisted on these arrangements in order to protect themselves from traders tempted to capture them as slaves. Traders sufficiently eager to acquire what the Africans had to offer had no choice but to choose this curious arrangement.

Around 750, ravenous at the thought of all that gold down south, the Arabs launched an expedition from Morocco to conquer the gold-bearing territories. This was one occasion when matters turned out badly for the Arabs. They failed completely in their objective, suffered serious casualties, and even failed to discover where the gold was coming from. Thereafter, they obtained their gold by means of trade instead of conquest.

Although the Arab and European traders in the Middle Ages occasionally offered the Africans merchandise or even the silver and copper coins that the Africans considered better money than gold, salt was the product most desperately in demand. Humans can never do without salt, but the people in the territories that produced the gold must have had an unusually intense and insatiable need for it. They were so unfortunate as to live in

one of the few spots in the world where the nearest sources of salt were far distant in a land where nobody could travel faster than ten miles a day.

Substantial sources of salt did exist about 1000 miles to the north, where the salt miners, many of them black-skinned slaves, worked under extremely harsh conditions. They were nearly a 20-day journey from the nearest towns, were often blinded by desert winds, and on occasion even starved to death because of delays in the arrival of the traders who would bring them food and fresh water to swap for the salt.

Most of the salt was transported south in camel-driven caravans. At many points, however, where pasturage was so scarce that the camels could proceed no further, the great slabs of salt had to be broken into small pieces that were then placed on men's heads for the rest of the trip. One 15th-century traveler from Portugal described what happened next:

Each man carries one piece, and thus they form a great army of men on foot, who transport it a great distance . . . until they reach certain waters. . . . All those who have the salt pile it in rows, each marking his own. Having made these piles, the whole caravan retires half a day's journey. Then there come another race of blacks who do not wish to be seen or to speak. . . . Seeking the salt, they place a quantity of gold opposite each pile, and then turn back, leaving salt and gold.

This story is not just a matter of curiosity. It has a deeper meaning. Salt was so precious to the gold diggers that many of them would trade their gold only in return for salt. In many transactions, an ounce of gold exchanged for an ounce of salt. Bovill asserts that 'Salt was so infinitely the more important [compared to gold], that it is no overstatement to say that gold was valued by the Sudanese almost entirely for its purchasing power in salt. . . . It was the basis of their domestic, as it was of their foreign, trade, neither of which can be comprehended without an understanding of how starved they were of this essential to the well-being of man.' Look at it the other way, however. If an ounce of salt could acquire an ounce or more of gold, fetching the gold must have been an enormously profitable operation.

Thanks to the practice of dumb barter, the uncongenial geography of the goldfields, and the natural reticence of the natives, Europeans and Arabs were frustrated for centuries in their search for the source of the African gold. The whole area acquired a kind of mysterious glow among the peoples to the north.

During the 15th century, Europeans developed the custom of calling the gold-bearing areas Guinea (which the British persisted for a long time in spelling 'Ginney'). Indeed, the Portuguese, who were the first to explore the territory, received permission from the pope in 1481 to call their king 'Lord of Guinea', a title that survived until the 20th century. In 1662, the British began to use gold imported from West Africa by the African Company to mint a coin that they called the guinea.



Precious salt.

Controversy persists over the source of the name of Guinea, because no such place existed in Africa at that time. No doubt the word is a corruption of something that sounded like Guinea. A likely candidate is Ghana, but Bovill insists, convincingly, that Guinea is derived from the name of the trading post of Jenne, situated on a tributary of the Niger River about 300 miles southwest of Timbuktu, toward the gold-mining areas.

As our story has wound its way from golden palaces and religious icons, from bezants to dinars, from golden balls to golden tribute, and finally to the dumb barter of gold for slabs of salt in darkest Africa, a disturbing question comes to the surface: Where is value? For the Europeans, the Byzantines, and the Arabs, gold was the magical focal point of their material desires. Not so for the Africans.

To the Africans toiling for gold but starving for salt, the salt standard was a force far more powerful and durable than anything that the gold standard stood for in the sophisticated civilizations everywhere else on the globe. What must those poor diggers have thought of the funny people from the north country who swapped inestimable salt for stuff whose only role on earth was to give men pride and pleasure by letting them see its lustre?

The question reverberates into our own time.

The Legacy of

AS THE LONG STAGNATION THAT FOLLOWED THE FALL OF ROME YIELDED TO A MEANINGFUL PACE OF ECONOMIC DEVELOPMENT AFTER THE FIRST MILLENNIUM, THE SEARCH FOR THE MEANING OF VALUE BECAME A FOCUS OF INCREASING STUDY AND DEBATE IN EUROPE.

SIX

Coba, Babba and Uddol

A great debate developed among the scholastics and the monks in the universities over the definition of a 'just price'. St Thomas Aquinas himself, in the 13th century, acknowledged that 'It is true that money is subordinated to something else as its end; still, to the extent that it is useful in the quest for all material goods by its power, it somehow contains them all. . . . This is how it has some likeness to beatitude.' Gold is a fountain of beatitude.

Commercialization and trade cannot take place without money. Creating new monetary systems from the ground up is no simple matter, for nothing can function as money unless it comes in a format that will be acceptable to everyone who uses it. No decree establishing a system can work unless the arrangements match the values, traditions, and needs of the community. The history of money – and much else affected by money – is a long and tortuous tale of how people have tried to deal with these difficulties under widely varying circumstances.

When all money is hard – when all payments are in bullion and coin – the process is especially intricate, because supplies of gold and silver are determined by nature rather than by the humans who use them. Mines can be exhausted, countries can gain or lose from plunder, and supplies can move across borders when trade is unbalanced. But human decisions matter, too. People can hoard rather than spend their coins, which was a common practice in the political and economic turbulence of the Middle Ages. Gold is a hedge against the risks of chaos, and persuading people to bring their treasure back into circulation as money is no simple matter in a world where money in transit is often plundered by robbers or lost in shipwrecks, and the insatiable demands of the state are a constant threat.

The impact of frequent changes in the relative supplies of gold and silver complicated matters throughout the Middle Ages, and would continue to mess up monetary systems in both Europe and the United States until well into the second half of the 19th century. When the amount of available gold began to exceed the supply of silver, or vice versa, the prices that determined coinage rates of each metal at the mint would vary from the prices at which people could buy or sell the precious metals in the marketplace. Under those circumstances, one or the other of the metals was likely to disappear from circulation or to be exported to countries where the opposite state of affairs happened to exist.

Nevertheless, in spite of these obstacles, and without any theoretical or even much historical background to guide them, monarchs and their citizens during the Middle Ages, and even at certain moments before AD 1000, succeeded in evolving – indeed, inventing – monetary systems that have developed over time into the financial world around us today. None of these systems worked without disruption for very long, but reversion to the moneyless and tradeless society of the early days after the fall of Rome was never an issue.

During the first 1000 years after the fall of Rome, gold's role in Europe was much less important than it had been in Byzantium or in the domains of the Muslims. That difference was not a matter of choice for the Europeans. They simply had less gold available to them. Europe had no mines that could match the copious natural supplies that were available to the Byzantines and the Muslims. The Europeans also had an insatiable desire for the spices, silks, and – in the absence of central heating – furs and rugs that the peoples to the east were only too happy to sell them. As an unfortunate result, human slaves became one of Europe's primary exports, especially to the Muslims.

The Europeans left themselves with no choice but to reject the Byzantine fashion of covering everything in sight with gold. On many occasions, gold was in such short supply that religious ornamental objects such as crucifixes and chalices were sacrificed to the melting pots at the mints to be transformed into coins. David Hackett Fischer, in *The Great Wave*, cites a theologian, Fulbert of Chartres, who justified the practice of melting religious objects into coinage 'with the causuistry that it was better to sell sacred vessels to Christians than to pawn them in the hands of the Jews.'

You will not find the gold mosaics of the Byzantine churches on Romanesque and Gothic churches, which have austere interiors and only stone carvings on the exterior. Their color comes instead from stained glass and small work by goldsmiths on reliquaries, the chalices on the altar, and the cloaks and mitres of the higher-ranking priests. When, for example, the Benedictine Abbé Suger, the great architect and regent of France, began building the first Gothic cathedral at St Denis in 1137 as the resting place of France's patron saint, there was no way he could emulate Justinian's extravagance at St Sophia. Even the fragile work of the goldsmiths scandalized St Bernard. Suger was not about to



The creation of an illuminated manuscript using gold suspended in egg white or gum

back down: 'If the ancient law ordained that cups of gold should be used for libations and to receive the blood of rams,' he retorted, 'how much rather should we devote gold . . . to vessels designed to hold the blood of our Lord?' How, one wonders, would St Bernard have reacted if it had been he rather than Moses who descended from Mount Sinai to find his people worshipping the golden calf?

The Europeans did follow the Byzantines in the delicate use of gold known as chrysography, in which a small amount of powdered gold was suspended either in egg white or gum. In this form, the gold was then applied in the illustration of books as calligraphy, which the Europeans developed into art of exceptional beauty. The technique itself had first come into use as far back as the 2nd century AD via Egypt and Greece to satisfy the Roman demand for luxurious articles, but it was Charlemagne who launched the European art that has come down to us as the illuminated manuscript.

Charlemagne insisted on the highest standard for books produced during his reign and gave primary responsibility for that task to an English cleric, Alcuin of York. The most famous of the books produced under Alcuin's supervision were the Godescalc Gospels, which were written in 783 for Charlemagne, and the St-Méthard Gospel Books, both of which now reside at the Bibliothèque Nationale in Paris. The St-Méthard books were written entirely in gold calligraphy, illuminated with full miniatures in gold and silver on purple ground. The lettering was designed with great care, much of it adapted from Roman writing at the time of Virgil, with the letters formed deliberately and always taking the identical form. The cursive writing that we learn in school today is a direct descendant of Alcuin's golden script of 1200 years ago. We write faster in modern times, however: just one initial letter in chrysography took more than a full day to execute, which made carrying out these tasks a full-time job for the monk artists assigned to them.

The most significant development in the story of post-Roman money in Europe took place in Britain, which was at that time divided into a number of small kingdoms. Credit for this innovation goes to Offa (757–96), king of Mercia, a powerful ruler, and a contemporary of

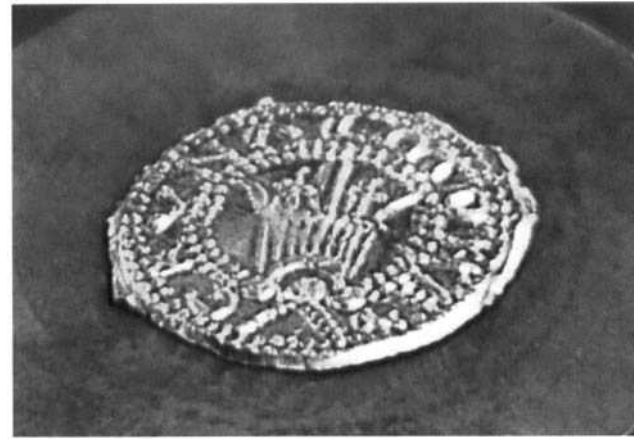
Charlemagne. Offa's domain extended through central England as far north as the Ouse and Trent rivers around today's York and Nottingham and as far south as Kent, Essex and Sussex. This territory was large enough, and sufficiently integrated into one large community by Offa, for some modest amount of trade to start. In addition, Offa had large armies to maintain, and armies in those days were made up of mercenaries who would not fight unless they had received in advance the money payments due them.

When Offa took over Kent, he found three outstanding designers and producers of silver coins, known as moneyers, who had the delightful names of Eoba, Babba and Udd. They sound like part of a Victorian poem for children, but the combination would also have made a fine name for an 8th-century London law firm. Eoba, Babba and Udd were master moneyers in silver and they inaugurated the longstanding leadership of Englishmen in that role. England has minimal local sources of gold, but Cornwall was rich in silver deposits whose output was now fashioned into a growing quantity of coins. The English did issue gold coins for about 70 years around ad 700, but they soon began to add silver alloy and then converted all their high-denomination coins into silver, with copper or brass for subsidiary coins.

The purity of the silver pennies produced by Eoba, Babba and Udd was so well maintained that the coins were soon circulating throughout Europe, even out to the Volga and the Don. Smaller denominations were created when people cut the pennies into halves or quarters. Later on, *shillings* would come into being; the word means 'a piece cut off'.

An abundant flow of Offa's pennies was soon coming out of the mints. Offa was so busy issuing pennies that he had to add 18 additional moneyers to his original three-man staff. The production of Offa's fine silver pennies ran into the millions – a powerful commentary on how rapidly the demand for money would increase as countries groped their way out of the Dark Ages. An even greater demand for coins lay just ahead, as the English had to arm themselves against Viking invasions and, from time to time, had to offer huge sums to these Scandinavian invaders in an effort to buy them off. By the year 1000, England's coinage was the most advanced in Europe, produced by a network of more than 70 local mints spread around the country.

In the year 800, not long after Offa had started minting his pennies, Charlemagne, king of the Franks and victor over the Lombards, traveled to Rome so that the pope could crown him emperor of the Holy Roman Empire as a reward. Charlemagne took the Byzantine emperors as his model by focusing on gold rather than silver. It was probably Empress Irene of Byzantium (whom he had contemplated marrying) who stimulated his interest in coinage, although he was also a friend and admirer of Offa. Charlemagne must have had much more gold available to him than other European rulers who either preceded or followed him for a long time afterward. He reopened the old gold mines in Saxony and Silesia and attracted goldsmiths from Byzantium to his capital at Aix. He worked on a golden desk with a map of the universe etched on its top. He had many villas,



ABOVE: *What one of
Offa's coins would have
looked like newly minted*

LEFT: *Charlemagne*

each of which had its local goldsmith. When he died, he was embalmed and buried sitting on a great gold and ivory throne that he had imported from Constantinople, together with a gold scepter, shield and sword. Plundering gold from beaten enemies, as always, was important as a source of gold for these luxuries. For example, when Charlemagne defeated the Avars in 796, an Asian tribe who had founded the first Mongol empire in AD 407, he needed 15 wagons, each pulled by four oxen, to carry the captured booty of gold and jewels.

All that splendor would have been incomplete without a gold coinage. Charlemagne set his pound equal to 20 shillings and 240 pence and a pound weight of 12 ounces – like the Romans before him and like the system the English were later to follow. Charlemagne's coinage was to enjoy only a brief life span, however, despite the longevity of his system of denominations and weights. His progeny spent as much time fighting

Charles V, when, in 1529, Francis I of France paid over 1.2 million escudos to Charles V of Spain to ransom his two sons, the process of counting and testing the coins took four months



among themselves as they spent in defending their domains, and his kingdom broke apart. Yet the difficulty was due only in part to his failure to establish a line of succession that would sustain the integrity of his domains. The evidence suggests that the process of recycling old gold for monetary purposes had reached a limit that could not be breached without the introduction of new gold supplies from some source beyond Europe.

Offa's silver coinage fared better than Charlemagne's gold coins, even though Offa's English kingdom also fragmented after his death. Offa's coins situated the penny as the core of the English monetary system: until about the end of the 13th century, 500 years after Offa's innovation, pennies were the primary means of payment. Offa's penny was so well established when the Normans arrived on the scene in 1066 that William the Conqueror rejected a policy of debasement for the English money.

When Richard I – Richard Lionheart – was captured by Leopold, duke of Austria, in 1192 on his way back from the Crusades, and subsequently 'sold' to the Holy Roman emperor, the ransom of 150,000 marks (the equivalent of £100,000) levied on the English people was transferred from England to the continent in the form of silver pennies. This was a pile of small change that would warrant a place in the *Guinness Book of Records*: £100,000 added up to 24 million pennies, enough money to employ over 40,000 skilled carpenters for a year. It is remarkable that the English were willing to make such a

heavy sacrifice at such an early stage of national awareness and for a king who spent so little time in England during his reign.

It is also difficult to imagine the sheer mechanics of transferring 24 million pennies. When, in 1529, Francis I of France paid over 1.2 million escudos to Charles V of Spain to ransom his two sons, the process of counting and testing the coins took four months, during which time the Spaniards rejected 40,000 coins as below standard. In a later time, 1662, 100 chests were required to handle the physical transfer of 500,000 large-denomination French coins.

Gold coins were so valuable in the Middle Ages that they did not circulate much among the common people. For the most part, gold coins were used in transactions by merchants and traders involved in foreign trade, by tax collectors, by the retinue of the monarch himself, and, as we have already seen, by monarchs as a means of buying off enemies and ransoming friends and family members. All these people exercised great caution to avoid accepting gold coins whose weight or composition was below required levels of purity, thereby rendering a public service for everyone else.

A favorite method of testing quality was the touchstone, which functioned in these times in precisely the same way as it had functioned more than 1500 years earlier under the Lydian kings in Asia Minor – a stone rubbed by gold objects and then compared against a set of needles containing varying proportions of gold and silver, gold and copper, and all three metals. Many merchants kept touchstones for this kind of rough-and-ready test. In cases of controversy, coins were taken to goldsmiths, who were skilled in the use of touchstones and their companion touch-needles; for over 700 years, the Goldsmiths Company of London has been the official arbiter of the purity of the British coinage.

The most important and reliable tests of purity were held at the Trials of the Pyx, in which a public jury of '12 discreet and lawful citizens of London with 12 skillful goldsmiths' presided over public examinations of coins freshly or recently issued by the Royal Mint. This ceremonial procedure probably began under the reign of Edward I in 1282; on the occasion of its 700th anniversary in 1982, the trial was attended by Queen Elizabeth II and the chancellor of the Exchequer.

Pyx derives from the Greek word for 'box' and refers to the container in which the officials stored the coins selected for testing at the trial. These coins were chosen on a random basis from the output of the mint – a practice still in use today in factories that check their output for uniform quality – after which they were meticulously compared with a special trial plate of the king's gold that was stored in a treasury room called the Chapel of the Pyx in Westminster Abbey. Some coins were melted down in an additional test of the purity of the gold.

These uniquely English Trials of the Pyx were serious affairs. The trials served a real purpose, because the goldsmiths and their jurymen had no conflicts of interest,





Present-day Trials of the Pyx in London. The 700-year-old ceremony of checking the weight of all the coins minted

performing their task in the open before the public, not in secret, for no reason except to assure the integrity of the currency. Today, we would refer to this as transparency. The whole process discouraged the monarch from debasement and encouraged people all over Europe to accept and execute transactions with English coinage.

The Trials of the Pyx are not the only evidence of the English intolerance for low-quality coins. Punishments were swift and unpleasant for moneyers and other employees of the mints who crafted poor coins or who were suspected of activities in the mints for their own benefit.

In 1124, with public confidence in English money having been destroyed by deterioration of the coinage, Henry I called before him all the mint-masters in the kingdom, about 200 men, and punished almost half of them by chopping off their right hands. This was a punishment that fitted a crime for which higher authorities had the ultimate responsibility. As Glyn Davies, a distinguished British economist and historian, observes, 'At least they were spared the stiffer penalties of being blinded or castrated or both, which were occasionally administered.' Similar types of punishment were by no means unusual until well into the 17th century.

This English tradition of 'sound' currency stood in sharp contrast with the irregularity of continental currencies. The resistance to going off gold that haunted the British from 1925 to 1931 had deep roots. In 1344, for example, when the weight of the English penny had been almost unchanged for 200 years, Edward III attempted to finance the great war against France with a small reduction in weight and followed with a deeper cut in 1351. The Statute of Purveyors enacted by Parliament in 1352 'expressed the hope that the king would no more tamper with the coinage than with the standards of weights and measures'.

Contrary to popular myths about the eagerness of the state to debase the currency whenever possible, the monarch had a strong vested interest in maintaining the purity of the coinage when coins were almost the sole medium for effecting transactions and for paying taxes and debts. Good coins with royal authentication stamped on them tended to exchange at a significant premium over their intrinsic value as metal, because they were so much more convenient than any other means of payment. This difference provided a source of profit, known as seignorage, to the kings who maintained a monopoly on the process of minting coins; anyone else engaging in that activity would live to regret it. The monarch's eagerness to earn seignorage in the Middle Ages explains why wholesale recalls of outstanding coins occurred in many countries at three- to five-year intervals, to be replaced with new coins of different design. The process was often a welcome one, as old coins constantly deteriorated from handling or were clipped by those who hoped to sell the odd pieces in the black market for money.



The Great

THE COMING OF THE YEAR 1000 WAS CAUSE FOR GREAT CELEBRATIONS THROUGHOUT THE CHRISTIAN WORLD, AND NOT JUST BECAUSE OF THE ARRIVAL OF THE MILLENNIUM. FOR SOME 500 YEARS, BARBARIANS BEYOND THE EASTERN AND THE NORTHERN BORDERS OF THE OLD ROMAN EMPIRE HAD REPEATEDLY CARRIED OUT VIOLENT RAIDS ON A HELPLESS LOCAL POPULATION.

Chain Reaction

In time, however, raiding lost its novelty and the former barbarians transformed themselves from outside predators into part of the scene. Like the industrious Lombards of northern Italy, they settled down, married and raised families. Warfare of one kind or another within Europe continued through the ages, but at least the terrifying toll of the barbarian invasions ultimately came to an end. The Normans who invaded England in 1066 were far more civilized than their rough ancestors, the Norsemen, who had earlier descended into France from Scandinavia.

These developments set the scene for major advances in the uses of gold, especially in the promotion of trade, commerce, and finance. In the process, gold would become the preeminent tool in the management of economic power. Gold's strategic role became so dominant over time that the struggle to obtain adequate sources of it would motivate monarchs and nations to great deeds and tragic treachery in the years ahead.

The proliferation of great cathedrals built throughout Europe during the first 300 years after the celebration of 1000 is dramatic evidence of how life was brightening up in Europe. Between 1100 and 1200 alone, France built more than 80 cathedrals, including Notre Dame and Chartres, to say nothing of 500 abbeys and 10,000 parish churches. Durham, Canterbury and Ely rose in England; Spain built Burgos, Toledo and Santiago de Compostela; in Italy and Sicily, cathedrals were completed in Venice, Florence, Siena and Palermo. The universities of Paris, Oxford, Bologna and Salerno were founded. Some of literature's most famous works appeared, including *El Cid*, the *Nibelungenlied*, the *Chansons de Geste* and the legend of King Arthur. And great kings governed – men such as Henry II in England, Frederick Barbarossa in the Holy Roman Empire, and Philip Augustus in France. The most popular role models were knights skilled in the arts of chivalry, and St Louis, who was known for his passionate religious and moral leadership.

The increase in population was the most important development. It was not just that fewer people were being killed. In a more peaceful environment, larger numbers of babies were born and survived. In Paris, for example, the population expanded from a little town clustered on the Île de la Cité in 1100 into a full-fledged city of some 50,000 people by 1215. Higher rates of population growth do not have to mean falling standards of living. In the 12th century, larger populations made greater specialization possible, permitted more people to spend time in study, the arts, and research, and stimulated all the networking benefits that cities with a diversified population create.

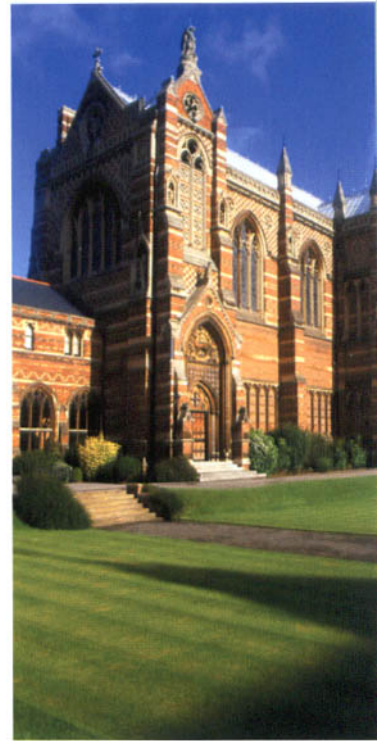
The optimism and vitality of this period culminated in the Crusades, the great adventure of the Middle Ages. From 1095 to around 1450, waves of Europeans – often women as well as men, and, in one instance, children – walked or sailed to Constantinople and to Asia Minor, professing to regain the Holy Land for Christendom. These expeditions

The Duomo in Florence



were, on occasion, a powerful positive expression of faith, but more often they were an escape from the tedium of small-town or rural existence, inspired by dreams of glory and even more vivid dreams of riches and treasures to be captured and carried back home.

During the 1300s, massacring infidels became a much less compelling objective as trade, commerce, and the interchange of intellectual ideas with the Arabs flourished, along with a flood of medical, scientific, mathematical and philosophical innovations – including the windmill and the compass – provided by the Arabs. In addition, Arab shipping and transportation routes opened up access to the silks and damasks, spices and lemons, and finely woven tapestries of the countries to the east. Princes and churchmen whose predecessors had been content with bare walls and floors covered with filthy rushes now insisted on having palaces with gilded vaults, with such furnishings as curtains, cushions, embroideries and floors covered with oriental carpets. Not everything learned from the Arabs had been developed by the Arabs themselves, but the Arabs had



Keble College, Oxford



accumulated and put to good use a substantial pool of knowledge from the Indians and the Chinese. It was this set of influences that provoked Marco Polo to take off in 1271 on his famous quest in that direction.

The Crusades imposed massive financial requirements on an unsophisticated financial structure. Quite aside from the costs of supplies and equipment, soldiers had to be fed, clothed, housed, and paid in coin acceptable in the occupied territories, where gold was the basis of all the currencies and where the armies consisted less of men motivated by religious zeal and knightly chivalry and more of adventurers and mercenaries. In addition, ransoms payable in gold were often demanded for captured prisoners. Ships that traveled full of soldiers and supplies in an eastward direction were willing to take on freight at very low rates for the trip back to Europe, rather than traveling empty, and this pattern encouraged large-scale importations of the attractive merchandise of Arabia, which in turn required payments, most often in gold.

Much of the gold used by the crusaders came from the Holy Land itself, which relieved the need to import gold from Europe. Professor Andrew Watson, in a 1967 paper for the Economic History Society, lists a wide variety of local sources, such as 'subsidies paid by the emperor of Constantinople to the Franks; tribute exacted from Arab potentates who bought off the Christians . . . booty, such as the 20 golden lamps weighing 20,000 mithqals, removed by Tancred from the temple of Jerusalem . . . taxes raised in conquered areas where the basis of the currency had long been gold.' Watson asserts that these sums were 'truly enormous, though they were often quickly spent'. In 1191, for example, the Templars bought the island of Cyprus for 100,000 golden bezants and then sold it at the same price to Guy de Lusignan. Raymond of Tripoli was ransomed at a cost of 150,000 bezants and the entire army of St Louis was redeemed from captivity for the sum of 800,000 dinars.

The Christian governments in the Levant were striking gold coins as early as 1124, using captured dies so that the money looked just like the local coinage, including the usual Arabic inscriptions praising Mohammed. The Christians continued to produce such coins for another 125 years, although in time a growing proportion of the coins was counterfeit, usually base metal such as copper plated with gold.

In 1250, Pope Innocent IV, scandalized less by the counterfeiting than by Christian mints issuing coins that honored the enemy, finally took action by excommunicating all those involved. Such drastic action was essential, because the Christian princes, more businesslike than spiritually inclined, had insisted on continuing to issue coins that had ready acceptance by the Muslims. In response to the pope's demands, the princes took the modest step of stamping Christian sayings while retaining the Arabic script.

Innocent appears to have been an appropriate name for this pope. One can only wonder why the Vatican was so slow on the uptake – or whether it had chosen to look the other way. Indeed, the identical sequence of events had occurred in Spain during the 11th and 12th centuries, and, in both gold and silver, throughout Europe.





Notre Dame, Paris

Innocent's effort was insufficient to satisfy St Louis, who was in the Holy Land on a crusade and added his authority to the pope's. An entire new coin appeared, the *Agnus Dei* (Lamb of God), which reflected both Louis's religious humility and his French pride: *Christus vincit, Christus regnat, Christus imperat* (Christ conquers, Christ reigns, Christ commands), the ritual acclamation of the French kings.

We now move westward across the Mediterranean to the Kingdom of Sicily, which was ruled from 1211 to 1250 by the Holy Roman emperor Frederick II, grandson of the great Frederick Barbarossa and one of the towering figures of the Middle Ages. Frederick was an enthusiastic producer of gold coins that he employed to project his economic power.

Frederick was born in Sicily in 1194, the same year his father had been crowned king of Sicily. Although a hypochondriac, Frederick was a valiant crusader, established an expert bureaucracy, opened Sicily to free trade, surrounded himself with the greatest intellectuals of the age, levied heavy taxes on the clergy and prohibited them from holding civil office, built handsome castles throughout Sicily and founded a university at Palermo to train public servants – the first European university with a royal charter. Not incidentally, he also left his mark on the gold currency of his time.

Frederick did an extraordinary amount of traveling and was almost constantly at war with the papacy. He launched the Sixth Crusade in 1227, returned to Sicily to do battle with Pope Gregory IX – who excommunicated him – and then went once more to the Holy Land, where he reclaimed Jerusalem for the Christian forces. Jerusalem fell to Muslim mercenaries later on and, despite another 200 years of crusading, was never again in Christian hands until the British General Allenby captured it in 1917. After declaring himself king of Jerusalem, Frederick traveled back to Sicily to defend his lands against another papal attack and survived a second excommunication (as a heretic, rake and anti-Christ).

When, to Frederick's relief, Gregory gave up the ghost in 1241, the new pope was Sinobaldo Fiesco, who adopted the title of Innocent IV. This was the Innocent (innocent?) who had so belatedly prohibited the Christians in the Holy Land from issuing coins with Arab inscriptions praising Mohammed. Frederick had expected this pope to be a friend, but that may have been innocence on his part, because Fiesco came from Genoa, Sicily's mortal enemy and determined competitor for economic dominance. Frederick and Innocent were soon involved in a vicious war with each other that included an assassination attempt on Frederick and the capture of his son, who spent the last 23 years of his life in prison. Frederick suffered a final defeat at Parma in 1248 and died suddenly two years later. Frederick's son-in-law was reduced to pawning the Sicilian throne to some enterprising Genoese businessmen in return for gold.

At the time of Frederick's death, Sicily was operating with two concurrent gold standards. One was the *tari*, which had Arabic roots and had been in use since the 9th century. Although the *tari* had gone through some debasement over the years, from the early 12th century onward it was stabilized at 16 ½ carats of gold. This was better than the purity of the Byzantine solidus at that time and established it as one of the most stable in Europe. The coins were stamped out in a variety of sizes and tended to circulate on the basis of their weight rather than their face value. The *tari* enjoyed such wide circulation that it became a kind of unit of account by which many items were priced.

Frederick II considered the *tari* unimpressive and too irregular for the homeland of a Holy Roman emperor of his exalted status. After a military victory against the Tunisians in 1231, he was assured of a substantial annual tribute in both gold coin and gold dust from the West African gold sources. Now Frederick's imperial mints began to strike a new gold coin called the *augustalis*. Robert S Lopez describes the *augustalis*, with its classical eagle imprinted on one side and the emperor's laureate head on the other, as 'a startling advertising medium' and a dramatic contrast to the formless *tari*. The *augustalis* was minted in 20 ½ carats and weighed 5.28 grams, which gave it greater value than the Arab dinar. This impressive coin soon eclipsed the *tari* and was in strong demand throughout western Europe and the Near East.

Genoa had long considered Frederick II and the Sicilians their arch-enemies. The Genoese nursed dreams of attaching Sicily to their own domains and had been intermittently at war with Frederick ever since 1238. Genoa derived immediate and significant benefits from the election of Pope Innocent IV in 1241 and Frederick's defeat in 1248 and his death in 1250. In particular, Innocent showered privileges on his hometown and proceeded to claim the Kingdom of Sicily for the Holy See.

Genoa compensated for lack of military power with aggressive economic policies. By 1250, the Genoese were enjoying a prosperous textile industry and widespread construction of new buildings. Genoa's huge shipyards produced most of the 1800 ships

Venice



that sailed for St Louis's crusade in 1248 under the command of Genoese admirals. Commercial enterprises of all kinds were making their appearance, and bankers and merchants from the major city-states of northern Italy were there to do business. Lopez states, 'The very technique of credit operations, which had constantly progressed during the last 100 years, displayed at this period a maturity not to be surpassed for many years to come.' The Genoese had lent large sums to both St Louis and Innocent IV and had been bankers to just about every important crusader. And then there was the opportunity opening up in the Holy Land for new gold coins as the Christian princes finally yielded to Innocent's insistence that they reform their coinage.

Meanwhile, increasing supplies of gold were becoming available to the Genoese. New gold mines opened up in Bohemia, but the primary source was the African gold that flowed mainly toward Genoa as a result of the favorable balance of trade that the Italians maintained with North Africa. Genoese records suggest that trade with the Levant was also turning favorable, bringing Islamic and Byzantine coins to the Italian shores for remelting into Genovese coins. Indeed, even China was complaining at that time of a loss of gold through foreign trade. Finally, a period of sustained prosperity probably led to the dehoarding of gold as well. We know that the Italians were supplying the English Treasury with gold in the middle of the 13th century.

As Englishmen in their time knew well and as Americans have learned in the years since World War II, good money adds its luster to the world's image of a nation. A high-value gold coin, sustained in purity, was an ideal vehicle for Genoa to extend the reach of its economic prestige. In 1252, two years after the death of Frederick in Sicily, when the price of gold in Europe happened to be unusually low compared with the price of silver, Genoa began to issue a 24-carat gold coin called the *genovino* or *genoin*. Oddly enough, their sour relationship led the Genoese to take the Sicilian coinage system as their model. They adopted the weight systems of the *tari* and then stepped up the quality of the coin from Frederick's by minting it in 24-carat gold. Both features would enhance the acceptability of the Genoese money in the Sicily that the Genoans so avidly coveted.

These coins weighed about 3.5 grams, a full gram less than Constantine's original bezants, but the 24-carat purity was a big attraction. A 3.5-gram solid gold coin would be the equivalent of about \$33 of 1999 purchasing power, but the purchasing power of gold in terms of goods and services was many times greater in the Middle Ages than it is today. The high contemporary value of these coins had more than economic significance: their high value reflected prestige and glory upon their issuers. This aura was further enhanced because such coins were not meant for the use of the *hoi polloi*; they circulated among the upper classes and most active merchants.

The chief motive of the Genoese in issuing the *genoin* was commercial, but they also understood that economic power and political power mutually enhance each other. Indeed, within ten years of the introduction of their golden currency, Genoese power had persuaded the Latin rulers of Constantinople that the proprietary trading privileges held

by the Venetians should be transferred to Genoa. The Genoese then used their base in Constantinople to extend their trade and influence into northern Persia, the Crimea and the farthest shores of the Black and Caspian seas. Soon they were venturing into the upper Nile and exploring the Sudan and the Niger River basin.

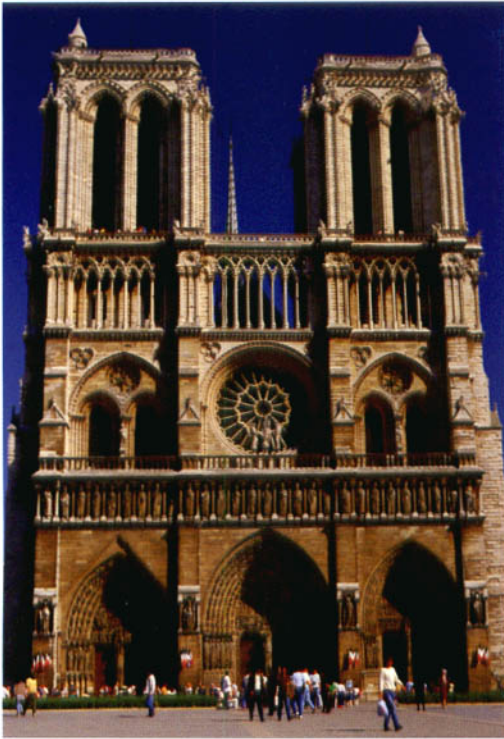
The needs of commerce explain why the first steps toward minting gold coins in the Middle Ages took place in such cities as Genoa and Florence that were centers of economic and financial activity instead of in the capitals of the nation-states, such as London or Paris or even Rome. Lopez asserts that the events of 1252

touched off one of the greatest chain reactions in monetary history. . . . The return to gold did more than provide symbols and tokens: it relieved the strain which economic growth was placing on a chronically inadequate currency. . . . It was the most spectacular token of the economic gains accumulated by the Catholic world during the preceding two or three centuries, and a tangible symbol of the initial superiority of the West over the East — for the Islamic world and Byzantium, which minted gold when Europe was content with silver, now debased their gold or ceased to strike it.

Lopez was accurate in using the metaphor of a chain reaction to describe what happened with the use of gold as money following the innovation of the *genoin*. Only a few months after Genoa had acted, the Florentines issued their *fiorino d'oro*, or florin, so called because it had a fleur-de-lys on one side. Perugia and Milan followed with gold coinages shortly afterward, and Lucca around 1273. In 1284, the Venetian ducat appeared, the most famous and successful of all the gold coins born in the course of the 13th century. When, in *The Merchant of Venice*, Shylock hears that his runaway daughter Jessica has 'in one night spent fourscore ducats', he cries, 'Thou stick'st a dagger in me! I shall never see my gold again! Fourscore ducats at a sitting! Fourscore ducats!' The ducat served as a standard of value throughout Europe and maintained its gold content until the Venetian Republic fell to Napoleon in 1797. All these coins weighed 3.5 grams and were 24-carat gold.

The issuance of new gold coins was by no means limited to Italy. Both Alfonso X in Castile and Henry III in England issued gold coins in 1257, and St Louis joined in at just about the same moment. Sad to relate, Henry's gold penny was set at an inappropriate value relative to silver and ended up a total failure. Henry began with his coin equal to 20 pence in silver, later raised to 24 pence, but the same quantity of goods that the gold could buy could be purchased more cheaply just by paying with silver. Within three months, complaints rose within the City that no one wanted to exchange the new coins for silver. Merchants had no use for them, and poor people would never spend that much in a single transaction, or perhaps in a year of transactions. No trace of these coins has been found since about 1280.

Nevertheless, the English merchants complained that they had to use foreign gold coins in their international transactions at values in terms of sterling that were



LEFT:
Notre Dame, Paris

RIGHT:
Ponte Vecchio, Florence

unfavorable to them. In 1343, Edward III tried issuing an English gold florin whose face value was above the market value for the gold it contained, but this, too, met with resistance everywhere, including among many of England's best customers abroad. When the Florentines refused to accept these coins, they used the quaint excuse that the coins lacked an image of St John the Baptist. These difficulties led Edward to issue a new gold coin the following year with a more appropriate weight, called the *noble*, whose sides celebrated Edward's great victories in the Hundred Years' War against the French: Crécy on land and Sluys at sea. The noble also had a checkered career in its early years, but in time it became the basic gold coin of England until well into the 17th century.

This laggard performance by the English in large part reflected their slow pace of economic and financial development compared to developments on the Continent. In dramatic contrast to its role in the world in the 19th and 20th centuries, 13th-century London was far less cosmopolitan than Paris or Augsburg or the Italian city-states. With a population of about 50,000 and the only city in England except York with more than 10,000 inhabitants, London was about half the size of Paris, Florence, Venice and Genoa and no larger than Bruges, Bologna or Palermo. The English, still feudal and rural, had not yet been consumed by the hard-headed, business-oriented calculations that led to the *genoin*, the florin and the ducat, in large part because the typical English transaction was far smaller than transactions on the Continent. Even as late as the middle of the 15th century, aliens still controlled about 40 per cent of English overseas trade; Florentine bankers financed the wars of Henry's son Edward I and bought the wool crop from Edward III. According to one authority, the Italian merchants 'achieved a financial despotism which London has never had, for during a considerable period [these merchants] seem to have been able to fix quite arbitrarily their own exchange rates.' The arrival of the 'nation of shopkeepers' still lay in the future.



The Disintegrating

THE SUMMER OF 1314 WAS UNCOMMONLY COLD AND WET IN EUROPE. CROPS ROTTED, HARVESTS WERE LATE, AND ALARMED AUTHORITIES PLACED PRICE CONTROLS ON FARM PRODUCTS AND FIREWOOD. ALL THESE WERE ROUTINE DISASTERS THAT HAD HAPPENED MANY TIMES BEFORE.

EIGHT

Age and the Kings' Ransoms

The awful weather of 1314, however, was just the beginning of a succession of catastrophes. Bad crops seldom happen two years in a row, but the weather in 1315 was even worse than during the previous year. Heavy and incessant rains caused flooding that smashed dikes. Rising rivers destroyed villages. Violent storms crashed onto the coasts. The tragedy stretched from Scotland to Italy and from the Pyrenees to the homes of the Slavs. Food prices rose over fivefold and starvation was widespread. Even that was not the end. The weather wreaked havoc once again in 1316, causing the worst famine in European history. People ate cats, rats, insects, and animal droppings, and then, lacking anything better, dug up the corpses in the burial grounds. Epidemics and violent crime were widespread. Bloody and public self-flagellation was common. Scapegoats – Jews, lepers, noblemen – were murdered without hesitation.

The Great Famine, as it came to be known, was only the introductory chapter to this appalling story. In 1347, the Genoese were defending their Crimean colony of Kaffa (known as Feodosia in modern times) against a besieging Tartar army that had recently swept in from the Far East and across the great expanses of Russia. The siege was not going well for the Tartars, so they decided to use a unique projectile to catapult over the walls and into the center of Kaffa: dead bodies of their own men who had just died from a particularly vicious form of the plague. Stricken with terror – and soon sickened – the Genoese abandoned Kaffa, fleeing in their galleys through the Black Sea and the Aegean toward Italy. When one of the Genoese ships arrived at Palermo in Sicily, its fleas and rats and dying humans launched what has come to be known as the Black Death.

The summer of 1314 was uncommonly cold and wet in Europe



Woodcut showing the Black
Death

This frightful affliction spread like wildfire through much of the world over the next two years. Population estimates are fragmentary and unreliable, but it is likely that the Black Death killed about a third of the people from India to Iceland, at least 20 million deaths. The population of Europe would not regain the levels of 1300 until the middle of the 16th century.

If all this were not bad enough, in 1348 – just as the Black Death was gathering momentum – a calamitous earthquake spread havoc and destruction from Naples to Venice, with aftershocks that smashed buildings and killed humans as far away as Germany and Greece. And nature was by no means the only force to deliver violence and death in the 14th century. Brutal political disruptions became endemic.

In 1303, the pope was taken prisoner by a Roman mob and died shortly afterward under mysterious circumstances; one historian refers to his dying 'of humiliation'. His successor was murdered. The next in line, a Frenchman by the name of Clement V, took the prudent step of removing the papacy to the papal enclave in his native land at Avignon in 1305, where the popes lived in high style for the next 73 years. Petrarch complained that even the papal horses were 'dressed in gold, fed on gold, and soon to be shod in gold if God does not stop this slavish luxury'. What was the source of all that gold? It was from the bunching of bequests from the wealthy who had died from the Black Death. It was at this time that the king of France, Philip IV, prohibited any exportation of gold from France. One writer suggested that this ruling, rather than fear of physical violence, was

the real motivation for the pope to move to Avignon. He argued that the papacy would have faced bankruptcy if it had remained in Rome and that it transferred itself to France in order to sustain its ample revenues from French sources.

In 1327, Edward II of England, an avowed homosexual, was slaughtered with a hot iron poker shoved up his rectum. Louis X of France, known as Louis the Quarrelsome, was deposed in 1316 after only two years on the throne. In 1332, even the melancholy Danes dissolved into anarchy. In the Holy Roman Empire, Guelfs went to war against Ghibbelines. In 1338, the Hundred Years' War between England and France broke out, piling organized state-sponsored killings on top of widespread mayhem on a private level. Workers' uprisings in Italy broke out in 1346–7. Shortly afterward, Rome was thrown into chaos by Cola di Rienzi's brutal uprising motivated by the dream of restoring the Roman republic of Cicero. In 1358, the Jacquerie produced a violent peasant revolt in France against war taxes, the burden of paying huge ransoms for captured royalty, and the pillage perpetrated by wandering mercenary veterans of the wars. In 1379, a year after insurrection in Florence, the weavers and merchants of Ghent mounted their own insurgency, going so far as to try to divert the river Lys and leading the contemporary historian Froissart to ask, 'What shall they say that readeth this or heareth it read, but that it was the work of the Devil?' In 1381, following a succession of new poll taxes, the English got their version of a fierce uprising by the peasants, led by Wat Tyler. Seventeen years later, Henry Bolingbroke deposed his cousin, Richard II.

The economic consequences of all these deaths and disruptions were strange, to say the least, especially the massacre of the Black Death. As the masses of human bodies disappeared, their physical possessions and monetary wealth remained behind. This grisly process left most Europeans far richer than they had been before tragedy struck. To turn a phrase, as the poor got fewer, the rest got richer. They would soon act accordingly.

According to one historian, the proportion of citizens in Albi, in southern France, with fortunes greater than 100 livres grew from 11 to 20 per cent of the population between 1343 and 1357, while the percentage of those with less than ten livres declined from 31 to 18 per cent. Many people died without the opportunity to write wills, leaving their wealth with no readily identifiable ownership. This led to a great demand for lawyers to settle the quarrels over questions of inheritance and succession, but it also led to opportunities for the enterprising to pick up unclaimed assets. In addition, as the supply of labor had shrunk dramatically, a scarcity of labor joined with the plethora of money to produce a sharp rise in wages and the incomes of working people.

Contrary to what one would expect under the circumstances, with so many workers leaving the farms to enjoy the temptations of the city, the price of food was remarkably stable. The loss in population was so enormous that it had an even greater impact on the demand for food than the reduced supply of food caused by the decline in the number of

people engaged in agriculture. With necessities taking a smaller proportion of total spending, the consumption of meat, butter, fish, wine and exotic spices began to expand even among people at the lowest end of the scale.

In the uncertain and turbulent environment, the incentive to save was minimal while the incentive to spend was irresistible. As late as 1375, a Florentine chronicler was indignant 'at the spectacle of the *popolo minuto* who refused to practice their old trades, dressed themselves in a manner unbefitting their station and insisted on the finest delicacies at their table'. In Britain, a petition from the House of Commons in 1362 blames rising prices on 'laborers [who] use the apparel of craftsmen, and craftsmen the apparel of valets, and valets the apparel of squires, and squires the apparel of knights'. William Langland, in *Piers Plowman*, attacked the worker 'who refuses to bear the burden of poverty patiently, but blames God and murmurs against Reason and curses the king and his council for making statutes [legalized ceilings on wages] to plague the workmen.'

The clergy were no exception. In 1351, Pope Clement VI asked his prelates, 'What can you preach to the people? . . . If on poverty, you are so covetous that all the benefices in the world are not enough for you. If on chastity – but we will be silent on this, for God knoweth what each man does and how many of you satisfy your lusts.'

While the cost of ordinary domestically produced agricultural products was relatively stable, exotic foods came increasingly into fashion, with predictable consequences for prices. According to Harry A. Miskimin, the price index of foreign goods such as herring, pepper, oil, sugar, almonds and saffron rose from 100 in the period 1261–1350 to 162 during the period 1351–1400. He also calculates that per capita expenditures on wine approximately doubled at the same time.

The swollen appetite for imported luxuries such as fancy foods and the increasingly popular clothing frills combined with a heavy burden of military expenditures to send the demand for both gold and silver surging. The supply of precious metals, however, failed to respond to the expanding demand. Metal shortages left the mints inactive for extended periods of time. From 1373 to 1411, the production of gold coins in England averaged only £9500 a year, about a tenth of the output prior to the Black Death. Mining sources also dried up as even record high pay was insufficient to attract men to the discomforts of working the gold mines. Ordinances against exporting 'good money' or precious metals accomplished no more than the regulations that required importers to use the revenues they earned to purchase domestically produced goods for export. The repetitive sequence of such orders suggests that they were difficult to enforce and frequently ignored.

Controls over both wages and jobs did not fare any better. For example, Edward III's Statute of Labourers, enacted in 1351, set maximum rates of pay at pre-plague levels, required all able-bodied men to work, and limited the mobility of workers between jobs and even their freedom of movement between villages. Repeated attempts to enforce these restrictions would ultimately lead to Wat Tyler's hotheaded rebellion in 1381.

One of the more curious – and equally fruitless – efforts to economize on gold included a multiplication of regulations with the odd name of sumptuary laws. The purpose of these laws was to economize on scarce gold by prohibiting people from using it lavishly as personal adornment – a dubious objective in the wake of the Black Death. This was a time, as Barbara Tuchman described it, of ‘frenetic gaiety, wild expenditure, luxury [and] debauchery’.

A statute of Edward III enacted in 1363 was typical of the 14th-century sumptuary laws. Edward set upper limits for the permissible extravagance of each class. Rustics were limited to blanket cloth and the coarse reddish-brown homespun called russet; grooms and servants were not allowed to wear gold in any form; gentlemen below the rank of knight were prohibited from dressing with any cloth of gold; knights were forbidden to wear gold rings. In 1380, the king of Castile went further by prohibiting all Spaniards except queens and princesses from wearing cloth of gold or gold jewelry.

Like the prohibitions against the export of gold, and doubtless for the same reason, the sumptuary laws were enacted over and over again. Gold, like liquor, satisfies too many needs to survive prohibition.

The Byzantine emperors used gold to persuade others to fight and kill on their behalf. The almost constant warfare of the 14th century put gold to the opposite use: for the payment of ransoms that would save lives. Most of the ransoms in the 14th century called for the movement of gold within Europe, but the risks of military defeat in all countries meant that monarchs had no choice but to hoard massive reserves of gold as insurance against the evil day when prisoners would have to be redeemed. In the ugly environment of the 14th century, ransoms were especially onerous.

Should we deplore the heavy price of redeeming prisoners? The higher the price that the victors could expect, the greater the incentive to constrain the bloody slaughter on the battlefields. The ransom business – and in many ways it was a business – must have saved many lives, especially among the upper classes of society.

The most spectacular example of capture and ransom involved the king of France himself, Jean II, who was known as Jean le Bon. Jean loved luxury to the extreme of having the court painter decorate his toilets. In a remarkable step for his time, he commissioned French translations of the Bible so that he could read it more easily. He spent so much money on himself and on trying to fight the English that he soon became an expert at debasing the currency: 18 alterations in the first year of his reign and 70 more over the next ten years.

Jean le Bon's son, the dauphin (who was also duke of Normandy), was shifty in his loyalty to his father. In April 1356, he hosted a dinner party in his castle in Rouen for his cousin and neighbor Charles le Mauvais, king of Navarre, hoping to organize a conspiracy to capture the throne of France. Charles le Mauvais was such a bad man that almost

anybody compared to him, like Jean, would have been characterized as 'le Bon'. Jean, who had advance notice of the meeting between Charles and the dauphin, burst in on the gathering in full armored regalia. He thereupon had some of Charles's entourage butchered, threw Charles into prison, and confiscated Charles's Norman estates.

Charles's brother and surviving associates appealed for English help to recover their estates. The English responded without delay and, under the command of the duke of Lancaster, were soon on their way into France from Cherbourg. In July, the prince of Wales, known as the Black Prince and one of the greatest fighters and commanders of his age (the 'Black' referred to his armor), landed at Bordeaux with 8000 troops and launched a series of devastating raids as he traveled northward through western France. Jean decided that he had no choice but to face his enemies in pitched battle. Confidently leading his army of 16,000 men, the largest army of the century, Jean marched toward the Loire to block the Black Prince's northward approach.

On 19 September 1356, the French army was overpowered by the Black Prince's forces at the Battle of Poitiers, despite superior strategic positions in the field and twice the number of soldiers. Seven hours into the battle, the English discovered Jean's unit and charged at high speed against it, 'like the wild boar of Cornwall'. Jean fought valiantly with one of his loyal sons beside him, but he lost his helmet and began to bleed from two wounds on his face. When voices cried, 'Yield, yield, or you are a dead man,' Jean handed over his glove to an enemy soldier and thus the king of France became a prisoner of war.

The king was by no means the only distinguished prisoner to be taken that day. The list included the highest-ranking French military commanders and over 2000 members of the nobility. The number of prisoners was greater than the English could handle. Most of the prisoners were instructed on their honor to come to Bordeaux with their ransoms by Christmas -- in the days of chivalry, such a request was a matter of routine. And still many of the English soldiers complained that their archers' aim had been too good, because the arrows that hit the French forces with such accuracy had deprived the victors of an even larger number of prisoners to hold for ransom.

The Black Prince took the French king to England seven months after the battle and installed him in high style at the Savoy Palace until the ransom was paid. But how big was the ransom to be? When the French rejected a preliminary settlement in 1358, the English responded by raising their demands. Meanwhile, the clock was ticking.

In March 1359, with just six months left before the truce negotiated at Poitiers was scheduled to expire, Jean signed the Treaty of London. His desperation is apparent in the conditions to which he agreed: in exchange for his release from captivity, he yielded all of western France from Calais to the Pyrenees plus a ransom of four million gold écus (gold crowns, the equivalent of more than £600,000), the ransom to be collateralized by 40 noble and royal hostages. If the French blocked the execution of this treaty in any fashion, Edward had the right to send his armies back to France -- at the expense of the French



*Chartres Cathedral at
Night*

king. Edward knew what he was doing by putting the financial burden on the enemy, for his wars in France were terribly costly. In one year alone, he borrowed 200,000 gold florins from his Italian bankers (on which he subsequently defaulted).

When the dauphin, serving as regent in his father's absence, received word of this total capitulation, he summoned the estates general to help him make the fearful choice between peace and renewal of the war. The response was immediate and unanimous: the treaty was intolerable, and war was to be declared on England.

The English promptly launched another protracted campaign in northern France, but this time the French resisted a pitched battle, resorting to a scorched-earth strategy instead. On 13 April, when the depleted and now ragged English army was camped near Chartres, an extraordinarily powerful hailstorm hit them, accompanied by cyclone-force winds and cloudbursts of freezing rain. According to Barbara Tuchman, 'In half an hour Edward's army took a beating that human hands could not have inflicted and that could hardly be taken as other than a celestial warning.'

It is the rare military commander who at one time or another has not heeded messages from supernatural sources. Edward III, tough though he was in many ways, decided at this point that discretion was the better part of valor. In any case, he retained plenty of bargaining power, because Jean was still his prisoner. He agreed to reopen negotiations, which were finally completed on 8 May 1360, at the nearby village of Brétigny. Jean's ransom was scaled back to three million gold crowns. The territorial concessions were also reduced, but they still amounted to about a third of France, a prize unmatched until Hitler invaded France 580 years later.

The treaty was explicit about the terms for the 40 hostages to be held as security against the payment of the king's ransom. The stipulations included the king's two

younger sons, his brother, the brother-in-law of the dauphin, and nine great counts. The English agreed to return Jean from London to Calais, upon payment of the first installment of 600,000 gold crowns on the ransom. At that point, ten of his fellow noble prisoners would also be liberated, but they were to be replaced by 40 wealthy members of the Third Estate – the bourgeoisie. The remainder of Jean's ransom was due in six semi-annual installments of 400,000 gold crowns, with one-fifth of the hostages to be released after each payment.

The ransom would have been a terrible burden on the French under any circumstances, but especially following the depredations of the Black Death and the havoc and destruction of war. The going was so difficult at one point that the French invited back the Jews whom they had ejected from France in 1306, offering them 20 years' residence subject to payment of 20 florins per head entrance fee and 7 florins annually thereafter. Jean himself contributed the handsome golden dowry he earned from marrying off his 11-year-old daughter to the rich tyrant of Milan, Galeazzo Visconti, a step that the chronicler Matteo Villani described as the king 'selling his own flesh at auction'.

The first installment of the ransom was made in October 1360. Edward then met with Jean in Calais, and the two monarchs swore together to keep the peace into perpetuity. After four years in captivity, the king of France was finally a free man. The moment was hardly one for rejoicing. Jean returned to a country that Petrarch, on hand as an ambassador from Visconti, described as 'a heap of ruins. . . . Everywhere was solitude, desolation, and misery.'

Nor is this the end of the story of Jean's ransom payments. Plague, which continued to reappear periodically, killed off some of the hostages in England. Other members of the group were attempting to use their own resources to buy their freedom. The ransom payments were soon in arrears. Ceded territories resisted the change in sovereignty. In 1363, convinced that his honor was in disrepute, Jean sailed back across the Channel the week after Christmas and restored himself to captivity in London, disregarding the urgent advice of his Council, his prelates, and his barons. He was received by the English with great ceremony and celebration, but he soon fell ill and died in April 1364. He was only 45 years old. A million gold crowns were still payable on his ransom.

In the end, less than half the ransom was paid, but even 1.5 million gold crowns was a colossal amount of money. It was equivalent to a full year's pay for approximately 6000 agricultural laborers or the cost of 300,000 sheep or 1.6 million gallons of ale, or to more than four times the total of all the poll taxes that would stir up a vicious rebellion almost 20 years later.

Those people who survived the endless agonies of the 14th century must have been convinced that the darkness of their age would never lift. Yet as this terrible century finally drew to an end and the new century dawned, conditions in Europe began to improve. Peace provided opportunity to put abandoned farmlands back into use, and the

lower prices for food that resulted led to an expanding population. After losing six million people from 1350 to 1400, the population of Europe grew by 15 million – about a third – during the next 50 years and gained another 9 million between 1450 and 1500. Improving food supplies also facilitated a return to urbanization, which in turn enabled commerce and industry to revive.

Progress was not uniform across Europe. Italy fared best among the major European countries, and there the greatest glory of the 15th century was in Venice, although Florence also became a great center of commerce, industry, finance – and art – during this period. Venice remained the most important station for the great volume of trade with the lands to the east, but the city was not just a group of charming islands in the Adriatic. By the end of the century, Venice controlled most of the cities within a radius of around 100 miles from St Mark's Square – including such centers as Verona, Vicenza, Ferrara and Bologna – as well as the Mediterranean islands of Corfu, Cyprus and Crete.

Venetian Masks





The medieval Venetians, as this mosaic attests, lavishly utilised gold to decorate their city

That kind of power gains gold. The territories remitted a million gold ducats a year to the Venetians, which paid for many of the palaces that are now so familiar on the Grand Canal, including the Ca d'Oro or House of Gold, whose exterior ornamentation was once generously covered with gold.

Probably the most revolutionary developments in the course of the 15th century took place in an area that had been a sideshow to the Middle Ages in Europe: Iberia. The marriage of Ferdinand of Aragon to Isabella of Castile in 1469 united Spain. Under their leadership, the Spaniards finally expelled the Moors and, while they were at it, the Jews. Ferdinand and Isabella also launched a commanding dynasty that would deploy Spanish

power across the entire European landscape and, in time, in the Americas as well. One daughter was married to the king of England, the other, Joanna the Mad, to the oldest son of the Holy Roman emperor. Poor Joanna earned her name because she carted her husband's cadaver around with her for many years after his death. The contribution of Ferdinand and Isabella and the Spanish to the history of the Americas needs no elaboration.

Meanwhile, the little country of Portugal was also stirring. The Portuguese had always been great sailors; as early as 1300, they had established a navy trained by Genoese and Venetians. John I, crowned in 1385, was an enlightened ruler who found ways to transform a third-rate nation of only one million people into a world power. John signed an alliance with the English that was to last into perpetuity and is still in effect; he then cemented the deal by marrying a granddaughter of Edward III, whose brother, Henry Bolingbroke, would soon usurp the English throne from Richard II.

European output of gold during the 15th century was smaller than ever relative to the needs of the times. According to one authoritative estimate, domestic gold production in Europe in 1400 was no more than four tons. In money, that works out to about one million ducats. It is estimated that the Venetians alone exported the equivalent of a ton of gold a year in the form of ducats, significantly reducing the available supply of gold. Economic historian Charles Kindleberger cites estimates that as much as 5 per cent a year of the coinage also disappeared due to ordinary wear, hoarding, shipwreck, and movement into plating for decorative purposes.

After some 3000 years of developing civilization, the total amount of gold in Europe in 1500 in all forms – coins, hoards, and every manner of adornment and decoration – could have been fashioned into a cube only two meters in each dimension. This modest supply meant that even small discoveries or transfers had a magnified effect on the gold market.

The economic historian John Day, in an essay titled 'The Great Bullion Famine of the 15th Century,' cites striking examples of the shortage of gold coins that developed during the fifteenth century as well as the fruitless efforts by governmental authorities to do something about it. 'In 1409, the Paris money-changers protested in chorus that they had no bullion for the mint at any price. The civil war years (1411–35) witnessed the rapid decline of the influential goldsmiths' guild of Paris for lack of metal, lack of clients, and because of new restrictions on the manufacture of gold and silver artifacts, which were intended "to prevent the destruction of the king's coinage".' An ordinance issued at the port of Bruges in 1401 required merchants to settle all foreign exchange transactions entirely in gold; the ordinance was repealed eight months later because so few paid attention to it. The mints of the Estates of Flanders were shut down from 1402 to 1410. The output of the Tower Mint in London, which averaged around £5000 in gold coins



Venetian ducat

in the 1460s, fell to £2000 from 1476 to 1485 and then virtually came to a halt over the next ten years. Silver coinage shows similar trends. Day estimates that total bullion reserves in Europe shrank by 50 per cent between about 1340 and 1460.

The dearth of both silver and gold provoked a reversion to barter in many communities, especially for local payments. Pepper, worth more than its weight in gold, was the most popular commodity enlisted for this purpose; German princes even called their bankers 'peppermen'. Although this kind of improvised money served a purpose, the import flow of such commodities as pepper was uneven, which made their prices uncomfortably volatile. A few bags of pepper unloaded in Amsterdam or London could quickly depress its price. Not so a few bags of gold or silver. As a result, paper currencies – essentially promissory notes issued by high-quality borrowers – began to circulate, but Day argues that 'The circulation remained overwhelmingly metallic. Even in mid-18th-century England, on the threshold of the industrial revolution, it was estimated that minted coin accounted for 90 per cent of all the money in circulation. . . . As late as 1861, Italian circulation consisted of 75 per cent coin.'

When money is in short supply, people try to economize on the amount they spend for goods and services. The usual result is a declining price level. That is precisely what happened during the 15th century. Reliable estimates indicate that prices for commodities throughout western Europe fell by anywhere from 20 to 50 per cent between 1400 and 1500. In Aragon, for example, the index of prices fell about 20 per cent. The price of English wheat fell by half between 1360 and 1500, while the price of rye in Frankfurt dropped even faster. Similar trends in the Low Countries and Italy demonstrate that this was a universal phenomenon in 15th-century Europe.

At the same time, however, the demand for gold was so great that its price moved in the opposite direction. In England, where developments were typical of trends throughout Europe, the price of gold climbed slowly but almost without interruption from 23 shillings an ounce in 1345 to 40 shillings by 1492. The resulting increase in the purchasing power of gold meant that the volume of commodities that an ounce of gold could buy doubled, at the very least, between the beginning and the end of the 15th century. As a consequence, this was one of the few periods in history when gold was spent instead of hoarded.

Gold has always been a valuable prize, but the alluring combination of falling commodity prices and rising gold prices promised tempting rewards for those who could find new sources. In that setting, the great explorations of the 15th century appear to have been an inevitable response.

Or was it inevitable? One might argue that the forces of raw economics were only an incidental cause of the passion for exploration across the seas in the 1400s. Perhaps these enterprising voyages to reach the ends of the earth were just one more manifestation of the spirit of the Renaissance, a new age that broke with the more rigid mindsets that

religion had imposed on the Dark Ages and the Middle Ages, a time that encouraged bold experimentation in art, culture, and science. Progress in navigation and the expansion of geographical knowledge were natural by-products of the important innovations during the Renaissance in mathematics, in measurement, and in perspective. The discovery of the world was what the Renaissance was all about. Between 1492 and 1500, the size of the world known to Europeans more than doubled; 25 years later, it had more than tripled.

At first glance, this explanation appears to make sense, but the contention that the great explorations could not have happened in an earlier age leads to two strange and counter-intuitive conclusions. First, if the surge in the purchasing power of gold had occurred during a less innovative era, there would have been no one like Henry the Navigator, Columbus, or Magellan – explorers who were unable to resist the temptation of extraordinarily profitable rewards from hunting for gold across the seas. Sailors would have gone along traveling the traditional routes as though nothing had happened to the market for gold. Second, if the great explorations were purely a consequence of the adventurous spirit of the Renaissance, these voyages would have taken place even if the price of gold was falling while the price of commodities was rising. Neither of these possibilities makes much sense.

Gold and jewel encrusted pot





BEFORE RAILROADS CROSSED THE UNITED STATES, YOU COULD REACH SAN FRANCISCO FASTER AND CHEAPER BY WATER FROM CHINA THAN BY LAND FROM ST LOUIS. THIS SIMPLE FACT EXPLAINS WHY THE GREATEST CHALLENGE TO WESTERN EUROPEANS SEEKING GOLD AND EXPANDED TRADE IN THE 15TH CENTURY WAS TO FIND A DIRECT SEA ROUTE TO INDIA AND THE FAR EAST.

The Sacred Thirst

Success in that venture would replace the strenuous and hazardous overland paths that for hundreds of years had carried cargo and humans on horseback or mule or on foot.

The Portuguese were the first to step out and lead the way to the discovery of the world. This was no coincidence. Portugal is such a narrow country that, relative to its small population, it ranks first in Europe in coastline facing the sea. Portugal was also one of the poorest countries in Europe at the end of the Middle Ages, which meant that even small, profitable discoveries could make a big difference. Political disturbances and monetary depreciation in the dark days of the fourteenth century had ruined the gentry and galvanized them to seek new fortunes in foreign undertakings. The whole environment provided the perfect setting for the single-minded zeal of Prince Henry the Navigator – the third son of John I – for using the seas to expand Portugal's power and influence.

Gold was the primary objective of the Portuguese explorations, but that was not all. All the 15th-century explorers claimed that they were on a new crusade to smite the infidels and convert them to Christianity. When the inhabitants of distant lands turned out to have dark skins, the Portuguese soon convinced themselves that taking these poor souls as slaves was a great convenience that facilitated the process of religious conversion while incidentally satisfying the pressing demand for low-cost labor. Slavery was thus an afterthought, a by-product of the search for metallic treasures, but it became increasingly important.

Prince Henry's immediate goal was to contain the threat to Portugal itself from the Moorish pirates who preyed on shipping in the western Mediterranean. These pirates, later known as the Barbary pirates, would still be a danger to trade and travel when John Paul Jones pursued them nearly 400 years later. Henry opened his campaign in 1415 by

capturing the Moorish city of Ceuta, just east of the Straits of Gibraltar on the north coast of Africa. Ceuta was an immensely wealthy city that served as the main Mediterranean port for the goods that Arab traders brought from Africa and Asia. In particular, Ceuta was the primary port for the caravans that brought African gold across the desert for shipment to Europe. As the Portuguese troops ravaged every corner of Ceuta, they found plenty of evidence of the golden resources of West Africa.

The capture of Ceuta opened up a compelling strategy to Henry and his men: if they could transport the output of the African mines by sea to their own shores, they could outflank the rest of Europe by circumventing the tedious and expensive voyage by camel across the Sahara desert to the northern Mediterranean trading posts. The arithmetic involved is worth setting forth in detail.

Depending upon its quality and endurance, a camel can carry between 120 and 200 kilograms of cargo across the desert for 8 to 12 hours a day at 2.5 to 5.0 miles an hour. Let us take the average camel – carrying 160 kilograms for 10 hours at, say, 3.5 miles an hour. That camel would cover 35 miles a day. The distance from the Mediterranean coast at Morocco to the gold country is nearly 2000 miles – roughly the distance from New York to Las Vegas, or from one end of the Mediterranean to the other – which means that the camel would have to spend about 55 days in travel. On completing the trip, the camel had to be given an extended rest period to recover from its labors. A man could usually manage four camels at a time. As the caravans numbered from as low as 300 camels to as many as 3500, the crews would vary from 75 to nearly 900 men; even the smaller caravans wending their way across the sands must have been quite a sight. A caravan of 1000 camels, each carrying 160 kilograms, would have transported a total of 160 metric tons of cargo.

A ship at sea moves more slowly than a camel, at approximately 60 per cent of the camel's speed. However, a ship travels 24 hours a day; the camel caravans had to travel at the pace of the slowest camel and for only a third to a half of a 24-hour day. Consequently, a ship can readily cover more than twice as much distance as a camel in one day. As the distance by sea from Gibraltar to the gold country is at most twice the distance across the desert, the ship has a modest advantage in travel time – even though, like the camel, it needs refitting after long voyages. The big advantage is manpower. The arithmetic above indicates that, at four camels per man and 160 kilograms per camel, a man was responsible for about 0.7 tons of cargo. On a ship, depending on the size of the vessel and the crew required, a man could be responsible for anywhere from 3 tons to 14 tons. The ship is vulnerable to loss in storms or to piracy, but camels could fall ill and entire caravans could be attacked by Berbers and other nomads.

These 15th-century ships incorporated the lateen sail, a remarkable technological innovation that had been developed in the eastern Mediterranean as early as the 2nd century AD but came into wide use only in the Middle Ages. Unlike conventional square

*Camel train in the Sahara*

sails, which were positioned horizontally and limited the movement of a ship almost entirely to running ahead of the wind, the lateen sail was triangular and positioned vertically to the length of the ship. It could swing from port to starboard and back again, which for the first time made tacking possible for larger ships, so they could take the wind on either side. This innovation significantly extended the range of the sailing ships without it, Columbus could never have discovered America.

Prince Henry and his men began by launching a systematic policy of aggression against Moroccan cities, first on the Mediterranean side and then on the western, Atlantic side. The payoff came in short order, in the form of a rapidly growing traffic in slaves, indigo, and sugar between Africa and Portugal. Gold began to flow as well, but the volume appeared to the Portuguese to be no more than a trickle compared with their grandiose expectations. They were convinced that somewhere down the west coast of Africa they would discover the *rio d'oro*, a river of gold in both a literal and figurative sense. They had to keep pushing farther.

It was not just a matter of sailing down the coast until they found the *rio d'oro*. Although the Phoenicians and a few later explorers had made some short voyages in that direction, no one had ever thoroughly explored the west coast of Africa by sea. The few reports that existed were unanimously scary. Arab sailors, who were used to plying the east coast of Africa, fed the Mediterranean folklore of boiling hot seas on the west side,

swarming with serpents waiting to grab human flesh from the decks of the ships. The lucky survivors of such a voyage would suffer the terrible fate of seeing their skin transformed from white to black. The winds were dangerous, the natives unfriendly, and the actual source of the gold an unsolved mystery.

Despite the perils, the prize was too seductive to be frightened away. Bit by bit, in successive voyages, the Portuguese navigators crept down the coast, braving poisoned arrows, building fortresses and capturing slaves (Moors would do when blacks failed to appear, or Moors cooperated by supplying the blacks). Even though progress was slow, the Portuguese mariners never lost faith that they would ultimately discover the elusive river of gold. Finally, around the middle of the century, the Portuguese caravels succeeded in rounding the thick western part of Africa where the coast faces southward. There was the area of Guinea, where the population was predominantly black rather than Moorish.

The participants in these adventures were colorful characters. One in particular stands out, a merchant of Venice by the name of Alvise da Cadamosto. Cadamosto arrived in Portugal in 1484 and approached Prince Henry for permission to participate in the African trade. Convinced that Venetians knew more about maritime trading than anybody else, Henry hastened to agree to Cadamosto's request. Cadamosto fully lived up to Henry's expectations: he was an expert in evaluating commercial prospects.

We are grateful to Cadamosto for having left a journal of his voyages that is invaluable for its information and irresistible for its charm. It was he, for example, who first reported back to Europe on the silent bartering of salt for gold along the Niger River. Yet, with all his travels into the interior – at one point he was 250 miles inland – and his ability to get along with suspicious natives, he was unable to unlock the secret of the source of West African gold.

By the early 1470s, the Portuguese had established a major trading post on the south-facing coast of West Africa, which they named San Jorge de Mina. Although they erected an imposing capital at San Jorge and carried on an active business with the natives to the north and to the west, the Portuguese never managed to become owners or even to participate in the operation of any of the African goldfields. The gold that returned to Portugal by way of San Jorge de Mina was acquired in a set of barter arrangements in which the Portuguese paid for it by trading salt, capes and robes, red and blue cloth, canvas, copper and brass pots and pans, coral, red shells and white wine. Business was good. By the early 1500s, approximately 700 kilograms of gold a year was moving from Africa to Portugal, a meaningful sum when all of Europe's total annual output was no more than about four tons and Portugal's was zero.

In August 1487, Bartholomeu Dias, an experienced Portuguese explorer of African waters, sailed from Lisbon in command of two caravels and a supply ship with orders to go around Africa to India. Six months later, Dias was the first European to moor his ship on the southeast coast of Africa. He continued on some way, with every intention of

*Christopher Columbus*

continuing on to India, but his men were impatient to go home, especially as the supply ship had fallen far behind. Dias had no choice but to turn back, which meant that Vasco da Gama would be the first to reach India in 1497 and start the process of establishing major Portuguese posts throughout Asian waters. Turning his ships around, Dias sailed once again past the cape at the bottom of Africa, which his king would later name the Cape of Good Hope, because it leads on to the passage to India.

In December 1488, Dias returned to Lisbon, 16 months after his departure. One of the people in the audience who came to hear him report in detail on his voyage was a Genoese seaman named Christopher Columbus, who took copious notes as he listened to Dias's presentation.

Columbus was the son of a weaver; like many other Genoese, he had gone to sea at an early age. He had been on a ship sunk in battle, had sailed to the eastern Mediterranean and possibly to Turkey, had been on a voyage to Iceland that had stopped at Ireland, and had sailed through many of the known waters on the African coast. Samuel Eliot Morison,



*Christopher Columbus
landing in the New World*

his most distinguished biographer, emphasizes that Columbus was one of the best navigators and seamen of his day. He had no doubts that the sea route he envisaged straight out into the Atlantic would not only replace the overland routes to Asia but by heading straight west would make much more sense than the tortuous eastward routes that the Portuguese were attempting to establish. Columbus was confident that he would find gold, but by shortening the time involved he would also significantly reduce the cost of the voyage to India and the rest of Asia, making possible the shipment of a substantially larger variety of goods. He was also deeply religious and dreamed of spending the gold he would earn from his pathfinding voyage on a crusade to win back the Holy Sepulcher from the Muslims.

Columbus lived for a long time in Lisbon, where he married a local girl and worked on occasion as a mapmaker. Convinced that Portugal would be the obvious country to back him, Columbus approached the throne in 1484, offering the first chance to sponsor and finance his voyage. Yet John II, the nephew of Prince Henry and grandson of John I,

turned Columbus down flat. The Portuguese, growing fat on their African arrangements and their newfound linkages to the Indies via the Cape of Good Hope, saw no need to take on another big risk. John was in any case put off by Columbus's insistent demand that the king make him a knight, admiral of the Ocean Seas, and viceroy and governor of all lands discovered, in addition to turning over 10 per cent of all profits from these lands – which specifically included gold. These were precisely the terms that Columbus would later manage to squeeze out of Ferdinand and Isabella.

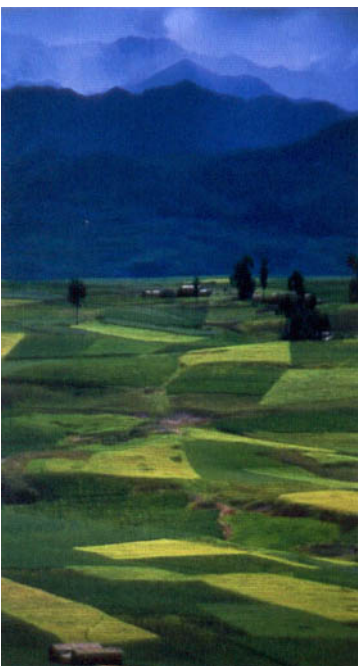
Still hopeful, Columbus turned to the Spanish monarchs in 1486. The commission they appointed to study his proposal took until late 1489 to decide that his idea was without merit: the trip would take too long, and they doubted whether any lands remained to be discovered. After that, Columbus was rejected by Henry VII of England, whose advisors ridiculed the whole idea as fantastic. Charles VIII of France also said no.

At that point, Columbus saw no choice but to give up the whole thing. Four European monarchies had turned him down, scorning his conviction that he was offering them the most direct, the most economical, and the fastest route to the Indies. He decided to return to mapmaking.

Queen Isabella, however, had never lost interest in Columbus's ingenious plan. The risks were obvious, but the rewards could be tremendous: Columbus's shorter route would give Spain the means to take the leadership in the Indies away from the Portuguese, while the gold that he promised would finance and support the dynasty that she dreamed of creating. So Isabella summoned Columbus back, even providing him with money for a new suit and a mule to carry him to her. At first the news was excellent, as Isabella's new commission did approve his proposal. Then came the bad news: the Grand Council rejected what they considered to be Columbus's exorbitant demands for titles and financial rewards.

Defeated and dejected, Columbus mounted his mule and headed back home. He had just begun the trip along the mule path when a horseman caught up with him and told him to return to the queen. An influential advisor had prevailed upon her to change her mind. That was in April 1492. Four months later, just before sunrise on 3 August, Columbus and his crew made confession, took communion, and boarded their ships. Columbus's command to weigh the anchors ended with the words, 'in the name of Jesus'. The rest is history.

Two days after touching land on San Salvador, which Columbus was certain was offshore from Japan, or Cipangu as the Spaniards called it, he sailed onward in search of his goal. He was confident that he would soon be able to confirm Marco Polo's observation that the palaces in Japan were roofed with gold – he carried a copy of Marco Polo's writings with him as a guidebook to the lands he expected to visit. The bits of gold that the natives wore as nose plugs only increased the sense of anticipation. As soon as he noticed that the Indians did not put much value on their gold, he hastened to offer them beads and caps in exchange.

*The Andes*

The natives Columbus encountered on San Salvador told him about a large island nearby named Cuba, which sounded enough like Cipangu to convince Columbus and his crew that their goal was now within reach. They landed on Cuba on 28 October, but they found no gold. Although they did discover tobacco, it held no interest for them; nothing except huge quantities of gold would satisfy them. From 12 October 1492 to 17 January 1493, when Columbus headed back to Spain, his diary mentions gold more than 65 times. Indeed, his entry of 13 October 1492, the day after the first landing, reports, 'I was attentive and took trouble to ascertain if there was gold.' He was encouraged by the dark skins of the natives, for Europeans had long believed that dark skins would be a sure sign of the presence of gold. While sailing along the Cuban coast, he noted in his diary, 'From the great heat which I suffer, the country must be rich in gold.'

Columbus was euphoric over his discoveries, but he as well as his early followers faced painful disappointments. The lands they had discovered were not the Indies after all, although Columbus still thought he was in Asia even during his third voyage six years after the first. Worse, the continental landmass they encountered appeared to be such an endless barrier that nobody could figure out how to get around it to reach the Indies, the only objective that made the whole dangerous business worthwhile. If the amount of gold on these lands had at least met their expectations, that would have been some compensation for the frustration of missing out on their ultimate objective. Gold there was, but surely no bonanza.

The marching orders were nevertheless certain. As King Ferdinand had commanded, 'Get gold, humanely if possible, but at all hazards – get gold.'

In 1510, a debt-ridden, disgruntled Spanish farmer from Estramadura named Vasco Núñez de Balboa decided to leave Hispaniola (now Santo Domingo) and join an expedition to Darien, the area where the Isthmus of Panama connects to the northern shores of Colombia. There was gold in central Hispaniola, where the Spaniards worked both the mines and the Indians so hard that by 1519 only 2000 of an original population of more than 100,000 remained and slaves were already being imported from Africa to do the mining. Nevertheless, rumors were rife in Darien about vast supplies of gold somewhere to the south, perhaps near a sea whose uncertain existence might lead to the gold. When Balboa arrived in Darien, he became close friends with an illiterate swordsman, also from Estramadura, whose name was Francisco Pizarro. Pizarro, like Balboa, was a man who did not flinch at danger if a venture promised a commensurate reward.

The move to Darien did nothing to solve Balboa's financial problems. One day in September 1513, still frustrated and in trouble with the law, Balboa was weighing some gold when a barbarian chieftain came up, scattered the glittering metal around the room, and cried, 'I can tell you of a land where they eat and drink out of golden vessels, and gold is as cheap as iron is with you.' This was all Balboa needed to prompt him to lead a great



Atahualpa

enterprise he expected would bring him to King Ferdinand's attention. He proceeded to gather together a group of 190 Spaniards, aiming to track down the rumored gold sources once and for all and also to resolve the mystery of the incomplete search for an ocean route to Asia. Despite three weeks of attacks by hostile Indians, insects, and snakes, Balboa's men hacked their way westward until they came to an abrupt escarpment. The Indians told them that the slope on the other side of the crest led down to a great sea.

A couple of days later, Balboa waded into the surf of the Pacific with drawn sword and claimed 'the great South Sea . . . with all that it contained', for the king of Spain. And then, setting a standard for cruelty to the Indians that many other Spaniards would emulate, Balboa and his men mercilessly plundered the copious treasure of gold objects that they found in the Indian villages. The elegance and dazzling sophistication of these semi-abstract objects evidently held no meaning for them. They were far more fascinated by the traces of gold ore they uncovered on the sandy shores of their South Sea.

This glittering achievement failed to solve Balboa's problems, and he seems to have been in chronic trouble with authority. Some time after the discovery of the Pacific, at a moment when Balboa was making plans to sail southward on his newly discovered sea toward Peru in search of more gold, the governor of Darien accused him of treason and ordered him to be beheaded. The governor, who had been sent out with 1500 men by the



king of Spain after receiving the electrifying news of Balboa's discoveries, happened to be Balboa's father-in-law; the executioner assigned by the governor to this task was none other than Francisco Pizarro.

Pizarro was an illegitimate child, abandoned by his mother on the church steps of the town where he was born. He grew up tough, a man of great endurance and strong leadership skills. At a time when most men rationalized their mistreatment of the Indians as motivated in some way to improve their lot and bring them the blessings of Christianity, Pizarro refused to mask his goals. After the victory in Peru, when a priest asked him to do more to convert the natives, Pizarro's response was, 'I have not come for any such reasons. I have come to take away from them their gold.'

He was a man of iron will and unshakable confidence in his own abilities, regardless of the obstacles placed in his way. Consider this astonishing fact: Pizarro's first decisive contact with the Incas did not occur until 1532, which was eight long years after the first exploratory expedition down the Pacific coast from Panama with one ship and 100 men. He even made two trips back to Spain during those eight years to firm up royal support and sufficient resources for the campaign in Peru.

William H Prescott, in his *Conquest of Peru*, first published in May 1847, takes a dim view of the hypocrisy of the Spaniards in justifying their terrible deeds in the name of Christ. Yet he cannot help but admire their courage, ingenuity and audacity in the face of the staggering odds against them.

After scaling the wild passes of the Andes, often on paths scarcely wide enough for a horse and flanked by sheer drops of thousands of feet into the abyss, 200 Spaniards managed to subdue an empire of at least 35 million people that included large portions of the modern states of Ecuador, Peru, Bolivia, Chile and Argentina. At the turning point in the campaign, Pizarro's little army overcame the resistance of 30,000 Inca soldiers trained to fight at altitudes of over 10,000 feet.



Machu Picchu



The Spaniards – hardened, brave and ruthless fighters – were surprised at the advantage they achieved by appearing godlike to the Indians. The pale faces, cannons and muskets, trumpets, horses, shining armor, and wagons were awesome and frightening to the Indians. Despite a society that was in many ways more highly structured, agriculturally productive, and artistically sophisticated than the Spanish, these Indians had never invented the wheel and their ubiquitous llamas could hardly match the speed, rideability and intelligence of the horse. Their only technological advantage was an extraordinarily effective relay system of highly trained runners who carried news and information up and down the majestic Andes on roads as good as Roman roads, functioning with such remarkable efficiency that they even delivered live fish from the coast to the nobility living high in the mountain valleys.

The decisive climax to what is a long story occurred in November 1532, when Pizarro and his men reached a watering place high up in the mountains called Caxamalca, where the Emperor Atahualpa, ‘the Inca’, or Child of the Sun, had taken up temporary residence. Atahualpa was well aware of the approach of the Spaniards and, in fact, had



Atahualpa's deal – the red line that would mark the height of the gold to be piled up in 'ransom'

The place where Atahualpa was imprisoned



sent emissaries to bid them welcome. The Spaniards had been especially interested in one of those emissaries because he came drinking chicha – the fermented juice of the maize from golden goblets that his attendants carried for him.

As the Spaniards looked down on the verdant valley and the little city of Caxamalca with its 10,000 inhabitants, they spotted the hot baths where the emperor and the princes came to take the cure. They also observed a less attractive feature: a mass of white covering several miles. These were the tents of the Inca's army, a spectacle that startled the Spaniards by their immense numbers. But it was too late to turn back.

The Conquerors, as Prescott calls them, found only empty streets as they entered Caxamalca. After a short distance, they came to an immense open plaza surrounded by low buildings containing capacious halls, presumably barracks for the Inca's soldiers. Pizarro was determined to occupy this area.

Pizarro immediately dispatched a small force toward the Indian encampment, commanded by his brother Hernando Pizarro and his senior colleague Hernando de Soto. (De Soto would later make his mark exploring in Florida for the Fountain of Youth.) In addition, there was one Indian member of this detachment, who had been taught enough Spanish to act as interpreter. The Spaniards called him Felipillo.

The Spaniards found the Inca seated in a spacious courtyard with attractive structures around it and a fountain in the center, surrounded by nobles and women of the royal household. He was about 30 years old, handsome, more robust than many of his countrymen, with a large head and bloodshot eyes that made him look fierce. Hernando Pizarro greeted Atahualpa and informed him that the Spanish commander and his men



Inca gold



Machu Picchu

were 'the subjects of a mighty prince across the waters . . . come . . . to offer their services, and to impart to him the doctrines of the true faith which they professed.' De Soto then invited the Inca to visit the Spaniards in their quarters on the following day; the Inca bluntly accepted the invitation. As a defensive measure, the Spaniards never dismounted from their horses, but this still left them in a position to respond eagerly to invitations to drink the sparkling chicha from golden vases of extraordinary size handed up to them by the dark-eyed beauties of the harem.

Hernando and his contingent returned to their comrades in a state of high anxiety over the evident strength and discipline of the military forces of the Inca. In addition, the level of civilization was much more impressive than anything they had observed in the lower regions of the country. Pizarro, however, was undaunted. He gave a rousing speech to his men, reminding them that 'If numbers, however great, were on the side of their enemy, it mattered little when the arm of Heaven was on theirs.'

Pizarro had concocted an audacious scheme that – if it worked – would give his side an overwhelming advantage despite the enormous differences in military strength: he would take the Inca prisoner in the face of his own army. This was a high-risk strategy, but he had no doubt that the huge disparity in manpower had put him in a situation so desperate that a more modest effort would have been doomed.



The next morning, Pizarro hid his troops throughout the buildings on the plaza, with his artillery of two small cannon in the fortress. He made certain that all the arms were in good order, that the armor was shining, and that the horses were garnished with bells to make maximum noise at the crucial instant of attack. Then mass was said. 'One might have considered them a company of martyrs,' Prescott observed, 'about to lay down their lives in defense of their faith, instead of a licentious band of adventurers, meditating one of the most atrocious acts of perfidy on the record of history!'

The Inca's royal procession appeared a few hours later but halted about half a mile from Caxamalca and began to pitch their tents. Pizarro sent a messenger to ask Arahualpa to join the Spaniards as soon as possible, as both dinner and entertainment had been provided for him.

Atahualpa swallowed the bait. He arrived with only a few warriors, and without arms. Was Atahualpa so absolute in his own empire that he had no fear of entrapment? Or did he simply figure that a small troop of only 200 men would never even contemplate such a brazen deed? Whichever it was, his lighthearted decision would seal his doom.

Atahualpa may not have brought his army with him, but he did not spare the numbers of the rest of his retinue; 5000 or 6000 people filled the square in Caxamalca. There were hundreds of menials, singing as they cleared the path the Inca would follow. Nobles

came in costumes of checkered red and white squares, while the guards and the Inca's immediate attendants wore a rich blue livery and a profusion of bright ornaments. The Inca himself was carried high on an open litter of gold and seated on a massive throne, also of gold. He wore a collar of enormous, brilliant emeralds and his hair was decorated with a variety of golden ornaments.

When he and all his people had gathered in the square, without a Spaniard in sight, Atahualpa wondered aloud where they had all gone. At that moment, the chaplain appeared, holding a Bible in one hand and a crucifix in the other. He was accompanied by Felipillo, the Indian interpreter. The chaplain announced that he had come to set forth to the Inca the articles of true faith, which he proceeded to do at great length. He finished by explaining the role of the pope, who had commissioned the Spanish emperor, 'the most mighty monarch in the world, to conquer and convert the natives in this western hemisphere. . . . [His] general, Francisco Pizarro, had now come to execute this important mission.'

Atahualpa exploded. 'I will be no man's tributary,' he announced. 'I am greater than any prince on earth. . . . For my faith, I will not change it. Your God, as you say, was put to death by the very men whom he created.' He halted to point to the sun, then just beginning to set behind the mountains, and added, 'But my God still lives in the heavens and looks down on his children.' He took the Bible from the shocked friar's hands, looked at it briefly, and then threw it on the ground, declaring, 'I will not go from here until your comrades have made me full satisfaction for all the wrongs they have committed.'

Inca gold



The priest ran to Pizarro and commanded him, 'Set on at once; I absolve you.' Pizarro waved a white scarf, a cannon boomed from the fortress, and his men, some mounted and some on foot, dashed into the plaza shouting their battle cry: 'St Iago and at them!' The Indians panicked. Stunned by the thunder of the artillery and muskets and blinded by the sulfurous smoke, they made no resistance as the Spaniards trampled them down with the horses and slashed into their defenseless bodies. Meanwhile, the Inca, still up on his heaving litter, saw his faithful nobles falling in a desperate attempt to protect him. Indeed, Pizarro himself ran to protect the Inca from excessively eager Spaniards and received a wound on his hand for his efforts – the only wound the Spaniards suffered that day. The massacre of the Indians continued for a long time until thousands of them had fallen. The precise number of dead is a matter of dispute, while the crowd of prisoners was beyond counting.

Some of Pizarro's troops wanted to put the prisoners to death, or at least to disable them by cutting off their hands. Pizarro refused and liberated all the prisoners except for a sufficient number to tend to the needs of the Spaniards.

As the Spaniards settled down to await reinforcements from the Spanish base on the coast, Pizarro used the time to become better acquainted with his captive. Atahualpa, on his side, closely observed the Spaniards. He soon discovered that they had an appetite even more potent than their repeated efforts to convert him to Christianity: the love of gold.

One day Atahualpa proposed a deal. If Pizarro would set him free, the Inca would arrange to have the room he occupied filled with gold as high as he could reach, all within two months; the gold would come from the royal palaces, temples, and public buildings. The area of the room was about 17 feet by 22 feet, with a height of 9 feet. Pizarro eagerly accepted the proposition. As Atahualpa stood on tiptoe, a red line was drawn at the height he indicated, a notary recorded the details of the agreement, and Atahualpa dispatched couriers to execute the task.

Pizarro also sent emissaries to the capital city of Cuzco, a difficult journey of over 600 miles across the mountains, where they found the great Temple of the Sun covered with plates of gold and royal mummies within, each seated on a gold throne. The Spaniards ripped 700 plates from the temple walls, each about the size of the lid of a chest and weighing around 41 ½ pounds. Before the Spaniards were through, they had packed 200 loads of gold to be carried back to Caxamalca on the backs of the humbled Indians. This was just a preliminary foray: a larger and more rapacious trip to Cuzco would take place later on.



Inca gold

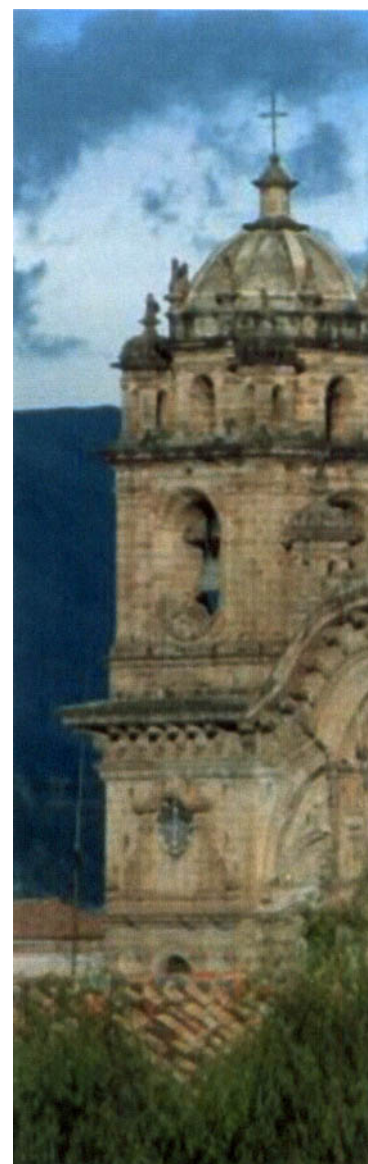


Pizarro's tomb

Meanwhile, gold was arriving from all over Peru, from the Inca's temples, palaces, and other public edifices, to satisfy his contract with Pizarro. The gold came in many forms -- goblets, ewers, salvers, vases in great variety, ornaments and utensils, tiles and plates, curious imitations of different plants and animals, and a fountain that sent up a sparkling jet of gold. Pizarro selected a small sample of these objects to ship back to the emperor, Isabella's grandson Charles V, who was known as Charles Quint. He inherited the throne of Spain via his mother, Joanna the Mad, and had also been elected emperor of the Holy Roman Empire -- a position his father's father had occupied. Only Napoleon and Hitler at their zenith have ruled over a larger area of Europe.

Except for the tiny lot that Pizarro sent to Spain, not a single piece of that heap of gold in Atahualpa's room has survived in its original form, but the small quantity of Peruvian goldwork that escaped the clutch of the Spaniards and has come down to us is breathtaking. Gold was so easily obtained at high purity from the river deposits in Peru that the art of working gold began there at an early time. By 500 BC, golden diadems, earrings, bracelets and plaques were being created. There are even earlier objects with clear Chinese and Vietnamese influences, which suggests that Asian sailors were finding their way across the Pacific at a time when Europeans were barely managing to paddle themselves around the Mediterranean.

The Peruvians at the time of the conquest were beating out thin gold into vessels and masks of great variety, complexity, and opulence. Among the more spectacular achievements were enormous beakers in the form of a human effigy, a difficult technical task with startling impact on the viewer. Some of these beakers show the head in an inverted position; one thereby drank out of the neck, indicating that these beakers



Cuzco Cathedral, Peru



probably represented the head of a defeated enemy with the user symbolically drinking from the enemy's skull. A woolen tunic has been found that contained 30,000 miniscule plaques of sheet gold. At the other end of the scale, the goldsmiths created sheets of gold with repoussé relief designs to cover walls, such as those that the Spaniards ripped from the temple walls in Cuzco.

Except for the small exhibit reserved for Charles V, all the accumulated treasure was transformed from adornment to money, as one article after another disappeared into the melting pots to be recast into gold bars of a uniform standard. Pizarro assigned this task to the Indian goldsmiths, the same men who had created many of these beautiful objects.

The work consumed a full month, but it produced 1,326,539 pesos d'oro, which Prescott calculated as the equivalent of \$15 million when he was writing his book in the 1840s. That would set the value in today's money as \$270 million, a handsome return for one's efforts under any circumstances, but that sum cannot convey the importance of this treasure in the far smaller economies of the 16th century. This calculation does not include the throne on which the Inca had made his tumultuous arrival – 190 pounds of 16-carat gold, or the equivalent of one year's output of the Peruvian gold mines. That prize Pizarro reserved for himself. If we convert the pesos d'oro into weight and express the result in tons, the Indians must have filled Atahualpa's chamber with nearly five tons of gold, which is more than the total annual output of gold within Europe at that time, or, even more impressive, the equivalent of 20 years of production by the Peruvian gold mines. In contrast, it is worth recalling that Justinian poured twice as much gold into St Sophia and that Jean II's ransom, at three million crowns, was more than double the mass of gold in Atahualpa's chamber. No wonder Justinian believed that he had surpassed Solomon and the French people rose up in revolt at the burdens imposed on them!

The account of the Inca has a hideous ending. Newly arrived Spanish troops saw little point in continuing to shelter Atahualpa and were strongly opposed to liberating him. Pizarro resisted the pressure at first but ultimately yielded. He put the Inca up for trial under charges of having usurped the throne, squandering public revenues, practicing adultery and idolatry, and attempting to instigate an insurrection against the Spaniards. The kangaroo court lost little time in finding Atahualpa guilty. After the sentence was pronounced, Atahualpa turned with tears in his eyes to Pizarro and asked, 'What have I done, or my children, that I should meet such a fate? And from your hands, too, you, who have met with friendship and kindness from my people, with whom I have shared my treasures, who have received nothing but benefits from my hands!' Pizarro turned away without reply.

On 29 August 1533, two hours after sunset, they lit the torchlights on the plaza and tied Atahualpa, chained hand and foot, to a stake surrounded by the fagots of his funeral pyre. The friar who had first lectured him on the blessings of Christianity appeared once again to hold up the crucifix before him and warn him of eternal damnation if he did not renounce his pagan religion and accept Christ. Atahualpa refused to yield. Finally, the priest promised Atahualpa that if he converted they would provide him with a quick death by garroting him rather than subjecting him to the extended agonies of the stake. Desperate, the Inca complied, accepting baptism with the name Juan de Atahualpa in honor of St John the Baptist, on whose day this unhappy event happened to fall. Then the executioner performed his gruesome task while the Spaniards muttered prayers for the salvation of the Inca's soul.

The end of the story of the Conquerors reads like a morality tale. Adam Smith decried 'the sacred thirst of gold' that drove the explorers and Conquerors into the New World,

and he was right. The quenching of that thirst led most of these men to a bad end, beginning with Balboa himself.

Pizarro's original company broke into factions that drowned the spirit of their great adventure in bloody internecine quarrels over leadership and spoils. After the enormous efforts they had invested in the conquest of Peru and the terrifying risks they had faced, many of the men never realized their dreams of returning to Spain with their golden wealth to live the easy life. Some lost their lives in battles with the Indians or in civil wars among themselves. Others lost their gold, because it was too heavy to carry in the constant fighting – like Ruskin's man on the sinking ship. And many lost it in betting for high stakes in gambling games with friends.

The history of the Pizarro brothers ends in the most bizarre tragedy. Hernando Pizarro returned to Spain with his treasure in 1540, where he was imprisoned at the behest of his enemies for 20 years and emerged an old and infirm shadow of the great soldier he had once been. Francisco Pizarro was assassinated in 1541, while having dinner in his own home in Lima, by conspirators from a group of dissidents. As the swords were plunged into his body, he cried, 'Jesu!' and kissed the cross he had traced with his finger on the bloody floor.

In time, after the Spaniards had grabbed every loose piece of gold and golden object they could find, the joy of plunder had been exhausted. Mining, which was serious business, had to take over. The Peruvian mines were great river gorges, shaped like caves and often reaching as far as 60 feet into the earth. Totally dark, these passages had room for only one man at a time, who crouched his way in, scraped out as much gold as he could from the rocks, crouched his way out, and was followed by another man to perform the same task.

Under the Incas, this hard work was supervised and carefully modulated to prevent exhaustion and to sustain the lives of the miners. Under the Spaniards, the merciless labor of the mines was devastating to the natives, as it was in every other golden venture of the Europeans in the New World. What an irony! Edward Gibbon, in his *History of the Decline and Fall of the Roman Empire*, emphasizing the importance of Spanish gold to the Roman Empire 15 centuries earlier, tells us that 'Spain, by a very singular fatality, was the Peru or Mexico of the old world. . . . The oppression of the simple natives [of Spain], who were compelled to work in their own mines for the benefit of strangers, form an exact type of the more recent history of Spanish America.'

Later on, when the Portuguese began to exploit the huge gold resources of Brazil, the death rate of the Indians was so high that the native population was decimated and great numbers of African slaves had to be imported to take their place. The descendants of these black slaves account for a significant part of today's population in Brazil. There was also the usual story of the transmittal of the white men's diseases, but the conditions of virtual slavery at the mines threw human life away as though it mattered not at all.





The Fatal

A MONUMENTAL MASS OF GOLD AND SILVER SAILED ACROSS THE ATLANTIC FROM THE NEW WORLD TO SPAIN DURING THE 1500S. ACCORDING TO ONE AUTHORITY, THE TOTAL EUROPEAN STOCK OF GOLD AND SILVER AT THE END OF THE CENTURY WAS NEARLY FIVE TIMES ITS SIZE IN 1492.

TEN

Poison and Private Money

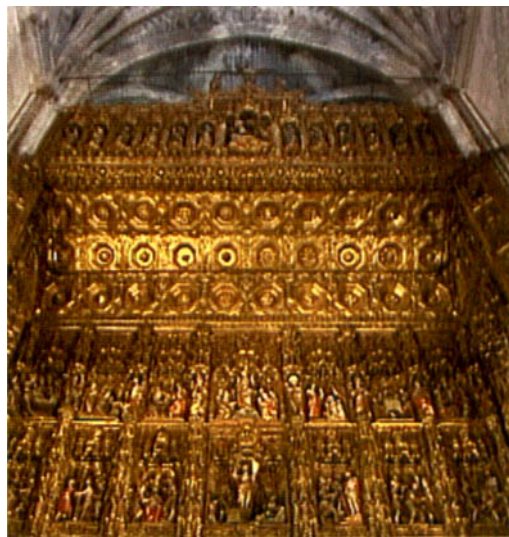
The volume was so enormous that the armed convoys that transported the treasure to Europe averaged about 60 ships; on occasion, the convoys included as many as 100 ships. Each of these vessels carried over 200 tons of cargo in the 1500s and around 400 tons on larger ships in the 1600s. In 1564 alone, 154 ships arrived at Seville to debark their cargo of treasure. At the end of the 16th century, the precious metals accounted for the bulk of the value of everything shipped from America to Spain.

As we trace the impact of all that gold on the European economy in the course of the 16th century, we shall see that the story has an ironic twist at the end. Gold had to face silver as a rival for most of history, but a serious rival to both precious metals was blossoming by the end of the 16th century – forms of paper money as debt instruments issued by private parties instead of by governments. All the excitement about gold in the 1500s was in essence a celebration of the past. While no one was paying much attention, the future was beginning to emerge.

The Spanish government was extraordinarily effective and efficient in accomplishing the complex task of moving the treasure across perilous and hostile seas. The gold and silver were loaded on ships at Vera Cruz in Mexico, Trujillo in Honduras, Nombre de Dios on the Atlantic side of Panama, and Cartagena in Colombia. The area of the Caribbean enclosed by these ports came to be known as the Spanish Main, a name that has lingered on in romances ever since. From there, the ships traveled to Cuban waters to join together in convoys, or *flotas*, for the long voyage to their home port of Seville. Two to eight armed galleons accompanied the *flotas* as protection against the pirates and buccaneers who roamed the seas waiting to pounce on treasure-filled vessels. The *flotas* also provided at



Seville Cathedral, the golden altar



least a hope of rescue, of cargo if not men, when heavy storms threatened to sink one or another of the galleons.

When vessels were forced by storms or fear of attack to put in at any port other than Seville, passengers were forbidden to land or to make any offer to trade in treasure. Once the cargo reached Seville, everything was transported under the tightest security to the House of Trade, where it was weighed and placed in special chests in a treasure chamber; both chests and chamber were provided with triple locks, with each of the three keys carried by three different officials of the House. The metal was smelted and refined right there. Some of it was then minted, but significant amounts of bullion were delivered to the creditors of the Crown, most of whom resided in other lands.

It staggers the imagination to consider the skill required to gather together, organize, control and maintain contact with that many sailing ships on a voyage of over 3000 miles of open ocean, with no handy means of communication such as the wireless systems or radar used on the convoys sent out 400 years later from the United States to Britain in World War II. By comparison, piloting 1000 camels across the wastes of the Sahara must have been a cinch. There was another big difference between the convoys of the 1500s and the convoys of the early 1940s. In World War II, the ships carried black gold – oil – plus food and armaments to defeat the Nazis; in the 1500s, the freight amounted to nothing more useful than shiny metal ingots. People knew what to do with the precious cargoes carried through the submarine-infested seas of the North Atlantic, but the Spaniards had a far more open-ended set of options for the precious metals. More often than not, they made the wrong choices.

Although losses to the Spanish *flotas* from pirates were a lot smaller than losses from storms at sea, the danger of attack was a constant worry. There was a rumor around Spain that the Holy Roman emperor Charles V – who was also Charles I of Spain through his mother, the daughter of Ferdinand and Isabella – cried with joy every time word came in of the safe arrival of a *flota*.

As the historian Kenneth Andrews describes the times, piracy was elevated to a preferred branch of policy by Britain, France, and Holland, each of which was seeking a piece of the action in America that the Spaniards and Portuguese had claimed exclusively



for themselves. Spain was also at war at one point or another with each of these countries during the 16th century. There were 13 organized English expeditions to the Caribbean, and doubtless many freelance junkets, just during 1570 to 1577. Nevertheless, despite all the romance about the pirates and their occasional dramatic successes, the records of the House of Trade contain many accounts of failed attacks and safe arrivals. Whole fleets were intercepted and defeated only three times, twice by the English and once by the legendary Dutch admiral Piet Heyn in 1628. More often, stragglers ran into difficulty. In March 1569, for example, 22 Spanish and Portuguese ships were brought into Plymouth, where the English happily relieved them of their precious cargoes.

The most serious and sustained threat to Spanish shipping came from Sir Francis Drake, who made a specialty of plundering gold from the Catholics of Iberia. His efforts made his crew rich, quite aside from the fortune he kept for himself and the even bigger sum that he turned over to the English crown. His hatred of the Spaniards was reciprocated: they referred to him as 'the master-thief of the unknown world'.

Drake kept at the task, off and on, for nearly 25 years. He even landed in Panama in 1572, aiming to capture the Atlantic-side port of Nombre de Dios and thereby interdict the north-south route of Spanish gold. A wound in his leg forced him to abandon that project, but he and his men succeeded in seizing a pack train loaded with gold en route to Nombre de Dios from Panama City on the Pacific, with £20,000-worth of coins to show for their efforts.

Drake's famous voyage on the *Golden Hind* during 1577-9 snared more than ten tons of gold, silver, and jewels from Spanish ships, first on the Atlantic side, and later, after



The Tower of Gold in Seville was constructed from the wealth created by the gold from the New World

sailing through the Straits of Magellan, on the Pacific side. Drake then sailed up the coast of California before crossing the Pacific. He landed at Point Reyes on the shores of Marin County on San Francisco Bay and claimed the territory in the name of Queen Elizabeth of England. In 1586, the Venetian ambassador to Madrid reported that Drake had landed on Santo Domingo, Puerto Rico and Cuba and had 'returned to England with 38 ships laden with much booty'. In 1595, the queen sent Drake back to Panama to capture Nombre de Dios and Panama City and hold them for ransom. This time he succeeded in taking Nombre de Dios, but he in turn was conquered by a fatal case of dysentery, along with many of his shipmates, and was buried at sea.

One might think that Spain in the middle of the 16th century should have been the richest nation in Europe by a wide margin. It was not. The impact of this immense and sudden addition to monetary wealth was felt throughout the rest of Europe and even out to the Far East, but in Spain no lasting payoff remained from the spectacular exploits of the Conquerors and the fountains of blood that flowed from white men and Indians. The gold came in one end and went out the other like a dose of salts.

How was it that the Spaniards were able to mismanage one of the greatest windfalls of all time? Why did so much of the fruits of history's first gold rush end up in the hands of others? Part of the answer to these questions was local, indigenous to the character of 16th-century Spain. Part, and perhaps a larger part, was the result of the dynamic and restless environment of the era, in which the rigid structure of Spanish society was ill-fitted to participate.

Once the gold began arriving in quantity, the Spanish were far more proficient at spending than at producing. The massive imports of gold and silver stimulated the spending skills at the same time that they stifled Spain's incentive to produce. Spain acted like a poor man who makes a great windfall at the gambling tables but comes to believe that the money is his destiny rather than a non-recurring event. This event was indeed non-recurring: copious as the shipments of gold to Spain may have been during the 1500s, they peaked in mid-century and dropped off sharply after 1610; silver shipments peaked around 1600 and fell into a steep decline after about 1630.

During the 16th century, five-sixths of outgoing cargoes from Spain, primarily to the colonies, consisted of goods grown or manufactured in other countries. Late in the century, the Cortes, or Parliament, declared, 'The more of [gold] that comes in, the less the kingdom has. . . . Though our kingdoms should be the richest in the world . . . they are the poorest, for they are only a bridge for [the gold and silver] to go to the kingdoms of our enemies.' Another Spanish observer, Pedro de Valencia, wrote in 1608, 'So much silver and money . . . always has been fatal poison to republics and cities. They believe money will keep them and it is not true: plowed fields, pastures, and fisheries are what give sustenance.' Instead of transforming the gold and silver into new productive wealth, the Spaniards paid the precious metals out to other countries and spent so much that debts to foreigners soared. As early as the 1550s, there was a popular saying that 'Spain is the foreigners' Indies,' because so much good Spanish money was being paid over to foreigners for 'puerilities' – baubles like bangles, cheap glassware and playing cards.

Spain had also committed a costly economic blunder in the Columbus-year of 1492, even though the decision had brought joy and pride at the time it was made. The Jews and the Muslims were both expelled in 1492. Some Jews remained by converting to Christianity, but the vibrant intellectual community that had contributed so much to Spain for hundreds of years rapidly disintegrated. Most Christian Spaniards at that time were peasants or soldiers, illiterate and without any knowledge of the simplest kind of arithmetic. The nobility were either idle or romantic warriors.

The Jews and the Muslims, in contrast, were highly educated, leaders in mathematical and scientific developments, and immune from the Christian strictures against usury. They were the skilled governmental administrators and men of business. The Muslims in particular had a long heritage of trading, importing and exporting. With their departure, Spain lost almost all of the native merchant class that was essential in a time of dynamic

economic development throughout Europe. Instead, Cadiz and Seville, the primary economic centers of 16th-century Spain, were filled with foreigners – Genoese merchants and bankers, German moneylenders, Dutch manufacturers, and purveyors of every kind of goods, services, and finance from all over Europe, even Bretons and people from as far away as the North Sea. Almost all of the massive borrowing by Spain in the 16th century was financed by foreigners.

The departure of the Jews and the Muslims was a loss in another sense. Because of its geographical location, Spain is not on the route that traders and travelers would take in going from one place to another. The line of countries from France eastward, and the projection of Italy and Greece down into the Mediterranean, were on the east–west crossroads of travel and trade through Europe. There was no need to cross Spain, unless you were coming from Africa, and even then Spain was not the only possibility. As a result, Spain tended to be more provincial and ingrown than the countries to the north and east; only Seville, Barcelona and Bilbao had any significant connections with the rest of Europe. The cosmopolitan flavor came from the Jews and the Muslims, who had many contacts in other lands dating back centuries. Their departure cut that link to the outside world, leaving Spain dependent on foreigners whose allegiance was elsewhere.

The greatest waste inspired by Spain's gold was not in the baubles or in the loss of commercial and financial sophistication. It was in the dreams of glory of the Spanish monarchs. Gold has always been associated with power. Once the kings of Spain realized how much new wealth the discoveries of gold in the American colonies would bring them, they convinced themselves that their wealth was great enough to bend the world to their will, especially in the fiery matter of Catholicism versus Protestantism. In the middle of the century, half of all business conducted in Spain was for the account of the king.

Charles V, who ascended the throne in 1516 at the death of his grandfather, Ferdinand, was determined to make Spain the dominant power in Europe. But Spanish power was not sufficient for Charles. He also wanted to follow in his other grandfather's footsteps and become emperor of the Holy Roman Empire. That job did not pass by inheritance; you could become emperor only through election by a group of Germans appointed by the pope and called electors. Francis I of France had identical ambitions. An intense bidding war to buy the votes broke out, in a no-holds bribery contest, with Francis backed by Genoese bankers and Charles by the Fuggers, the great Augsburg banking family. Charles won, but at a cost of 850,000 florins, which pushed him in debt to his ears. He proceeded to engage in 27 years of warfare with Francis I, with intermittent truces that were invariably violated and at one point that almost led to a personal duel between the two monarchs. Charles also claimed the Netherlands as part of his empire, and left his son, Philip II, to deal with the fruitless 80-year struggle to tame the Dutch and the Belgians, during which most of the fighters on the Spanish side were mercenaries who would fight only for payment in 'good money' – that is, gold or silver. Philip, in turn,



The Spanish Armada

was so bold as to try to topple Elizabeth of England in 1588 with his ill-fated venture known as the Spanish Armada, to say nothing of his sporadic campaigns against the Turks, who had launched an aggressive move into the Balkans and eastern Mediterranean.

These adventures had to be financed. The 37 million ducats of external debt accumulated by Charles V during the 40 years he was king of Spain exceeded by two million ducats the total value of the precious metals assigned to the crown that reached Seville from America in those years. In 1572, the cost of the war in the Netherlands was running at an annual rate of 14.4 million florins, but the Spanish were able to come up with only 7.2 million florins in all of 1572 and 1573. By July 1576, Philip owed the troops 175 million florins. Stretched beyond his means, he ordered payments to his creditors stopped, confiscated two shipments of silver that he owed them, and forced the creditors to convert most of his debts into long-term loans – nearly wiping out the Fugger banking house in the process. Philip's bankruptcy led his army of mercenaries to break apart in mutiny and desertion. It was said that at one point the captain general did not have enough money for lunch. Philip thereby introduced the Western world to the relatively rare but shattering phenomenon of default by a sovereign or, in today's parlance, by a sovereign state. Spain would go on to repeated financial crises in 1596, 1607, 1627 and 1647.

Meanwhile, a lot was going on throughout Europe, not just in Spain. Despite the continuous depredations of warfare and religious turmoil, these unpleasant developments played themselves against the background of the High Renaissance, when artistic and scientific achievements reached extraordinary levels. Leonardo, Tintoretto, Raphael, Palladio, Cellini, Michelangelo, Titian, Dürer, Cervantes and El Greco were all active during the 1500s. The great cathedral of St Peter's rose up on the banks of the Tiber, well splashed with gold in its interior. Copernicus and Galileo were exploring the solar system, while businessmen for the first time took the giant step of using double-entry bookkeeping. This was also a period when Latin was being replaced by vernacular languages, which facilitated communication among the great mass of individuals, including some very rich ones, who had neither been to universities nor joined the clergy.

The single most important event of the century occurred in 1517 when Martin Luther posted his 95 theses on the door of the church in Wittenberg. The Reformation tore through Europe like a hot poker, transforming beliefs and revolutionizing artistic styles while ripping across political and dynastic relationships. The Reformation was in some instances the cause of wars, but war was in any case an almost constant way of life in the 1500s.

England was at war for a total of 14 years. In 1545, at war with Francis I and simultaneously threatened by an invasion from Scotland, Henry VIII had 120,000 men under arms – and drawing pay from his treasury. Henry was forced to borrow money at interest rates as high as 16 per cent and even seized all the lead in the kingdom to be sold

for export. In a vast privatization plan that bears some resemblance to those carried out in many countries during the 1990s, Henry sold off valuable properties that he had grabbed from the monasteries and the churches when he veered into Protestantism after his divorce in 1533. As a final step, Henry resorted to debasement of his currency.

No matter how painful and expensive the wars were, the English spent much less time at war than the Spanish and the French, and the difference probably explains the relatively rapid economic development in England during the reign of the Tudors. The Spaniards fought with France for nearly thirty years. The major struggle was over who would dominate Italy, but that contest was in addition to Spain's ill-fated Armada against England and their brutal campaign to subdue the Netherlands. Religious wars were also fought within nations, many continuing at an unrelenting pace well into the 17th century. The result was repeated repudiation of debts by both the Spanish and the French.

The Europeans did not fight only among themselves. Just beyond the eastern Mediterranean, the Turks launched a sequence of campaigns against Europe that would continue with only brief interruptions for over 100 years. In 1529, the Turks were at the gates of Vienna for the first time. They ravaged Italy and Sicily in the 1530s. They were at war with Venice from 1537 to 1540 and again from 1545 to 1564, but suffered a major naval defeat at Lepanto in Greek waters in 1571.

Most of the military activity within Europe resulted from the ambitions of the great dynasts of the age - Charles V in Spain and Francis I in France. Henry VIII (1509-47), who was eager to establish and maintain the legitimacy of his own dynasty in England, acted as a kibitzer and occasional participant in the struggles between Charles and Francis, continuously playing one against the other. Henry's choices were limited in the early years, as Charles was the nephew of his first wife, Catherine of Aragon, but Henry also played at making alliances with Francis.

France was not among the lucky countries to discover gold in the New World, but France did gain gold from trade and from picking off Spanish galleons on the way to Seville from America. Francis I (1494-1547) was a great believer in the tradition that gold was essential for public relations, for ostentatious display, and for messages of power. His tastes were by no means unusual for an age in which Flemish and Burgundian embroidery was lavished with gold thread, when gold was becoming increasingly visible in church adornment, when nobles' ranks were distinguished by the weight of the gold chains they wore around their necks (Henry VIII had a 'masseye gold cheyne' of 98 ounces), and when knights went into battle wearing doublets strewn with gold and gems.

Francis was an enthusiastic patron of the arts, and when he arranged for Benvenuto Cellini to be released from a Roman prison to come work at the French court, he declared to him, 'I will choke you with gold!' Cellini proceeded to produce a salt cellar so sumptuously decorated with gold and jewels that Francis cried out with astonishment when he saw it.

Francis now considered himself the most powerful monarch in Europe. Nevertheless, in



order to cover himself against Charles V's increasing threats to capture Italy for the Holy Roman Empire, Francis decided to make an alliance with Henry VIII. It is worth mentioning that Charles succeeded in his goal of throwing the French out of Italy. The battle of Pavia in 1525 was a total defeat for Francis, who ended up Charles's prisoner, marking the second occasion on which a king of France became a prisoner of war. Charles was no model of chivalry, as Edward III had been when he captured Jean II: Francis languished for a year in a dank cell in Madrid, where he passed the time writing songs and poems.

Henry and Francis encountered each other face to face in 1520 at Guynes, in the neighborhood of Calais, at a summit meeting that accomplished even less than most summit meetings. Francis was unaware that Charles V had traveled to London for a secret meeting with Henry just prior to Henry's departure for France, which is one reason that the meeting at Guynes ended up as more pomp than circumstance, with the lavish overlay of ceremony and display obscuring empty confrontations on matters of substance.

Henry crossed the Channel aboard the *Henri Grace-de-Dieu*, the Royal Navy's largest ship, accompanied by enough smaller vessels to transport a retinue of 4500 for himself and 1200 for Queen Catherine, to say nothing of 3000 horses and a wide variety of associated equipment. On his arrival at Calais, Henry's chancellor, Cardinal Wolsey, rode on a mule with golden stirrups to the French camp to proclaim the arrival of the English.

The encounter between the two posturing monarchs has come to be known as the Meeting on the Field of Cloth of Gold. The title is apt. The ground was not literally covered with cloth of gold, but the participants lavished gold all over their costumes, and the 2800 tents that Francis supplied for the occasion were covered with so much cloth of gold shimmering in the sunshine that the observer sensed himself immersed in gold.

Five days after Henry's arrival, at the precise moment announced by a gun firing a salute, the two kings and their elaborate entourages began to move toward each other to meet at the appointed spot at Guynes. The French archers rode with weapons sheathed in gold, followed by the marshals of France all shining in cloth of gold. Then came 200 nobles clad in uniforms of gold and crimson. Francis himself wore a cassock of cloth of gold, while his horse was decked out with gold filigree.

The English were not to be outdone in gaudy golden display. Wolsey was accompanied in the procession by 50 giant men carrying gold maces with knobs as big as a man's head. Gold was so much in everyone's consciousness that one of Henry's aides commented that the beard Henry grew for the occasion 'looks like gold'.

The entire setting was marked by fluttering flags and faux palaces, and included two fountains that spouted red wine the whole time. Henry even prompted a spontaneous wrestling match with Francis, a twisting tussle in which Henry ended up on the grass, purple with rage. Elaborate jousts on horseback and archery contests alternated with meals of 'cygnets [swans], venison, pike, heron, pies of pears, custard and fruit . . . kid, sturgeon, peacock, quails, pheasant, egrets'. No wonder the chef was called Merryman.

Withal, Henry and Francis were at war with each other only three years later. Had all the gold flaunted at their summit blinded them from reality? Now they had to use their gold for grimmer purposes.

Profound economic changes were at work in Europe and, in a secondhand kind of way, in Asia as well. The behavior of prices and the demand for money in Europe changed so dramatically in the course of the 1500s that economists refer to this period as the 'Price Revolution of the 16th Century'. The Price Revolution, the ceaseless warfare, the rapid growth in international trade, and expanding economic relations with trading partners thousands of miles away in the Far East galvanized methods of doing business and transformed the character of financial transactions. Regardless of the difficulties that monarchs may have had with their finances in the 16th century, affairs in the private sector reached a far more sophisticated level than at any time in the past.

The Price Revolution defined the tone of the entire century. A pattern of rising prices was first visible in Italy and Germany from about 1470, the low point for the decline in prices that had set in following the Black Death in 1349. Then, like another kind of plague, inflation infected Europe in a series of steps. It took hold in England and France during the 1480s, extended to Iberia in the next decade, and appeared in eastern Europe in the early 1500s. Although prices did not rise in every single year, for agricultural prices in particular are characteristically volatile because of weather variations, the low point reached in each decline tended to be higher than the previous low, while each high point tended to set a record on the upside.

Anyone who has lived through an inflationary period can testify that inflation is always unsettling because it clouds the future with uncertainty, but the shock of sustained inflation to the inhabitants of Europe in the 16th century was shattering. They had no prior experience with it, no good economic theory to explain it, and no established rules of behavior or policy with which to manage it. There had always been brief episodes of inflation in response to crop failures, but the Price Revolution persisted for more than 100 years before it tapered off at long last. No other inflation in history has been so stubborn.

The price increases were most rapid in raw materials, especially food. In England, wood, livestock, and grain rose five- to sevenfold from 1480 to 1650; manufactured goods merely tripled. A 700 per cent increase over 170 years compounds at only 1.2 per cent a year, but, with wages rising less than half as fast as the prices of necessities, it was the tenacity and the duration of the inflationary pressures that shook people. The purchasing power of money and of labor incomes deteriorated at what appeared at the time to be an alarming rate.

What caused this deterioration? A weighty literature has accumulated to record the debates on this matter. The bottom line of the controversy is that no one cause can be held responsible for the long duration of the Price Revolution. The economic historian

Glyn Davies describes it as 'strange and profound'. Contemporary observers in the 16th century engaged in plenty of dispute. Just a few of the causes mentioned in the literature of the time include the decline of agriculture, ruinous taxation, depopulation, market manipulation, high labor costs, vagrancy, luxury and the machinations of businessmen like the Genoese.

Inflation has always appeared during wartime, when spending leaps ahead and production of peacetime goods and services tends to fall. Tacitus wrote that '*Pecunia nervus belli*' (Money is the sinew of war), and there was not a single year of complete peace on the European continent during the 100 years from 1551 to 1651. The fiscal problems of financing these wars were intensified by the character of the 16th-century tax systems, which put almost all the burden on the lower classes. As it was the lower classes who fell furthest behind in the inflationary process, government revenues lagged even as inflation-cum-warfare was constantly driving government expenditures higher. Jumbo fiscal deficits and exploding government indebtedness were the inevitable consequences. Two resulting financial innovations were Spain's *asientos* and France's *Grand Parti*, both of which were forms of borrowing in the capital markets – the modern convention – which supplemented the traditional method of privately negotiated debts that piled up on the accounts of the bankers in Italy, Germany and Holland.

There was another method of royal finance that was by now an old trick: increasing the supply of money through devaluation of the currency. In 1523, the Spanish Cortes urged Charles V to reduce the gold content of Spanish coins to curtail the distressing outflow of their highly valued coins to other countries. That way, they could mint a larger number of coins with the same amount of gold. Charles waited until 1537 to take the step; the magnitude of his needs is apparent from his decision to make this move even after Pizarro and others had provided him with what looked at the time like a bottomless pit of gold bullion. Other rulers followed suit. Henry VIII's policy from 1542 to 1547 was so blatant that his move became known, with the uppercase letters, as The Great Debasement. Henry's debasement was a direct result of the war with France in the 1540s, when, as described by one historian, he 'worked the Mint for all it was worth'.

Monarchies were not the only urgent spenders in this environment. Inflation generates its own self-sustaining urgency. As goods become more valuable to people than money, inflation encourages hoarding by consumers and especially by businessmen and farmers. More elaborately, hoarding takes the form of speculation, in which people buy in advance of their needs or try to corner the market, either to beat anticipated price increases or to resell the goods at a higher price later on. All of this intensifies the upward pressure on prices and then encourages even more hoarding and speculation.

But what about the impact of American treasure on the Price Revolution? Adam Smith had no doubt about that: 'The discovery of the abundant mines of America seems to have

been the sole cause. . . . there has never been any dispute about the fact or the cause of it.' At first glance, it appears obvious that the flood of new money coined from the treasure of the New World must have been the driving force that supported inflation for so long. Population may have outstripped the supply of food, but babies do not normally come into the world with silver spoons in their mouths. If overpopulation were an automatic cause of inflation, countries like India and Bangladesh would have led the world in inflation, while countries with slower-growing populations would have steady or declining prices. The facts fail to fit that hypothesis by a wide margin. In some instances, population growth in excess of food supplies may be a necessary cause of inflation, but it is hardly sufficient. Where does the increased population get the wherewithal to pay the higher prices?

The question suggests an answer: money supplies must increase. That reasoning supported Adam Smith's didactic conclusion that the abundant mines of America were responsible for the price inflation. Smith's view stemmed from a remarkable piece of economic research in 1568 by a French observer named Jean Bodin, who went all the way back into ancient history to demonstrate how rising amounts of gold and silver were associated with higher prices. He pointed out that the prodigious inflow of precious metals from America had landed in Spain and that prices in Spain were higher than in France and Italy: 'Spain is rich, haughty, indolent. . . . It is . . . the abundance of gold and silver that causes, in part, the dearness of things.'

Bodin is the spiritual father of monetarism, an important branch of economic theory spelled out most authoritatively by Nobel laureate Milton Friedman, who has asserted that inflation is always and everywhere a monetary phenomenon. When prices in general are rising, the buyer has to spend more money for the same basket of goods and services. That is, inflation cannot continue unless it is financed in some fashion. If buyers cannot find the extra money they need to maintain the same level of purchasing, they will have to cut back and buy fewer things, thereby limiting the ability of sellers to keep raising prices. Monetarists therefore contend that the Price Revolution of the 16th Century would never have endured for such a long period of time had it not been nourished by the increased money supply produced from the New World's gold and silver bullion.

Yet fitting the facts to that monetarist theory is not as easy as Bodin made it appear. Not all the treasure remained in Europe as money. Hoarding, as always, kept some of it out of circulation. Gorgeous ornamentation in the churches took a share. And, as we shall see, a significant amount was shipped off to Asia, never to return.

Furthermore, although prices started rising around 1470 and were climbing throughout Europe by 1500, American gold did not arrive in Spain in any significant quantity until 1520; the Peruvian discoveries took place after 1530, and the big silver discoveries did not begin to bear fruit until about 20 years after that. The relationship continues to be confusing after 1600. The imports of gold and silver to Seville appear to

have peaked about 1590, to have remained at a high level for another 30 years or so, and then to have fallen off precipitously from around 1620 to the end of the century. Yet prices kept on climbing, at rates that seemed to bear no relation to the arrivals of fresh supplies of the precious metals. In England, for example, prices doubled between 1600 and 1650.

However, the Seville data are questionable, because an increasing supply of gold and silver was unloaded downstream at Cadiz or in Lisbon, and illegal diversions from official routes were also growing. Gold is too easy to smuggle for official statistics to be trustworthy. Analysis of informal kinds of information concludes that the flow of precious metals to Spain actually increased after 1600. The huge Brazilian gold mines began shipping to Portugal after 1700, but by that time the Price Revolution had played itself out. Even so eminent an authority on monetarism as Anna Schwartz, one of Friedman's principal colleagues, has described the experience of the Price Revolution as a 'contradiction of the basic hypothesis' of monetarism.

Other economists take issue with monetarism's focus on just one economic variable. They prefer to turn the argument upside down. The Price Revolution of the 16th Century, viewed from this vantage point, was not the result of an increased supply of money in the form of the precious metals; rather, the rising price level magnified the demand for money, arousing the Spaniards to redouble their efforts to bring in gold and silver from America. Seen from this perspective, expanding money supplies are not the cause of inflation but the result of it.

Whichever and whatever was the cause of the Price Revolution, the inflow of treasure must have contributed to its persistence. A dramatic example of this phenomenon appeared in the course of the long struggle between Francis I and Charles V.

After Charles captured Francis at Pavia and imprisoned him in Madrid, he forced Francis to sign a ruinous treaty that was to be secured by Francis's two oldest sons, seven and eight years old, as hostages. Goodbye to the French claim to Lombardy! The children remained prisoners for four years, until they were ransomed by Francis's promise to pay two million gold crowns to liberate them. Payment began with 1.2 million crowns sent in a boat that crossed the river at the French-Spanish border at the precise moment as the boat carrying the princes headed in the other direction. The exchange was delayed until four months had been spent counting the coins.

This enormous transfer of gold put a painful squeeze on the French nobility, clergy, and taxpayers while stimulating a wave of spending in the Spanish economy. As a result, prices in Spain were soon higher than prices in France. The resulting disparity in prices produced such a surge in French exports to Spain – everything from wheat, wine, and brandy to hammocks, candles, and canvas – that the transfusion of gold and silver was soon running in reverse as monies returned to the French side of the border.

The most curious aspect of this whole sequence of events was what happened to the value of gold itself, and silver along with it. The precious metals were no more immune than anything else to the inexorable influences of the law of supply and demand. The European supply of gold in the 16th century expanded rapidly, as the deluge of gold imports from America joined with the flow of gold from new mines and improved mining technology in eastern Europe and especially in Hungary.

Consequently, although the price of gold rose along with everything else in the course of the Price Revolution, the gold price changes were much more subdued. For example, the price of gold in England, reported in shillings, climbed from 40 to 60 shillings an ounce between 1492 and 1547, a rise of 50 per cent, then was stable for the next 50 years, and then had a further modest increase to 74 shillings by 1611. That was a total advance of 85 per cent – way below the increases in wages, clothing, or food. Although the English had no gold resources of their own in the Americas or anywhere else to draw upon, there is reason to believe that the supply of gold in England increased at a rapid pace nevertheless, thanks to huge gains of treasure through piracy and war.

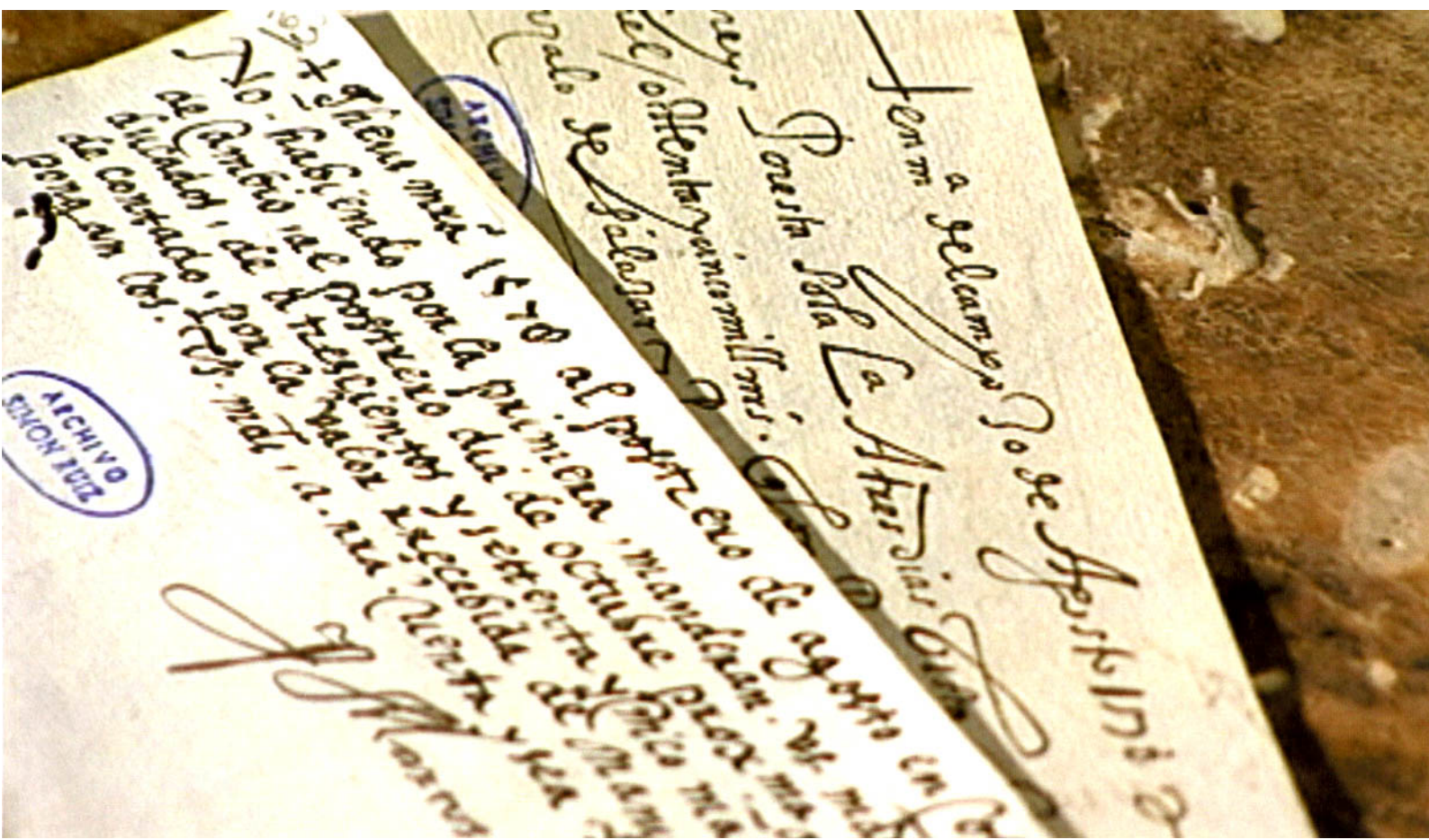
The Price Revolution and the discovery of the copious gold resources of America were abrupt shifts from the past, but additional significant economic innovations were at work during the sixteenth century. Indeed, the trade fair, a traditional institution, now began to play a more important role in the economic scene, initiating a mutation in the role of gold that has continued to the present day. The expansion of this institution is notable for its civilized character, which made a dramatic contrast to the warfare, religious controversy, pillage and plunder that persisted throughout these centuries.

Beginning in the Middle Ages, the trade fair developed into an essential institution for doing business – displaying wares, buying them, and selling them in a world where most towns were tiny places, without a bank on every street corner and without a supermarket just a five-minute drive from the house. It was also a world without telephones, Federal Express, the Internet or news services that could quote or advertise the prices of goods, financial instruments, and foreign exchange. Without central gathering places, merchants could not supply themselves or were limited to only one or perhaps two local sources; large customers could not locate the merchandise – and, more often, the bankers – they needed; the multiplicity of monies and of credit instruments could not be settled to pay the obligations that accumulated in the course of thousands of transactions. In today's world, the annual Frankfurt book fair, the colorful expos in Las Vegas for the high-tech industry, and the longstanding Leipzig fair for industrial machinery are pale shadows of this vital and essential institution of earlier times. Furthermore, most of these modern fairs meet annually, whereas the fairs that concern us here met at least twice a year; Lyons, one of the major locations, held its fair four times a year.

Unlike local situations, where merchants exchanged wares with their neighbors, transactions at trade fairs would permit a merchant to buy without necessarily having something to sell, or vice versa. One-sided purchases often require financing, because the buyer is not making an offsetting sale. The format of the fairs thus became increasingly elaborate, and during the 16th century, financing the purchases at the fairs became as important as merchandise itself. In many instances, the transactions were financial without regard to any movements of goods. Although fairs took place in many centers in Europe, both geographical and political factors determined the locations of the key towns where the major players met and where kings provided special protection and facilities for foreigners; at Lyons, the great preponderance of foreigners were from Florence, Milan, Lucca and Genoa. At the great fairs, the merchants and financiers were dominant, with representatives of municipal or royal institutions playing a subordinate role. The popularity of the fairs rose and fell as trade patterns and political hegemonies varied over time – the Champagne area, Antwerp in the 15th century, then Geneva, from which a French king lured the fair to Lyons, and then to the eastern French town of Besançon and finally Piacenza. In Piacenza, the fairs were known as Bisenzone, an Italianization of Besançon.

Many booths at the fairs were occupied by the money changers; in the fair of Medina del Campo in Spain, trading in promissory notes drawn in different nations' currencies, called bills of exchange, was the only activity. The money changers must have had a busy

*Medieval bills of exchange
– the forerunners of
modern banknotes*



time indeed. One authority lists 48 different kinds of gold coins circulating in Europe in the 16th century, including 11 from Italian cities, 9 from the Netherlands, 6 from England and smaller numbers from Spain, France, Portugal and Hungary.

Merchants struggling to deal with this multiplicity of monies were the targets of many popular jokes. In Chaucer's 'Shipman's Tale', the merchant is so involved with his counting board that he leaves word not to be disturbed, no matter what. A young monk takes advantage of this situation to make advances on the merchant's lusty wife.

Nevertheless, the money changers were much less involved with developments in coinage than with the increasing substitution of paper-money instruments for the inconvenience and complications of payment in coin. The principal vehicle for these kinds of payment was the bill of exchange, an instrument developed by the Italians in the 13th century, perhaps earlier. This was a remarkable financial innovation that lent itself to a wide variety of uses and formats.

Here is a simple example of how a bill of exchange worked.

Two transactions take place: Franco in Italy buys wool from Berthold in Flanders, while David in Flanders buys wine from Carlo in Italy. Franco, however, does not pay Berthold directly, and David does not pay Carlo directly. Instead, Carlo 'draws' a bill of exchange on David, a sheet of paper declaring that David owes him such-and-such an amount of Italian money for the wine. Carlo sells



Medieval helmet

this bill to Franco, which means that Franco's purchase of the bill has satisfied Carlo. In order to pay Berthold for the wool, Franco now sends that bill to Berthold, who turns around and sells it to David, which means that David's purchase of the bill has satisfied Berthold. Thus, both shippers, Berthold and Carlo, have been paid, but by the other fellow's customer rather than by their own: Franco has paid Carlo instead of Berthold, while David has paid Berthold instead of Carlo. The wine, the wool, and the bill that Carlo has drawn on David move across the borders, but no money goes from Italy to Flanders or vice versa.

This is an oversimplified explanation, but it indicates the essence of the process. In reality, there is no reason to believe that each transaction will precisely equal the other, or even that Franco and Carlo or Berthold and David will readily find each other. To settle up these differences, a lively market built up in the trading of bills of exchange. In 1585, for example, bills drawn on merchants and bankers in Amsterdam were trading in Antwerp, Cologne, Danzig, Hamburg, Lisbon, Lübeck, Rouen and Seville.

In these markets, dealers rather than principals would buy the bills and then would settle up the balances among themselves; dealers often acted as bankers by advancing payments to suppliers of merchandise and collecting later from the buyers of the merchandise. By settling differences rather than gross amounts and by making the business one in which a large number of dealers participated, these bill markets significantly reduced the need for coins to settle bilateral differences. On one occasion, a million livres tournois changed hands without a single penny being disbursed. That entire process, however, could not have functioned as well without the institution of the trade fairs, where the dealers and money changers could meet with one another, buying and selling bills back and forth, and working out their payments in foreign exchange as Italians settled with Flemish dealers, the Flemish settled with the English and so on.

Remarkable changes developed from these arrangements. Merchants no longer had to travel to settle up their accounts, and when they did travel they went to centers with trading posts where transactions could be settled most efficiently. Consequently, the centralized operations of the fairs attracted an increasing volume of financial transactions. Merchant firms became more diversified as a result and in time turned into the great family firms that grew up in this age, such as the Fuggers in the Holy Roman Empire, the Medici of Florence, and later the Rothschilds and Baring Brothers.

The whole concept of money was being transformed. The traditional public money of the prince in the form of coins stamped or engraved as official government issue now shared the money circulation with private money in the form of credit instruments that served as means of payment in transactions involving both merchants and bankers. When an individual in the modern world engages in the dominant form of doing business by writing a check instead of paying with paper currency, that is private money at work. That arrangement first developed during the 15th and 16th centuries with the growing use of

the bill of exchange and the trade fairs where transactions were cleared and settled and where foreign exchange trading became a major activity.

The private money had to be expressed in some kind of denomination, just as people keep money in bank deposits or write checks today denominated in dollars or sterling or euros. Nobody issues a check denominated in a given number of specified gold coins or weight of gold bullion, any more than someone in the 16th century who drew a bill of exchange would denominate it in a number of coins or weight of bullion. Money in the private world had to be expressed in terms of a unit of account, such as dollars or euros, which was a convenient numéraire for defining the size of the transaction and the local money used by the parties to settle up. A unit of account is an abstract concept – you cannot see the dollars that a check transfers, nor can you feel them, bite into them, or weigh them. The only concern of the owner of private money is that the prince regulates the supply of public money so that the integrity of the unit of account is stable instead of withering away in the fires of inflation.